

with lipids and during metabolic transformations of lipoproteins in circulation. Elucidation of the conformational changes at the atomic level will be a major challenge dependent on the success of high-resolution analysis of apolipoprotein and lipoprotein structures.

- (3) High-resolution determination of the three-dimensional structure of intact apolipoproteins and lipoproteins, under physiologic conditions. Accomplishing this goal will require not only major advances in the crystallization, X-ray, and NMR methodologies, but also the isolation or preparation of highly homogeneous reconstituted or native lipoproteins.
- (4) Further developments in computer modeling of reconstituted or native lipoproteins will require powerful molecular dynamics algorithms, and much more extensive experimental information on the folding and topology of the apolipoproteins. Especially promising are mass spectrometric methods combined with specific or random chemical cross-linking of apolipoproteins to establish spatial proximity.

Abbreviations

apo	apolipoprotein
CM	chylomicrons
HDL	high-density lipoproteins
K_d	equilibrium dissociation constant
LCAT	lecithin cholesterol acyltransferase
LDL	low-density lipoproteins
Lp(a)	lipoprotein(a)
PC	phosphatidylcholine
PL	phospholipid
SR-BI receptor	scavenger receptor BI
T_m	main phase transition temperature
VLDL	very low-density lipoproteins

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Assembly and secretion of lipoproteins

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1. Overview of lipoprotein secretion into the circulation

Lipoproteins are secreted into the circulation from hepatocytes of the liver and enterocytes of the intestine. All plasma lipoproteins share a common structure consisting of a neutral lipid core of triacylglycerols (TGs) and cholesteryl esters surrounded by a surface monolayer of phospholipids, unesterified cholesterol and specific proteins called apolipoproteins (Fig. 1 and Chapter 18). The primary function of plasma lipoproteins is to transport hydrophobic, water-insoluble lipids in the circulation. The TG-rich lipoproteins deliver TGs made in the liver and intestine to other tissues in the body for either storage or utilization as an energy source. High-density lipoproteins are thought to remove cholesterol from peripheral tissues (i.e. tissues other than liver) and deliver it to the liver for excretion in bile in a process referred to as 'reverse cholesterol transport' (Chapters 16, 20). A high level of plasma lipids is a strong risk factor for development of cardiovascular disease (Chapter 22). The steady state level of lipoproteins in the cir-

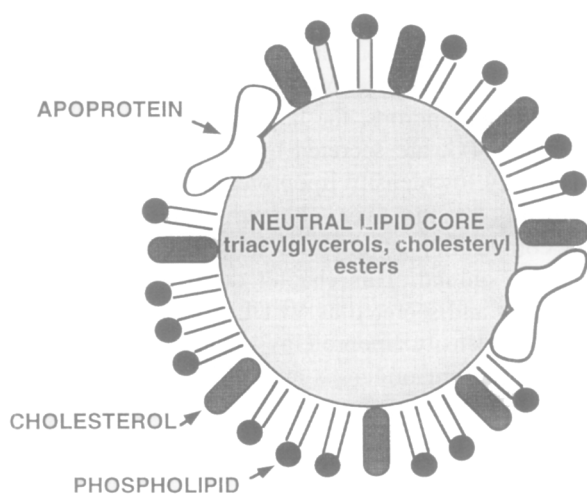


Fig. 1. Generalized structure of a lipoprotein particle. Lipoproteins are approximately spherical particles that consist of a neutral lipid core (triacylglycerols and cholesteryl esters) surrounded by a surface monolayer of amphipathic lipids (unesterified cholesterol and phospholipids) and specific apoproteins. Courtesy of Dr. J. Segrest, University of Alabama, with permission.

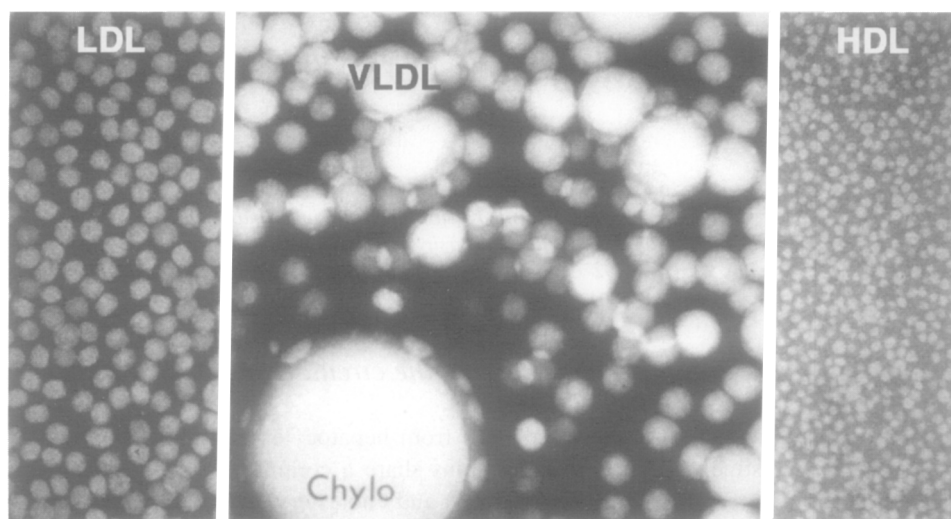


Fig. 2. Negative staining electron micrographs of human plasma lipoproteins (diameters 10–1000 nm). The largest particles [chylomicrons (Chylo) and VLDLs] contain a higher ratio of lipid to protein, and are therefore less dense, than low-density lipoproteins (LDLs) and high-density lipoproteins (HDLs) which contain relatively more protein. Photograph courtesy of Dr. R. Hamilton, University of California at San Francisco, with permission.

culation results from a balance between their rates of secretion into, and removal from, plasma. This chapter focuses on the mechanisms by which lipoproteins are assembled and secreted from the liver and intestine. The synthesis of the constituent lipids of the lipoproteins is described in Chapters 8 (phospholipids), 15 (cholesterol) and 10 (TGs).

The major plasma lipoproteins (Fig. 2) are usually classified according to density (table 1 in Chapter 18). Since lipids have lower buoyant densities than proteins, lipoproteins that have a high ratio of lipid to protein have lower density than lipoproteins having a lower ratio of lipid to protein. Chylomicrons, the largest and most lipid-rich particles, whose major lipid component is TG, are secreted by the intestine and are abundant in plasma only after a meal. Very low-density lipoproteins (VLDLs), which are also rich in TG, are secreted mainly by the liver, although some are also of intestinal origin. The size and lipid composition of chylomicrons and VLDLs vary according to the nutritional status of the animal. This chapter discusses how lipids are assembled with apolipoprotein (apo) B and secreted as VLDLs and chylomicrons. Intermediate-density lipoproteins and low-density lipoproteins (LDLs) are generated in the circulation by lipolysis of TGs within chylomicrons and VLDLs. High-density lipoproteins are particles of diverse composition that are formed in the circulation (Chapter 20).

Apo B is an essential component of chylomicrons, VLDLs and low-density lipoproteins. Unlike the 'exchangeable' plasma apolipoproteins (e.g. apos E, A1, A2, C), apo B does not exchange among lipoproteins and is present in plasma only when associated with lipids. In addition to apo B, VLDLs and chylomicrons contain apo E and apos C, and chylomicrons also contain some apos A1 and A2.

In many respects, the secretion of apo B from cells follows a pathway identical to that of typical secretory proteins. The mRNA encoding apo B is translated on ribosomes bound to the endoplasmic reticulum (ER) and the protein is translocated across the ER membrane into the lumen. A unique difference between apo B and other secretory proteins is that apo B must be non-covalently associated with lipids before secretion. Newly formed apo B-containing lipoproteins undergo vesicular transport from the ER through the Golgi stacks. Secretory vesicles, containing lipoproteins, are formed in the *trans*-Golgi network and subsequently fuse with the plasma membrane releasing apo B-containing lipoproteins from the cell. Evidence that apo B transits the same secretory route as other secretory proteins was provided by electron microscopy studies of rat liver in which lipid droplets the size of VLDL were detected in the lumen of the ER and Golgi [1]. In cultured rat hepatocytes, synthesis of an apo B molecule takes 7 to 15 min and apo B appears in the culture medium 30 to 40 min later. Pulse-labeling experiments indicate that the slow step in passage of apo B through the secretory pathway is movement of the protein out of the ER (R.A. Davis, 1987).

2. Structural features of apolipoprotein B

Full-length human apo B, designated as apo B100, is one of the largest single polypeptide chains known with 4536 amino acids (M_r 513,000) [2]. In humans, the liver secretes exclusively apo B100 whereas the intestine secretes primarily a truncated form of apo B100, called apo B48 (M_r 264,000), along with some apo B100 (J.P. Kane, 1983). Rodent livers, in contrast, secrete both apo B100 and apo B48. A convenient 'centile' nomenclature for apo B variants is commonly used in which apo B species are designated by a number indicating the size of the molecule relative to full-length apo B100. Therefore, in this shorthand scheme, apo B48 refers to the N-terminal 48% of apo B100 and apo B15 refers to the N-terminal 15% of apo B100.

Apo B100 is an amphipathic protein. Some of the complex molecular interactions required for assembly of this protein with lipids can be inferred from its unique domain structure. Indeed, apo B is the only protein known to require association with a lipid microemulsion for secretion. Each VLDL particle contains a single molecule of apo B (J. Elovson, 1988). Computer-based structural analysis of apo B100 predicts a pentapartite structure (Fig. 3) (R. Nolte, 1994) [3] consisting of a globular N-terminal domain followed by alternating amphipathic β -sheets and α -helices. As much as 40–50% of apo B100 is β -sheet structure, which is concentrated in two large regions. At least 25% of the molecule consists of α -helices which are too short to span a membrane. It is likely that the surface of a VLDL particle does not contain sufficient phospholipid molecules to completely cover the surface and, consequently, apo B is probably in direct contact with some TG in the particle core. The β -strands are thought to be the major motif mediating association with lipids. Since the amphipathic β -sheets of apo B have both polar and non-polar faces, the prediction is that the non-polar face interacts with neutral lipids, such as TGs, whereas the polar face is exposed to the aqueous environment.

A general correlation exists between the length of the apo B molecule and its ability

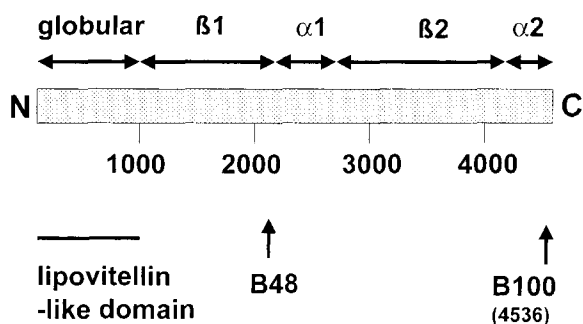


Fig. 3. Schematic representation of the pentapartite structure of apo B100. The N-terminal globular domain and the alternating α and β domains are indicated above the representation of the polypeptide. Numerical values below the polypeptide indicate an amino acid residue scale. Human apo B100 contains 4536 amino acids; the size of apo B48 is indicated. The N-terminal 1000 amino acid residues of apo B100 are homologous with the lipid-binding pocket of lamprey lipovitellin.

to associate with lipids. Naturally occurring mutations in the human apo B gene, in the group of diseases known as hypobetalipoproteinemia, show that secretion of some truncated forms of apo B as lipoprotein particles is severely impaired [4] (S.G. Young, 1993). In cultured rat hepatoma cells transfected with cDNAs encoding C-terminally truncated variants of human apo B100 (from apo B15 to B94) the size and density of the secreted apo B-containing particles are dependent upon the length of the apo B. For example, apo B18 is secreted essentially without neutral lipids whereas apo B28 is secreted in particles the size and density of high-density lipoproteins. As the size of the apo B molecule increases, from B37 to B48 to B53 to B72 to B94, the apo B associates with progressively more lipid in larger and less dense particles [5]. Thus, the sequence of apo B between apo B17 and B41 is necessary for assembly of TG-rich lipoproteins. Sequences in the C-terminus of apo B29 bind phospholipids and diacylglycerols but only a little TG. TGs, however, associate with the region between apo B29 and B32.5, and a massive accumulation of TGs occurs between B32.5 and B41 where ~ 1 molecule of TG binds for each 2 amino acids of apo B (D.M. Small, 2000).

The four alternating α -helices and β -sheets (Fig. 3) are common features of apo B100 in all vertebrate species examined. Amphipathic β -strands are also found in the egg yolk protein, lipovitellin. The X-ray crystal structure of lamprey lipovitellin reveals a 'lipid pocket' joined by three anti-parallel β -sheets. Based on the prediction that human apo B100 contains a similar lipid-binding pocket at the N-terminal domain of the molecule, a model has been proposed in which this pocket provides the initial site of association of TGs with apo B. Sequential accumulation of TGs within this cavity, in concert with expansion of the pocket, is thought to occur by unfolding of the amphipathic β -strands of the β 1-domain (J. Segrest, 1999). Not explained by these studies or models, however, is the reason why apo B48 assembles into chylomicrons, the largest and most TG-rich particles, whereas apo B100, which is approximately twice as large as apo B48, forms the smaller VLDLs.

3. *Transcriptional regulation of apo B synthesis*

3.1. *DNA elements regulating apo B transcription*

Under most metabolic conditions, changes in apo B mRNA levels do not influence how much apo B is secreted. Moreover, changes in apo B secretion, over at least a 7-fold range, occur without alteration in the amount of apo B mRNA (J. Scott, 1989). Consequently, it is generally accepted that apo B secretion is primarily regulated co- and post-translationally.

The liver and intestine produce the majority of apo B, but in mice the production of apo B in the yolk sac is crucial during fetal development. It is thought that apo B plays a role in transferring lipid nutrients to the developing mouse embryo (R.V. Farese, 1999).

Interestingly, apo B is also expressed at low levels in the hearts of humans and mice (S.G. Young, 1998). The presence of apo B and microsomal triacylglycerol transfer protein (which is also required for lipoprotein assembly, Section 6.4) in the heart presumably leads to the secretion of apo B100-containing lipoproteins by cardiac myocytes. Recent experiments suggest that secretion of TG-rich lipoproteins by the heart is a mechanism for regulating the amount of TGs stored in the heart and thereby protects the heart from a detrimental accumulation of lipids [6].

The human apo B gene spans 43 kb, is located on chromosome 2p and consists of 28 introns and 29 exons. Over half of the apo B coding sequence is encoded by one exceptionally large exon, exon 26, which contains 7572 bp. The apo B mRNA transcript is 14 kb and is remarkably stable, having a half-life of 16 h (J. Scott, 1989). Some elements responsible for expression of the human apo B gene have been defined. The gene promoter (bp -898 to +1) contains a TATA box. Within the promoter region are sequences that bind the liver-specific transcription factors HNF-3 and NF-1. Positive and negative *cis*-acting DNA sequences required for transcription of apo B in the liver were identified using transient transfection assays of promoter-reporter constructs in cultured human hepatoma (HepG2) cells and intestinal (CaCo2) cells. The majority of enhancer activity was localized to 443 bp within intron 2 [7]. By footprint analysis this enhancer region was shown to bind two nuclear transcription factors, HNF-1 and C/EBP. An additional, weaker, enhancer was found within intron 3, and a reducer element was identified in the 5'-flanking region between bp -3211 and -1802. However, when these promoter elements were linked to reporter genes and expressed in transgenic mice, a more complex picture emerged. In mice, both the proximal promoter (bp -898 to +1) and intron 2 enhancer were required for expression of a β -galactosidase reporter in liver, yet surprisingly no intestinal expression occurred even though the same constructs induced expression in the intestinal cell line, CaCo2. Moreover, the reducer element (bp -3200 to -1802) that was identified in transient transfection studies in HepG2 cells, did not decrease expression in livers of transgenic mice. Consequently, while CaCo2 and HepG2 cells are frequently used as experimental models of the intestine and liver, respectively, it is important to verify in intact animals results obtained from cultured cells.

Recently, factors controlling the intestinal expression of apo B have been described.

A high level of intestinal expression of human apo B in transgenic mice was achieved with a construct containing ~80 kb of the 5'-flanking sequence [8]. The region of the gene driving expression of apo B transgenes in mouse intestine was subsequently localized to a 3 kb segment over 55 kb upstream of the transcriptional start site [9,10]. This region contains putative binding sites for the transcription factors HNF-3 β , HNF-4 and C/EBP β ; the corresponding segment of the murine apo B gene exhibits a high degree of sequence conservation. Interestingly, elements of the apo B gene that had been previously demonstrated *in vivo* to confer expression in the liver (i.e. intron 2 enhancer and 5' upstream liver enhancer) do not play a role in intestinal expression of apo B transgenes in mice.

3.2. Apo B mRNA editing

Apo B100 contains 4536 amino acid residues and in humans is the only form of apo B produced by the liver. In contrast, the small intestine of all mammals, and the liver of some species, synthesize apo B48, the N-terminal 2152 residues of apo B100. Apo B100 and apo B48 are generated from a single structural gene. Subsequently, apo B48 is produced by an unusual RNA editing process in which a single cytidine at position 6666 of human apo B mRNA is deaminated to a uridine [11]. A stop codon is thereby introduced and the truncated variant, apo B48, is generated. The *cis*-acting elements directing this site-specific deamination are located within approximately 25 nucleotides in an AU-rich region flanking the target cytidine. A region of 11 nucleotides 5 bases downstream of this cytidine, is particularly important for editing and has been called the 'mooring sequence'.

A 27 kDa protein from enterocytes is required for apo B RNA editing [12]. This protein, called apobec-1 (for 'apo B editing complex-1'), is highly homologous to other cytidine deaminases but does not, by itself, edit the RNA. Rather, apobec-1 is the catalytic subunit of a multi-protein complex containing auxiliary proteins. Apobec-1 is a homodimer that is present in both the cytosol and nucleus and apo B48 is formed only in tissues that express apobec-1 mRNA (i.e. human and rodent intestine and rodent liver). However, apobec-1 is also expressed in some tissues that do not make apo B, suggesting that this enzyme also acts on other mRNAs in addition to apo B mRNA. This idea is supported by the finding that transgenic mice over-expressing apobec-1 develop hepatocellular carcinomas and liver dysplasia (T.L. Innerarity, 1995).

Apo B48-containing lipoproteins are removed from plasma more rapidly than apo B100-containing lipoproteins since apo E, which is added to apo B48-containing lipoproteins after secretion, is a high-affinity ligand for the LDL receptor (Chapter 21). It was reasoned, therefore, that if apo B100 secretion were blocked, by inducing the complete editing of apo B mRNA so that only apo B48 was produced, fewer circulating apo B-containing lipoproteins would be present and, consequently, the animals would be less susceptible than normal to diet-induced hypercholesterolemia. As predicted, when the apobec-1 gene was over-expressed in livers of atherosclerosis-prone mice via adenovirus-mediated gene transfer, plasma apo B100-containing lipoproteins and cholesterol were greatly decreased and the extent of atherosclerosis was reduced (B. Teng, 1994).

On the other hand, targeted disruption of the murine apobec-1 gene abolished apo B editing in all tissues and resulted in a complete deficiency of apo B48 in serum (N.O. Davidson, 1996) demonstrating that there is no functional duplication of the apobec-1 gene. Apobec-1^{-/-} mice are fertile, healthy, and serum levels of cholesterol and TG are normal showing that the apobec-1 gene is not essential for normal life. Moreover, mice that express only apo B48 (by replacement of the apo B48-editing codon with a stop codon) show no obvious defects in growth, reproduction or function (S.G. Young, 1996). A question that remains is: why is apo B100 produced at all?

4. Models used for studying apo B and VLDL secretion

The most commonly used models for studying the assembly and secretion of apo B-containing lipoproteins are hepatoma cell lines (HepG2 human hepatoma cells and McArdle 7777 rat hepatoma cells) and primary hepatocytes from rats, mice or hamsters. Each model has its limitations. In addition, several transgenic and gene-targeted mouse models have been developed.

Hepatoma cell lines are convenient laboratory models since they are easily maintained and manipulated under defined conditions. However, although hepatoma cells have retained many properties of hepatocytes, several key liver functions, including some important aspects of lipid metabolism, have been altered. One advantage of HepG2 cells is that they are of human origin and, like human liver, secrete apo B100 but not apo B48. However, the basal rate of apo B secretion in HepG2 cells is low, and the apo B-containing lipoproteins are not true VLDLs but contain less lipid than VLDLs. Thus, the use of HepG2 cells for studying VLDL assembly is not ideal. McArdle 7777 hepatoma cells are of rat origin and, like rat liver, secrete both apo B100 and B48. When incubated with oleate some of the secreted apo B is in VLDLs, but the majority of apo B is secreted in more dense, partially lipidated, lipoproteins. One advantage of McArdle cells is that they are excellent cells in which to express heterologous cDNAs, such as those encoding truncated and mutated forms of apo B.

Primary hepatocytes isolated from livers of rats, mice and hamsters are also commonly used for studying VLDL assembly and secretion. Although these cells must be freshly isolated for each experiment, they retain most properties of native hepatocytes, at least for 24 h. Unlike human hepatocytes, primary rat and mouse hepatocytes secrete both apo B48 and apo B100 but these particles are the size, composition and density of VLDLs. Hamster primary hepatocytes are also useful since they secrete VLDLs that contain apo B100, but not apo B48 (K. Adeli, 2000). However, stable transfection of primary hepatocytes with cDNAs is not possible.

Several gene-targeted mice have been effectively used to study apo B metabolism [4]. Expression of human apo B in mice fed a high-fat diet resulted in a 20-fold increase in the plasma content of apo B-containing lipoproteins and extensive atherosclerosis. Since the secretion of apo B from hepatocytes of these mice was only slightly higher than in wild-type mice the lipoprotein accumulation and atherosclerosis were attributed primarily to defective lipoprotein clearance. Another line of transgenic mice was produced in which Arg-3500 of apo B (the proposed site of binding to the LDL

receptor) was mutated. These mice were unable to clear apo B-containing lipoproteins from the plasma and developed severe hypercholesterolemia.

Generation of apo B knock-out mice was problematic since most apo B^{-/-} embryos died during development. Nevertheless, mice lacking the murine apo B gene, but expressing human apo B, were generated. During the suckling period the offspring were indistinguishable from wild-type mice but because the human apo B gene was not expressed in the intestine (Section 3.1) the mice did not produce chylomicrons and suffered severe fat malabsorption and retarded growth [8].

The disease hypobetalipoproteinemia is caused by point mutations in the apo B gene that result in production of low levels of poorly lipidated, truncated apos B. Several mouse models of this disorder have been generated. For example, transgenic mice expressing apos B70, B81 or B83 have low levels of plasma apo B-containing lipoproteins and develop severe neurodevelopmental abnormalities [4].

5. Covalent modifications of apo B

Several co- and post-translational modifications of apo B occur. Human apo B100 contains 25 cysteine residues of which 16 exist as intramolecular disulfide bonds; 14 of these are clustered in the N-terminal region. Disulfide bond formation likely occurs co-translationally by the action of protein disulfide isomerase, a luminal ER protein that coordinates disulfide bond formation and protein folding. As discussed in Section 6.4, protein disulfide isomerase is one subunit of the microsomal TG transfer protein heterodimer that is required for efficient secretion of apo B-containing lipoproteins. To determine if disulfide bond formation in the N-terminus of apo B were required for assembly of lipid with apo B, individual cysteine residues were replaced with serines or alanines. When either or two cysteine pairs in this region was eliminated, lipid assembly and secretion of apo B were greatly reduced (G.S. Shelness, 1997; Z. Yao, 1998) implying that disulfide bonds in this region are critical for the correct folding and secretion of apo B.

Plasma apo B100 contains 8–10% by weight carbohydrate and at least 20 potential *N*-linked glycosylation sites. Both *N*- and *O*-linked glycosylations occur in apo B100. *N*-linked glycosylation begins co-translationally in the ER lumen but it is not yet clear whether or not glycosylation is required for apo B secretion. Treatment of chicken hepatocytes with tunicamycin, which inhibits formation of *N*-linked oligosaccharide chains, did not reduce the amount of apo B secreted (D.M. Lane, 1982). However, when HepG2 cells were incubated with tunicamycin, apo B100 secretion was reduced and the intracellular degradation of apo B (Section 7) was increased. Moreover, the apo B100-containing particles that were secreted had a normal buoyant density suggesting that *N*-linked glycosylation is not required for assembly of lipids with apo B100 (L. Chan, 2001). An alternative explanation, however, is that the decreased secretion of apo B100 resulted from inadequate lipidation of apo B so that defective particles underwent pre-secretory degradation.

Several studies have shown that secreted apo B contains multiple phosphorylated serine residues (R.A. Davis, 1984) and that the phosphorylation is increased in diabetic,

compared to non-diabetic, rats (J.D. Sparks, 1990). The physiological relevance of this phosphorylation is not known. Apo B phosphorylation has been suggested to occur in the Golgi apparatus in agreement with the finding that the Golgi contains kinases that phosphorylate secretory proteins (L.L. Swift, 1996).

Human apo B secreted by HepG2 cells is also covalently acylated with palmitic acid on at least one cysteine residue (J.M. Hoeg, 1988). Mutation of Cys-1085 in human apo B29 expressed in McArdle hepatoma cells abolished palmitoylation at this site. Interestingly, the secreted lipoproteins containing the mutated apo B29 were associated with less TG than those containing apo B29 with palmitoylated Cys-1085 [13]. These studies suggest that palmitoylation of apo B might play a role in assembling neutral lipids with apo B. However, no information is available on how many cysteines of apo B are palmitoylated, which palmitoyltransferase is involved or the topology of the palmitoylation reaction.

6. Regulation of apo B secretion by lipid supply

6.1. Fatty acids and triacylglycerols

In most instances, apo B is synthesized constitutively and in excess of the amount secreted so that the rate of apo B synthesis does not regulate how much apo B is secreted. The assembly of apo B-containing lipoproteins requires the synthesis of apo B, TGs, cholesterol, cholesteryl esters and phospholipids. When insufficient lipid is available for assembly of apo B into stable lipoprotein particles, excess apo B is not secreted but is degraded intracellularly, primarily by the proteasome and also by undefined proteases within the lumen of the secretory pathway (Section 7).

When the fatty acid oleate is added to HepG2 cells the synthesis of TG and phospholipids is stimulated and the amounts of apo B and TG secreted are increased [14]. Thus, an increased lipid supply enables a larger proportion of newly synthesized apo B to be translocated across the ER membrane and enter the secretory pathway. In agreement with the idea that lipid supply can regulate apo B secretion, the oleic-acid-induced increase in apo B secretion is blocked by Triacsin D, an inhibitor of TG synthesis. The rate of apo B secretion is not, however, solely a function of the rate of TG synthesis since glucose stimulates both TG synthesis and secretion, but does not increase apo B secretion. Furthermore, when oleic acid is supplied to primary rat hepatocytes, in contrast to HepG2 cells, both the synthesis and secretion of TG are increased whereas apo B secretion is unaltered. Therefore, the difference in response of apo B secretion to oleate in HepG2 cells versus primary hepatocytes cannot be ascribed to differences in TG synthesis.

Early work suggested that two pools of TG exist in hepatocytes: a large cytosolic storage pool that turns over slowly, and a small microsomal pool that turns over rapidly. The source of TG used for assembly with apo B appears to be primarily (perhaps ~70%) the cytosolic storage pool rather than the pool made from de novo synthesis [15]. Recent data support a model in which cytosolic TG is hydrolyzed, perhaps by the recently discovered TG hydrolase in the ER lumen [16]. The mechanism by which

an ER luminal lipase accesses cytosolic TG is not clear but the small amount of TG (~5% of total membrane lipids) in the ER membrane is a potential source of this TG. The lipolysis products (diacylglycerols and monoacylglycerols) are thought to be subsequently re-esterified within the ER lumen to produce TG that is assembled with apo B.

Operation of this lipolysis–re-esterification cycle also requires fatty acids, as well as a mono- or di-acylglycerol acyltransferase activity, in the ER lumen for re-synthesis of TG. Until recently, all TG was thought to be synthesized by diacylglycerol acyltransferase-1, a microsomal enzyme (Chapter 10). Surprisingly, however, diacylglycerol acyltransferase-1 knock-out mice retain the ability to synthesize TG and have normal fasting serum TG levels [17]. These data indicate that secretion of TG-rich lipoproteins does not require diacylglycerol acyltransferase-1. Consequently, another gene product has been implicated in synthesizing TG for VLDL assembly: diacylglycerol acyltransferase-2 [18]. Consistent with this idea, diacylglycerol acyltransferase-2 is highly expressed in liver. Confirmation of this hypothesis awaits generation of diacylglycerol acyltransferase-2-deficient mice.

The type of fatty acid supplied to hepatocytes also influences the secretion of apo B-containing lipoproteins. Compared to oleic acid, the (n-3) fatty acids eicosapentaenoic acid (20:5) and docosahexaenoic acid (22:6), found in fish oils, decrease plasma TG levels in humans and decrease the secretion of apo B-containing lipoproteins from rat hepatocytes and hepatoma cells. Consistent with these findings, (n-3) fatty acids also stimulate apo B degradation (E.A. Fisher, 1993).

6.2. *Phospholipids*

Phosphatidylcholine is the major phospholipid on the surface monolayer of all lipoproteins including VLDLs. In the liver, phosphatidylcholine is synthesized by two biosynthetic pathways: the CDP-choline pathway and the phosphatidylethanolamine *N*-methylation pathway (Chapter 8). Choline is an essential precursor of phosphatidylcholine synthesis via the CDP-choline pathway. When cells or animals are deprived of choline, this biosynthetic pathway is specifically inhibited and secretion of TGs and apo B, but not other proteins, is reduced by ~70% even though the phosphatidylcholine content of ER and Golgi membranes is essentially normal [19]. Thus, phosphatidylcholine synthesis is required for VLDL secretion. Interestingly, choline deficiency inhibits the secretion of only apo B variants larger than apo B28 (i.e. only those that assemble a neutral lipid core) indicating that association of apo B with neutral lipids and/or VLDL secretion depend on phosphatidylcholine synthesis. The mechanism by which decreased synthesis of phosphatidylcholine inhibits apo B secretion is by pre-secretory degradation. Translocation of apo B into the ER lumen is not impaired since the number of apo B-containing particles within the ER lumen is normal. In contrast, the number of apo B-containing particles in the Golgi lumen is decreased by ~70% implying that defective VLDL particles are degraded in a post-ER compartment of the secretory route (Section 7).

An alternative pathway for phosphatidylcholine synthesis, which is quantitatively significant only in the liver, is the methylation of phosphatidylethanolamine to phos-

phatidylcholine via phosphatidylethanolamine *N*-methyltransferase. Male mice in which the methyltransferase gene was disrupted (D.E. Vance, 1997) have greatly reduced plasma levels of TG and apo B100, but not apo B48. Moreover, the secretion of apo B100 and TG from hepatocytes derived from these mice is similarly impaired (A. Noga, 2002). Thus, phosphatidylcholine synthesis is required for normal apo B secretion.

Phospholipid turnover also appears to play a role in VLDL assembly (G.F. Gibbons, 1996). A significant proportion of TG in VLDLs contains fatty acids derived from the deacylation of phospholipids. Moreover, the assembly of VLDLs has been proposed to require the calcium-independent phospholipase A2 (Z. Yao, 2000).

In addition to phosphatidylcholine, VLDLs contain smaller amounts of other phospholipids (e.g. phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol) as well as sphingomyelin and ceramide. Inhibition of sphingolipid synthesis by >90% by addition of fumonisin B to rat hepatocytes (Chapter 14), does not, however, inhibit VLDL secretion. Therefore, normal amounts of sphingolipids are not required for VLDL assembly/secretion (A.H. Merrill, 1995).

6.3. Cholesterol and cholesteryl esters

Sterol response element binding proteins (SREBPs) are master regulators of fatty acid and cholesterol synthesis at the level of transcription (Chapters 6, 7, 15). The transcriptionally active forms of SREBPs are produced from precursor proteins by a sterol-dependent proteolytic cleavage [20]. SREBPs co-ordinately increase the expression of genes involved in synthesizing fatty acids (fatty acid synthase, acetyl-CoA carboxylase, stearoyl-CoA desaturase) and cholesterol (3-hydroxy-3-methylglutaryl-CoA reductase), as well as the microsomal TG transfer protein (Section 6.4). All these components are required for VLDL assembly. The ability of the liver to secrete VLDLs is also linked via SREBP to the cholesterol/bile acid pathway through changes in the hepatic level and metabolism of cholesterol (R.A. Davis, 2001).

Cholesteryl esters are relatively minor constituents (5–15% of total lipids) of VLDLs but the amount of cholesteryl esters relative to TGs increases when rats are fed a high-cholesterol diet (R.A. Davis, 1982). There are conflicting data on whether or not the availability of cholesterol and/or cholesteryl esters directly influences apo B secretion. Inhibition of cholesteryl ester formation in hepatocytes decreases apo B secretion in some studies but not others. A severe reduction in the cholesteryl ester content of hepatoma cells reduces apo B secretion whereas an increased cellular content of cholesteryl esters above normal does not stimulate apo B secretion. Cholesterol is esterified by acyl-CoA:cholesterol acyltransferase (ACAT) which is encoded by at least two different genes. In mouse liver and intestine the majority of cholesteryl esters are made via ACAT-2. In animal studies, ACAT inhibitors reduce plasma cholesterol probably primarily by decreasing intestinal cholesterol absorption rather than by reducing VLDL secretion. In support of this idea, targeted disruption of the ACAT-2 gene in mice greatly reduces intestinal cholesterol absorption and results in complete resistance to diet-induced hypercholesterolemia (R.V. Farese, 2000). However, apo B-containing lipoproteins are made in normal amounts in ACAT-2^{-/-} mice despite

the absence of essentially all hepatic ACAT activity. Interestingly, plasma VLDLs of these mice have smaller diameters than those of wild-type mice. Thus, normal amounts of cholesteryl esters might be required for assembling large VLDLs. In mice with a targeted disruption of the ACAT-1 gene, hepatic cholesterol esterification activity and plasma cholesterol levels are normal [21].

Based on experiments with ACAT^{-/-} mice it has been suggested that ACAT-1 functions primarily in providing cholesteryl esters for storage in cytosolic droplets whereas ACAT-2 is linked to secretion of cholesteryl esters into lipoproteins. In agreement with this concept, the two ACATs have been proposed to have different membrane topologies with the active site of ACAT-1 facing the cytosol and that of ACAT-2 facing the ER lumen (L. Rudel, 2000). It is not yet clear, however, how these findings apply to humans since ACAT-1 accounts for the majority of ACAT activity in *human* hepatocytes whereas ACAT-2 is the major ACAT in human intestine.

6.4. Microsomal triacylglycerol transfer protein

Lipids are assembled with apo B into VLDLs in the ER lumen (J.E. Vance, 1993). During this process TGs are concentrated into the core of the particle. A microsomal luminal protein, the 97 kDa microsomal triacylglycerol transfer protein (MTP), that has the ability to transfer TGs between membranes *in vitro*, is proposed to transfer TG to nascent apo B particles [22]. MTP is present in liver and intestine as a soluble heterodimer with protein disulfide isomerase (55 kDa) (J.R. Wetterau, 1990). The latter is a ubiquitous ER luminal protein that catalyzes formation of disulfide bonds during protein folding. The 97 kDa MTP subunit confers all lipid transfer activity to the heterodimer. In *in vitro* assays, MTP catalyzes lipid transfer from donor to acceptor liposomes with substrate specificity of: TG > cholesteryl esters > diacylglycerols > phosphatidylcholine. The lipid transfer reaction displays ping-pong, bi-bi kinetics implying that MTP transfers lipids by a 'shuttle' mechanism. Although MTP contains no C-terminal KDEL sequence (this sequence retains soluble proteins within the ER lumen), association of MTP with protein disulfide isomerase, which does contain a KDEL sequence, is thought to retain MTP within the ER lumen. The tissue and subcellular location of MTP, and its preference for transporting neutral lipids, suggest that MTP is involved in loading nascent apo B particles with TG.

6.4.1. MTP deficiency in humans and mice

The hypothesis that MTP is required for secretion of apo B-containing lipoproteins from the liver and intestine was strengthened by the discovery that individuals with the rare genetic disease abetalipoproteinemia lack detectable MTP protein and intestinal MTP lipid transfer activity (J.R. Wetterau, 1992). These patients suffer from fat malabsorption and despite a normal apo B gene, their plasma apo B is barely detectable. Specific inhibitors of MTP lipid transfer activity have been developed based on the premise that MTP might be a target for reducing atherosclerosis by inhibiting production of apo B-containing lipoproteins. MTP inhibitors lower plasma cholesterol levels by up to 80% in rats, hamsters and rabbits (J.R. Wetterau, 1998). However, heterozygosity for MTP deficiency in humans does not result in altered plasma lipid or lipoprotein levels

suggesting that MTP is not normally rate-limiting for lipoprotein production unless MTP activity is greatly reduced.

The function of MTP has been investigated in genetically modified mice [23,24]. Complete elimination of the murine MTP gene is embryonically lethal, probably because apo B-containing lipoproteins are required for transferring lipids from the yolk sac to the developing mouse embryo. However, the murine MTP gene was inactivated specifically in the liver by generating mice harboring a 'floxed' MTP gene and subsequent Cre-mediated recombination. In these mice, plasma apo B100 levels were reduced by >90% but, surprisingly, apo B48 levels were reduced only slightly. Moreover, in hepatocytes derived from these mice apo B100 secretion was eliminated whereas secretion of apo B48 was almost unaltered. However, the apo B48-containing lipoproteins contained less lipid than those from wild-type mice. In addition, ultrastructural analyses revealed that, compared to wild-type hepatocytes, MTP^{-/-} hepatocytes contained very few VLDL-sized, lipid-staining particles within the ER or Golgi lumina. Moreover, MTP^{-/-} mouse livers contained numerous lipid droplets in the cytosol, a hallmark of impaired TG secretion. These observations suggest that (i) MTP is required for transferring TG into the lumen of the ER, and (ii) apo B100 secretion requires MTP whereas secretion of apo B48 is either less sensitive to MTP deficiency or does not require MTP. On the other hand, over-expression of MTP in mouse liver via recombinant adenoviruses increased the secretion and plasma levels of apo B100 and B48, as well as TG (D.J. Rader, 1999).

It should be noted that although MTP is assumed to function as a lipid transfer protein, based on its ability to transfer lipids between membranes *in vitro*, a lipid transfer function of MTP in providing lipid for VLDL assembly has not been directly demonstrated in intact cells or animals.

6.4.2. *Studies on the function of MTP in cultured cells*

Recently, investigators have extensively studied the role of MTP in the assembly of apo B-containing lipoproteins in hepatocyte-derived cells, as well as non-hepatic cells that have been genetically modified to express apo B and MTP. MTP and apo B have been shown by co-immunoprecipitation experiments to interact physically during lipoprotein assembly via specific sites that have been identified on MTP and apo B (C.C. Shoulders, 1999; G.S. Shelness, 1998; T. Grand-Perret, 1999). The association between MTP and apo B is transient and increases when lipoprotein assembly is stimulated upon addition of oleic acid, and decreased when TG synthesis is inhibited by Triacsin D (H.N. Ginsberg, 1996). Apo B also co-immunoprecipitates with other ER luminal chaperone proteins such as calnexin, calreticulin, GRP94, Erp72 and BiP which presumably facilitate the correct folding of apo B (H. Herscovitz, 1998). MTP itself has also been implicated as a chaperone for apo B during VLDL assembly (H.N. Ginsberg, 2001).

There is considerable controversy regarding the precise role of MTP in assembling lipids with apo B [25]. MTP has been suggested to play a role in several steps along the VLDL assembly pathway: translocation of apo B across the ER membrane and into the ER lumen; co-translational assembly of lipids with apo B; transfer of the bulk of TG into the core of VLDLs; movement of TG into the ER lumen. Experimental data both support and refute the participation of MTP at each of these steps. Some of the conflicting data likely arise from different experimental models used, e.g. hepatoma

cells versus primary hepatocytes, cultured cells versus genetically modified mice, human versus rodent models.

A general consensus is that lipid addition to apo B begins during translation and translocation of apo B across the ER membrane (E.A. Fisher, 1998; D.A. Gordon, 1996; S.-O. Olofsson, 1998). Experiments with cultured cells indicate that MTP stimulates, but is not absolutely required for, co-translational lipidation of apo B. Short apo B variants, such as apo B18, are secreted from hepatoma cells in poorly lipidated form independent of MTP, whereas secretion of longer (i.e. >apo B23), more highly lipidated apo B variants is stimulated by MTP. These findings suggest that the TG-transferring property of MTP is important for VLDL assembly. Moreover, the requirement of MTP is much greater for the secretion of apo B100 than for apo B48 suggesting that there might be different mechanisms involved in assembly of these two types of lipoproteins. In spite of a suggested chaperone function for MTP, this protein is apparently not *required* for translocation of apo B across the ER membrane since (i) apo B41-containing lipoproteins are secreted from mammary-derived cells lacking MTP activity (D.M. Small, 1995), (ii) apo B48 is lipidated and translocated equally well into the lumen of dog pancreatic microsomes (which lack MTP) and liver microsomes (which contain MTP) (A.E. Rusñol, 1997), and (iii) an almost normal amount of apo B48, albeit poorly lipidated, is secreted from MTP^{-/-} hepatocytes (S.G. Young, 1998).

Whether or not MTP plays a role in supplying the bulk of TG for VLDLs is controversial. Several studies indicate that MTP is not directly involved in this process. One suggestion is that MTP transports TGs into the ER lumen to form luminal TG droplets which are subsequently incorporated into apo B-containing lipoproteins. This proposed role for MTP is derived from observations that (i) MTP^{-/-} livers of mice lack VLDL-sized, lipid-staining particles that are normally present in the ER lumen of wild-type mice, and (ii) ER luminal lipid droplets, the size of chylomicrons, accumulate in enterocytes of mice lacking intestinal apo B (R.L. Hamilton, 1998). However, a role for MTP in transferring TG to apo B, either by shuttling TG monomers to apo B or by mediating the fusion between partially lipidated apo B particles and a luminal TG droplet, has not been unequivocally demonstrated. Nor is it clear how the proposed 'shuttle mechanism' for MTP can be reconciled with a role in transferring such large amounts of TG to apo B.

Based on available experimental evidence several models have been proposed for the assembly of apo B with lipids. In one popular model, called the 'two-step' model (Fig. 4A), MTP transfers lipid to apo B during translocation of the protein into the ER lumen to form partially lipidated apo B. These small particles then fuse with a TG droplet in the ER lumen to produce VLDLs. As discussed above, MTP might be involved in forming the luminal TG droplet and/or in fusion of the droplet with the small apo B-containing particle. However, the biophysical mechanism by which a TG droplet would fuse with a small apo B-containing lipoprotein is not clear. A second model of assembly of apo B-containing lipoproteins is one in which lipid is sequentially added to apo B that is either undergoing translocation across the membrane, or that is associated with the luminal face of the ER membrane (Fig. 4B). During times when lipid supply is low, small, lipid-poor apo B particles would be formed and secreted, but when lipid supply is abundant, fully lipidated VLDLs would be produced.

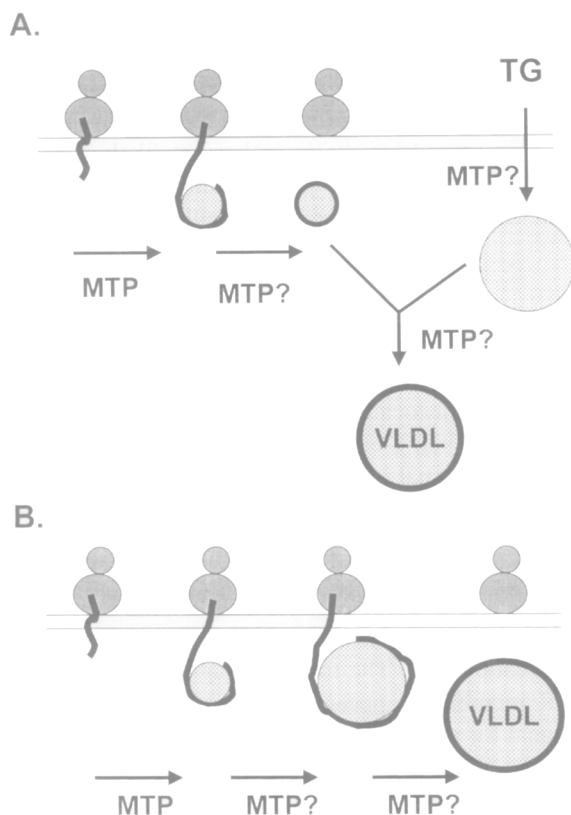


Fig. 4. Two models proposed for assembly of apo B-containing lipoproteins in hepatocytes. (Panel A) The 'two-step' model. Shown from left to right is the co-translational translocation of apo B into the ER lumen during which small amounts of lipid are assembled with apo B in a MTP-dependent step. The partially lipidated apo B particle is then released into the lumen and fuses with a luminal triacylglycerol (TG) droplet to form VLDL. Alternatively, MTP might 'shuttle' TG monomers to the particle. It is proposed that MTP mediates formation of a TG droplet in the ER lumen for assembly with apo B. (Panel B) The concerted assembly model. From left to right is shown the co-translational translocation of apo B into the ER lumen. Lipid is sequentially added to apo B during translocation until a VLDL-sized particle is formed. In the event of limited lipid supply, small, partially lipidated apo B particles would be secreted. The bulk of TG might be transferred either from TG within the ER membrane or from a luminal TG droplet, as indicated in panel A. The role of MTP in some steps of both schemes (indicated by ?) is still controversial.

Many issues remain concerning the role of MTP in the assembly of apo B-containing lipoproteins. (i) The existence of ER luminal TG droplets has not been unambiguously demonstrated by isolation of these droplets from hepatocytes. (ii) The mechanism of formation and stabilization of TG droplets in the ER lumen is not understood. (iii) It is not clear why some luminal TG droplets lacking apo B would not be secreted by default from hepatocytes. (iv) The partially lipidated apo B-containing particles that are formed co-translationally have not been directly shown to be precursors of fully lipidated VLDLs. (v) It is not clear why MTP deficiency abrogates the secretion of apo B100 but not apo B48.

6.4.3. *MTP gene expression*

The promoter region of the MTP gene contains elements predicted to bind transcription factors (e.g. HNF-1, HNF-4 and AP-1) that might dictate the cell type-specific expression of MTP. Hepatic MTP mRNA levels are increased in hamsters fed a high-fat diet. The activity of the human promoter is increased by cholesterol, most likely via the SREBP pathway (Chapter 15), and expression of the MTP gene is inhibited by insulin. However, it is not yet known if these alterations in gene expression regulate the secretion of apo B-containing lipoproteins.

7. *Translocation and intracellular degradation of apo B*

Typical secretory proteins co-translationally translocate across the ER membrane without interacting with the membrane bilayer or being exposed to the cytosol. However, translocation of apo B into the ER lumen is unusual in several respects. First, complete translocation of apo B occurs only when apo B is associated with lipid. Second, an unusual translocational pausing occurs during which translation continues and domains of newly synthesized apo B become exposed to the cytosol (V.R. Lingappa, 1990). Translocational pausing of apo B has been proposed to be mediated by multiple 'pause-transfer sequences' which are similar, but non-identical, sequences of ~10 amino acids that direct stopping and starting during translocation [26]. The translocating chain-associated membrane protein, TRAM, in the ER membrane appears to mediate translocational pausing (V.R. Lingappa, 1998). The signal for resumption of translocation after pausing is not known but has been suggested to be association of lipid with apo B. Others have argued that translational, rather than translocational, pausing occurs.

7.1. *Proteasomal degradation of apo B*

A third unusual property of apo B is that its translocation across the ER membrane is linked to proteasomal degradation (Fig. 5A). Pulse-chase studies in hepatocytes show that 40 to 60% of newly synthesized apo B is not secreted but is degraded intracellularly [27]. When lipid supply is restricted, or when MTP is inhibited, apo B does not efficiently translocate into the ER lumen but becomes exposed to the cytosol and is degraded. However, this degradation does not determine how much apo B is secreted because when degradation is inhibited by *N*-acetyl-Leu-Leu-norleucinal, apo B accumulates in microsomes whereas apo B secretion is not increased (R.A. Davis, 1994). This arrest in translocation and cytosolic exposure of apo B result in apo B being ubiquitinated and targeted for degradation by cytosolic proteasomes [28] in a process facilitated by the cytosolic chaperone hsp70 (E.A. Fisher, 2001). Several ER luminal and membrane proteins are known to undergo proteasomal degradation by a pathway in which the completely translocated protein undergoes retrotranslocation into the cytosol and is subsequently ubiquitinated and degraded by proteasomes. In contrast, the ubiquitination and proteasomal degradation of apo B are thought to be initiated while apo B is attached to the ribosome and still within the translocation channel (E.A. Fisher, 2001).

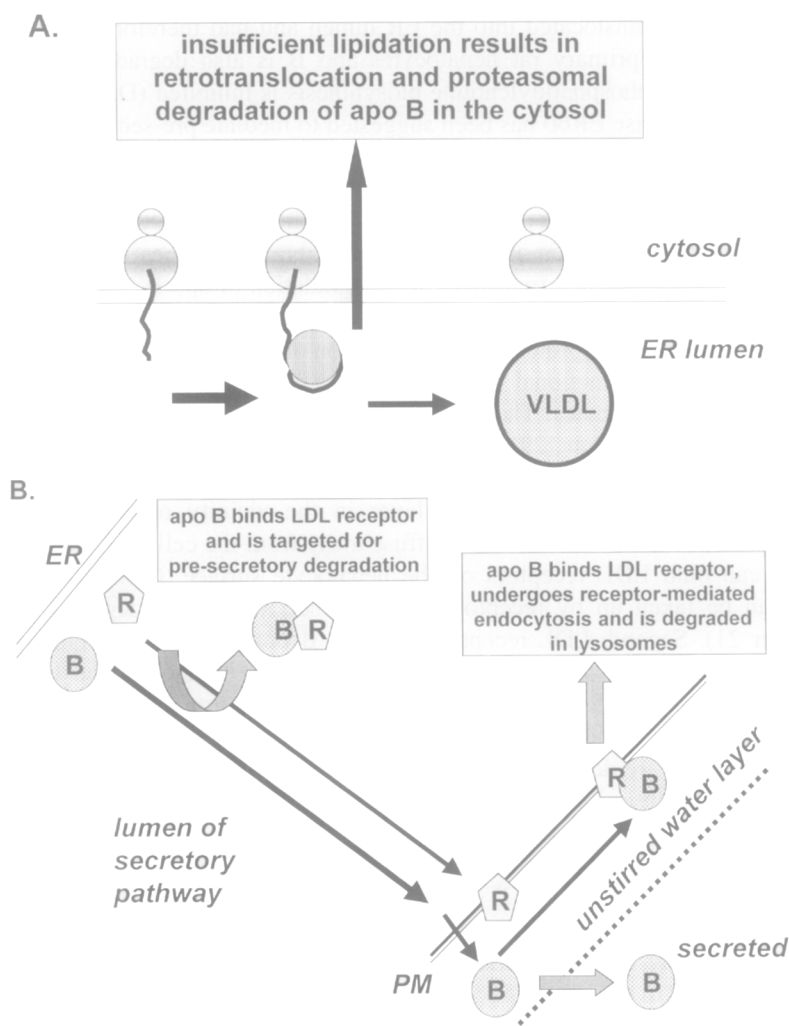


Fig. 5. Intracellular degradation of apo B. (Panel A) When lipid supply is limited, or when MTP is inhibited, lipid-poor apo B particles are formed and apo B is exposed to the cytosol, becomes ubiquitinated and degraded by the proteasome. Sufficient supply of lipids results in VLDL being secreted. (Panel B) Apo B is also degraded non-proteasomally. Nascent apo B particles, indicated as 'B' are transported via the secretory route to, and across, the plasma membrane (PM) and enter the 'unstirred water layer' adjacent to the cell. Some particles are fully secreted whereas others bind to cell surface LDL receptors, indicated by 'R', are internalized by receptor-mediated endocytosis and undergo lysosomal degradation. The LDL receptor is also proposed to mediate the pre-secretory degradation of apo B by binding to apo B100 within the lumen of the secretory pathway and targeting apo B100 for degradation by as yet unidentified luminal proteases.

7.2. Degradation of apo B within the secretory pathway

Apo B is also degraded within the secretory pathway. This site of degradation was demonstrated in HepG2 cells in which apo B was degraded by a luminal protease after

the protein had completely translocated into the ER lumen and had therefore escaped proteasomal degradation. In primary rat hepatocytes apo B is also degraded within the secretory pathway when phosphatidylcholine biosynthesis is inhibited (D.E. Vance, 1993). The ER luminal protease ER60 has been suggested to mediate pre-secretory apo B degradation (K. Adeli, 1997). Degradation of apo B within the secretory pathway is thought to provide a quality control mechanism for eliminating defective particles before their secretion. In contrast to the proteasomal degradation of apo B, which occurs in *response* to inefficient translocation across the ER membrane but does not control how much apo B is secreted, the post-translocational degradation of apo B can directly control how much apo B is secreted.

A clue to the mechanism of non-proteasomal degradation of apo B came from studies in which hepatocytes lacking functional LDL receptors were found to secrete more apo B than did wild-type hepatocytes. The LDL receptor was subsequently implicated in two facets of apo B degradation [29] (Fig. 5B). First, when fully assembled apo B-containing particles are exported across the plasma membrane of hepatocytes, the nascent lipoproteins enter the unstirred water layer in the vicinity of the plasma membrane. These particles do not immediately diffuse away from the cell (E.A. Fisher, 1990) but instead can bind LDL receptors on the hepatocyte surface. Consequently, the lipoproteins can be taken up by receptor-mediated endocytosis and degraded in lysosomes (Chapter 21). Second, LDL receptors and apo B both transit the secretory pathway en route to the plasma membrane. The ligand-binding domain of the LDL receptor is oriented towards the ER lumen and therefore has the opportunity to bind apo B intralumenally and target it for degradation before it can be secreted. Approximately equal amounts of apo B100 are degraded via the LDL receptor using the re-uptake and pre-secretory mechanisms [29]. Apo B100 is more susceptible than apo B48 to LDL receptor-mediated degradation because apo B100 contains the LDL receptor-binding domain that is absent from apo B48. Small amounts of apo B48 are also degraded because these particles associate with apo E which binds to the LDL receptor.

This mode of presecretory degradation of nascent apo B was also invoked as a possible explanation for why apo B secretion is impaired in apo E^{-/-} mice, whereas apo B secretion is increased in transgenic mice and hepatoma cells that over-express apo E. The basis for this conjecture was that apo E binds avidly to the LDL receptor and thus might compete with apo B for receptor binding, both within the secretory pathway and on the cell surface. The prediction, however, was found not to be correct because when apo E production was eliminated in LDL receptor^{-/-} mice, VLDL secretion was still 50% less than in LDL receptor^{+/+} mice (L.M. Havekes, 2001).

8. Assembly and secretion of chylomicrons

When dietary fat enters the intestine TGs are hydrolyzed to monoacylglycerols and fatty acids which diffuse across the microvillus membrane. TGs are subsequently re-synthesized in the enterocyte and large amounts of TGs are packaged with other lipids and apo B48 into chylomicrons that are secreted into mesenteric lymph. The wide variation in sizes of chylomicrons (35 to >250 nm) reflects the supply of TGs to the

intestine and affords a mechanism by which TG secretion can be readily altered in response to diet.

A cell culture model frequently used to study intestinal apo B secretion is the human colon carcinoma cell line, CaCo2, which secretes both apo B100 and apo B48 (E.J. Schaefer, 1987). In many respects, the intestinal assembly of chylomicrons is similar to the hepatic assembly of VLDLs, although there are some differences. Both apo B48 and MTP are required for normal chylomicron assembly. ER luminal TG droplets, the size of chylomicrons, but lacking apo B, were observed in enterocytes of mice lacking intestinal apo B synthesis, suggesting that chylomicron assembly might utilize these apo B-free, ER luminal TG droplets (R.L. Hamilton, 1998). Interestingly, in CaCo2 cells almost no newly synthesized apo B is degraded intracellularly. Moreover, ubiquitinylation of intracellular apo B is not detectable although CaCo2 cells do possess the machinery for adding ubiquitin to proteins for proteasomal degradation (L. Chan, 2000).

One unique feature of chylomicron assembly is revealed in individuals with chylomicron retention disease (Anderson's disease). These people secrete apo B100-containing VLDLs from the liver but do not secrete apo B48-containing chylomicrons from the intestine. This rare disease is characterized by hypocholesterolemia and lipid malabsorption [30]. Since the intestinal apo B editing mechanism is intact and apo B48 is synthesized normally, a defective gene, different from genes involved in VLDL assembly, appears to be responsible for this disease.

9. Assembly of lipoprotein(a)

In addition to its presence on VLDLs and LDLs, plasma apo B is also found in humans, some primates and hedgehogs (but not rodents) covalently bound via a single disulfide linkage to a glycoprotein called apo(a). Apo(a) is synthesized in significant quantities only in the liver and associates with apo B of LDL to form lipoprotein(a). In human plasma, the concentration of lipoprotein(a) varies from <1 to >100 mg/dl. In general, a high level of lipoprotein(a) is an independent risk factor for development of coronary artery disease (A.M. Scanu, 1992).

Apo(a) contains tandem repeats of sequences that are very similar to the 'kringle' motifs of plasminogen suggesting that apo(a) arose by duplication of the plasminogen gene. A cysteine residue (Cys-4057) in one kringle domain forms a disulfide linkage with Cys-4326 of apo B100. Apo B100 of many mammals cannot form a disulfide linkage with human apo(a). For example, in transgenic mice expressing human apo(a), this apo(a) circulates in plasma free of apo B-containing lipoproteins. A step-wise model for assembly of lipoprotein(a) has been proposed. Initially, a non-covalent interaction occurs between a lysine residue (Lys-680) in the N-terminus of apo B100 and a lysine-binding site in apo(a) (M.L. Koschinsky, 2001). Next, a non-covalent interaction between the C-terminal region of apo B100 and amino acids 4330–4397 of apo(a) occurs, followed by formation of the disulfide bond between apo B and apo(a) (S.P.A. McCormick, 2000). Most data support a model of lipoprotein(a) assembly in which apo(a) associates with apo B of LDL in the plasma after secretion, rather than within hepatocytes prior to secretion.

10. Future directions

The past few years have been marked by an explosion of data on lipoprotein metabolism obtained from genetically modified mice. These studies have been extremely useful for elucidating mechanisms of assembly and secretion of lipoproteins, particularly because there are limitations to studying this process in cultured cells. However, one must also acknowledge the limitations of using mice as models for studying human lipoprotein assembly since lipoprotein metabolism in mice is distinct in several respects from that in humans.

Several important questions remain regarding the mechanisms of lipoprotein assembly. It is surprising that although the MTP gene and protein have been so extensively studied, and although we know that an absence of MTP severely impairs the secretion of apo B-containing proteins, we still do not know exactly how this protein participates in lipoprotein assembly. In *in vitro* assays, MTP transfers lipids between membranes. Consequently, it is assumed that MTP transfers TGs to apo B. However, this lipid transfer function of MTP has not yet been demonstrated in cells. Another proposed role for MTP is in the transfer of TGs across the ER membrane to form luminal TG droplets, but how this process occurs and the composition of the droplets, are not known. There are currently no data explaining why animals make apo B100 in addition to apo B48 since apo B48-only mice appear normal. Nor do we understand why apo B is made in large excess of its requirement for lipoprotein secretion or why hepatocytes degrade a large fraction of newly synthesized apo B intracellularly in an apparently wasteful process. In addition, proteases involved in apo B degradation within the lumen of the secretory pathway remain to be identified. Several post-translational covalent modifications of apo B occur (glycosylation, phosphorylation and palmitoylation) yet we do not understand the biological relevance of these modifications. The mechanisms that regulate chylomicron assembly are poorly understood. For example, the defective gene responsible for Anderson's disease, in which intestinal apo B48/chylomicron secretion is severely impaired while secretion of apo B100 from the liver is normal, has not yet been isolated nor has its function been identified. The stage is now set for resolution of these issues within the next few years.

Abbreviations

ACAT	acyl-Co A : cholesterol acyltransferase
apo	apolipoprotein
ER	endoplasmic reticulum
LDL	low-density lipoprotein
MTP	microsomal triacylglycerol transfer protein
SREBP	sterol response element binding protein
TG	triacylglycerol
VLDL	very low-density lipoprotein

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Dynamics of lipoprotein transport in the human circulatory system

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1. Overview

1.1. Functions of the major lipoproteins

Plasma lipoproteins are soluble complexes of lipids with specialized proteins (apolipoproteins). Their function is to deliver lipids from the tissues where they are synthesized (mainly the liver and intestine) to those that utilize or store them. The apolipoproteins solubilize and stabilize the insoluble lipids of the lipoprotein particles, and prevent the formation of aggregates. Many apolipoproteins have additional functions in plasma lipid metabolism. Some are ligands for cell surface receptors, and determine the tissue-specific delivery of lipids. Some are cofactors for plasma lipases. Others regulate lipid reactions in the plasma, as competitive inhibitors of lipid uptake or metabolism (see table 3 in Chapter 18). It is the protein composition of lipoproteins that in large part specifies their metabolism in the plasma compartment. Conversely, the apolipoprotein content of lipoprotein particles alters during recirculation, as changes in the lipid composition of the particles modify the affinity of apolipoproteins for their surface. The interaction of these processes largely specifies the delivery of lipids to different tissues. Lipids delivered via plasma lipoprotein particles, in addition to neutral acylglycerols, phospholipids, and free and esterified cholesterol, include fat-soluble vitamins and antioxidants. Lipid binding to apolipoproteins is in most cases via the hydrophobic faces of amphipathic helical domains (J.A. Gazzara, 1997). Apolipoproteins have little tertiary structure, which gives them flexibility on the surface of the lipoprotein, as the diameter of the particle responds to the loading or unloading of lipids. Amino acid sequences functional in receptor binding or enzyme activation usually include clusters of charged residues.

While blood plasma contains the highest levels of lipoprotein particles, most lipoproteins, with the exception of the largest, triacylglycerol-rich particles, can cross the vascular bed, though their concentrations in the extracellular space are significantly lower. Interstitial lipoprotein particles interact directly with the surface of peripheral cells, delivering and receiving lipids. This recirculation is completed when interstitial fluid is collected into the main trunk lymph ducts, and returned to the plasma.

1.2. 'Forward' lipid transport

Functionally there are two main classes of lipoproteins. The first consists of particles whose main role is to deliver lipids (mainly triacylglycerols) from the liver or intestine to peripheral, extrahepatic tissues. These particles contain apolipoprotein B (apo B) together with a changing admixture of other lipids and proteins. In chylomicrons, which are secreted from the small intestine, this triacylglycerol originates from dietary long-chain fatty acids. These are re-esterified in the intestinal mucosa before being incorporated into lipoproteins. They contain a single molecule of a truncated form of apo B (apo B48). After loss of most of their triacylglycerol during recirculation in the plasma compartment, the chylomicron 'remnants' are cleared by the liver. Very low-density lipoproteins (VLDLs) secreted from the liver, contain one molecule of the full-length form of apo B (apo B100). Following the loss of most of their triacylglycerol to peripheral tissues, some VLDLs are returned to the liver, endocytosed and catabolized. Others remain in the circulation as intermediate density lipoprotein particles (IDLs). These still contain significant amounts of triacylglycerol and most of their original content of cholesteryl ester and free cholesterol, together with apolipoproteins B and E. After further modification by plasma lipases, most of the apo B100 particles remain in the circulation in the form of low-density lipoproteins (LDLs). After a plasma half-life of about 2 days, LDL are endocytosed, mainly by the liver. Their protein is degraded; their sterol content can be recycled into newly secreted lipoprotein particles, or degraded to bile acids.

Functionally, VLDL (density < 1.006 g/ml), IDL (density 1.006–1.019 g/ml) and LDL (density 1.019–1.063 g/ml) particles represent a continuum of decreasing size and increasing density created by the lipolysis of triacylglycerol. The traditional density limits of these fractions, shown in fig. 1 of Chapter 18, reflect this continuum. The density of each fraction depends mainly on its weight ratio of triacylglycerol (density 0.91 g/ml) to protein (density 1.33 g/ml). In terms of apolipoprotein composition, VLDL particles contain apo B100 and apo Cs with or without apo E, IDLs contain apo B100 and apo E but not apo Cs, and LDLs contain only apo B100.

1.3. 'Reverse' lipid transport

The second major class of lipoprotein particles carries lipids (mainly free and esterified cholesterol) from peripheral tissues to the liver. These high-density lipoprotein particles (HDLs) contain 1–4 molecules of apolipoprotein A1 (apo A1), together with other apolipoproteins that specify the metabolism and delivery of these lipids. HDL-dependent lipid transport is often defined as *Reverse cholesterol transport (RCT)* (C.J. Fielding, 1995).

Several sources of cellular cholesterol contribute to RCT. A part reflects peripheral sterol synthesis, despite the downregulation of this pathway by cholesterol in circulating plasma lipoproteins, mainly LDL. A second part represents the recycling of lipoprotein cholesteryl esters, mainly in LDL, internalized via endocytosis at peripheral LDL receptors; these are also highly downregulated, under physiological conditions, by LDL. Probably the major part of RCT responds to the selective cellular uptake of

preformed lipoprotein free cholesterol, independent of LDL receptors. This enters recycling endosomes returning to the cell surface. Cholesterol from all these sources transfers to HDL for further metabolism, including esterification, outside the cell. Some free cholesterol transfers within the circulation to HDL from other plasma lipoproteins.

Part of the HDL cholesteryl ester formed is transferred to apo B lipoproteins prior to their uptake by the liver. The remainder, mostly cholesteryl ester, is selectively internalized (that is, without the rest of the lipoprotein particle) from HDL by hepatocytes, and by steroidogenic tissues.

HDLs accumulate lipids from the peripheral tissues, and return them to the liver. Newly formed HDLs have high density and little lipid. Their density decreases as they accumulate lipid in the circulation. The classical subfractions of HDL [HDL-3 (density 1.12–1.219 g/ml), HDL-2 (density 1.063–1.12 g/ml), HDL-1 (density < 1.063 g/ml)] reflect this functional and structural continuum.

2. Lipoprotein triglyceride and lipolysis

2.1. Initial events

The structure of newly synthesized intestinal apo B-containing lipoprotein particles (chylomicra) is described in Chapter 19. Each consists of a triacylglycerol core containing a small proportion of cholesteryl esters, stabilized by a surface film made up mainly of phospholipid, some free cholesterol, and one molecule of apo B. Triacylglycerol and cholesteryl and retinyl esters in chylomicrons are derived almost entirely from dietary cholesterol, vitamin A and unesterified fatty acids; chylomicron phospholipids and free cholesterol are made in the enterocyte. Editing of full-length apo B transcripts (see Chapter 19) generates apo B48, which contains only the terminal 2152 residues of full-length apo B100. Since apo B does not exchange between lipoprotein particles during recirculation, apo B48 is an effective marker for chylomicron particles, and dietary triacylglycerol (E. Campos, 1992). Mice in which the editing enzyme was knocked out, when fed the same triacylglycerol load as control mice, were significantly less efficient in secreting chylomicron particles [1]. Dietary triacylglycerol accumulated in the intestinal mucosal cells. This finding indicates that apo B editing may have evolved along with dietary fat consumption to optimize the synthesis of large triacylglycerol-rich particles.

Chylomicrons are co-secreted with apo A1 (the intestine is the major source of this apolipoprotein in human subjects). This apo A1 is lost spontaneously to HDL as soon as chylomicrons reach the circulation. The transfer is independent of triacylglycerol lipolysis. At the same time, apo E and apo C proteins move to the surface of chylomicrons from reservoirs within the plasma population of large spherical HDL particles.

VLDLs secreted from the liver include a single molecule of full-length apo B100 containing 4536 amino acids (Chapter 19). The triacylglycerol-rich core of VLDLs contains significant levels of hepatic cholesteryl esters. Studies with isolated rat livers indicate that the incorporation of cholesteryl esters into VLDLs is necessary for their

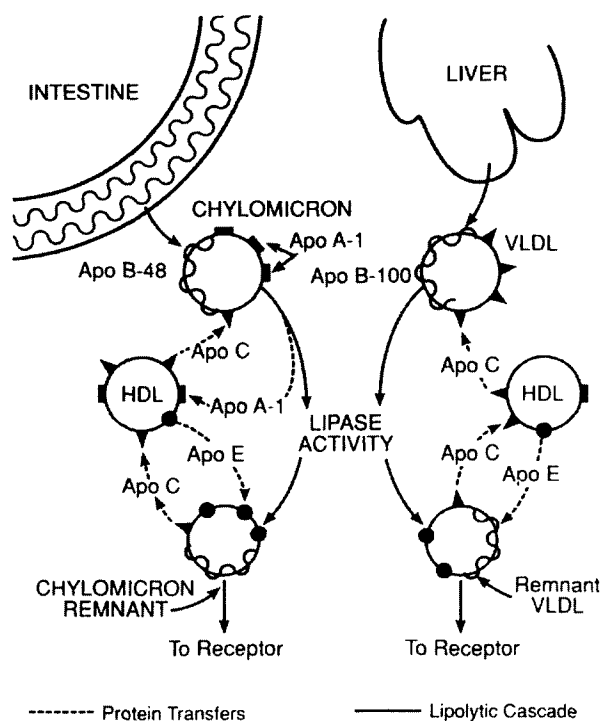


Fig. 1. Transfer of plasma apolipoproteins to newly secreted chylomicrons and VLDL. These small proteins play a critical role in optimizing the reaction rate of these triacylglycerol-rich particles with peripheral lipases and receptor proteins.

successful secretion into the perfusate. The phospholipid moiety of newly synthesized VLDL is enriched in phosphatidylethanolamine, in comparison with circulating VLDL. Newly synthesized plasma VLDLs contain apo C apoproteins but little apo E. As in the case of chylomicrons, enrichment of VLDL with apo E and additional apo Cs takes place in the plasma compartment.

These preliminary events in the circulatory system occupy about 5 min. In the case of both chylomicrons and VLDLs, the product is a triacylglycerol-rich apo B-particle functional to deliver triacylglycerol fatty acids to the peripheral tissues (Fig. 1). The purpose of this time lag is probably to allow these lipoproteins to distribute through the plasma compartment, prior to the inception of hydrolysis. A chylomicron or VLDL fully activated for lipolysis contains 10–20 molecules of apo C2, the cofactor of lipoprotein lipase (LPL). Titration of apo C2 content vs the rate of lipolysis indicates that 2–3 apo C2 molecules per chylomicron or VLDL are needed for maximal activity. Apo C2 and other apo Cs leave VLDL and chylomicrons as lipolysis proceeds, the triacylglycerol core shrinks, and surface phospholipid and proteins are transferred away to other lipoproteins, particularly HDL. Because apo C2 is present in initial excess, lipolysis rates are maintained until a major part (~80%) of initial triacylglycerol content of the particles has been lost.

2.2. *The structure and activation of lipoprotein lipase (LPL)*

LPL hydrolyzes the 1(3)-ester linkages of triacylglycerol of chylomicrons and VLDLs whose surface contains apo C2. The primary product of LPL-mediated lipolysis is 2-monoacylglycerol. After spontaneous isomerization of this lipid, LPL has activity against the 1-monoacylglycerol formed. Limited further lipolysis by plasma and platelet monoacylglycerol hydrolases also takes place. Monoacylglycerol is also readily internalized by vascular cells. As a result the end-products of LPL-mediated triacylglycerol hydrolysis are unesterified fatty acids, monoacylglycerol and glycerol. Fatty acids originating from LPL activity are cleared by adipose tissue and re-esterified under postprandial conditions and stored. Under fasting conditions, hormone-sensitive lipase promotes the release of unesterified fatty acids from adipocyte triacylglycerol back into the circulation (Chapter 10). Fatty acids from LPL-mediated lipolysis in muscle tissue are mainly catabolized to two-carbon subunits as part of oxidative metabolism.

LPL gene transcription is stimulated by sterol response element binding protein-1 (SREBP-1) (Chapter 15) and by Sp-1, and inhibited by Sp-3. The regulation of LPL expression is tissue-specific. During fasting, adipocyte LPL expression is reduced, while expression in muscle cells is increased. Postprandially, expression is upregulated in adipocytes, and decreased in muscle cells. Tissue-specific expression of LPL is mediated by the transcription factor PPAR γ via its PPAR/RXR heterodimer [2]. This mechanism is related to the need to supply fatty acids to muscle for oxidative metabolism under conditions of scarcity, and to direct excess fatty acids to adipose tissue for storage postprandially, conditions where circulating glucose levels provide alternative substrate for muscle cells.

The secreted human LPL protein has 448 amino acids. It is functional as a dimer. LPL is a member of a triacylglycerol lipase protein family (Chapter 10) others of which include hepatic lipase, which like LPL is released into the plasma by heparin, and pancreatic lipase. Pancreatic lipase and several related fungal lipases have been crystallized. LPL is ~30% homologous in primary sequence to pancreatic lipase, whose X-ray coordinates have been used to model LPL structure. Other information on structure–function relationships in LPL has been obtained from the site-directed mutagenesis of key amino acids of receptor- and heparin-binding sites that are absent from pancreatic lipase (Fig. 2). LPL is a serine hydrolase whose active site triad is made up of the S₁₃₂, D₁₅₆ and H₂₄₁ residues. Consistent with other lipases in this family, the primary sequence of LPL predicts a polypeptide ‘lid’ (residues 239–264) which opens when LPL binds to its lipoprotein substrate. A short sequence of hydrophobic amino acids in the C-terminus (residues 387–394) has also been implicated in LPL binding to triacylglycerol-rich lipoproteins. Other data implicate residues 415–438 in both substrate interaction and dimer stability. Heparin binding by LPL was thought earlier to be mediated mainly via five basic residues in two adjacent clusters (R₂₇₉, K₂₈₀, R₂₈₂, K₂₉₆, R₂₉₇). Additional basic residues (K₄₀₃, R₄₀₅, K₄₀₇) were recently implicated (R.A. Sendak, 1998). The involvement of additional sequences at the C-terminus (residues 390–393, 439–448) has also been described (Y. Ma, 1994). These, while not directly heparin-binding, amplify binding by the other domains.

LPL is present within intracellular pools in adipocytes and muscle cells, but the

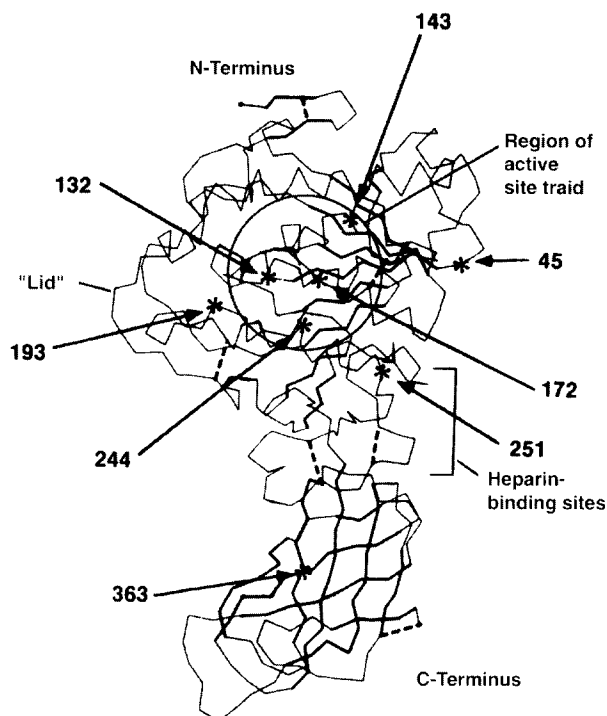


Fig. 2. Structure–function relationships in LPL. Because of the overall sequence similarity between LPL and pancreatic lipase, structural features and the locations of selected amino acids in LPL (which has not been crystallized) have been superimposed on the three dimensional structure of pancreatic lipase. (Modified from Faustinella et al. (1991) *J. Biol. Chem.* 266, 9481–9485, with permission).

functional fraction of LPL is at the vascular endothelial surface, where it is bound by heparin-like glycosaminoglycans. The products of the reaction of chylomicrons and VLDL with endothelial LPL, lipoprotein remnants, continue to circulate in the plasma compartment (see below). Small amounts of LPL are also present in the circulation, especially postprandially. LPL binds to several members of the LDL receptor protein family (specifically, LDL receptor-like proteins-1 and -2, and VLDL receptor protein) to induce receptor-mediated lipoprotein catabolism [3]. Some triacylglycerol clearance may occur via this endocytic route, but the predominant role of receptor binding by LPL is likely to lie in the endocytosis and degradation of LPL itself. Consistent with this, mice in which genes encoding the VLDL receptor, or both VLDL and LDL receptor proteins, were knocked out, had normal levels of plasma triacylglycerol. The receptor-binding domain in the LPL primary sequence lies within the C-terminal region of the protein. It includes K₄₀₇, but is distinct from the lipid-binding domain, which includes W₃₉₃ and W₃₉₄.

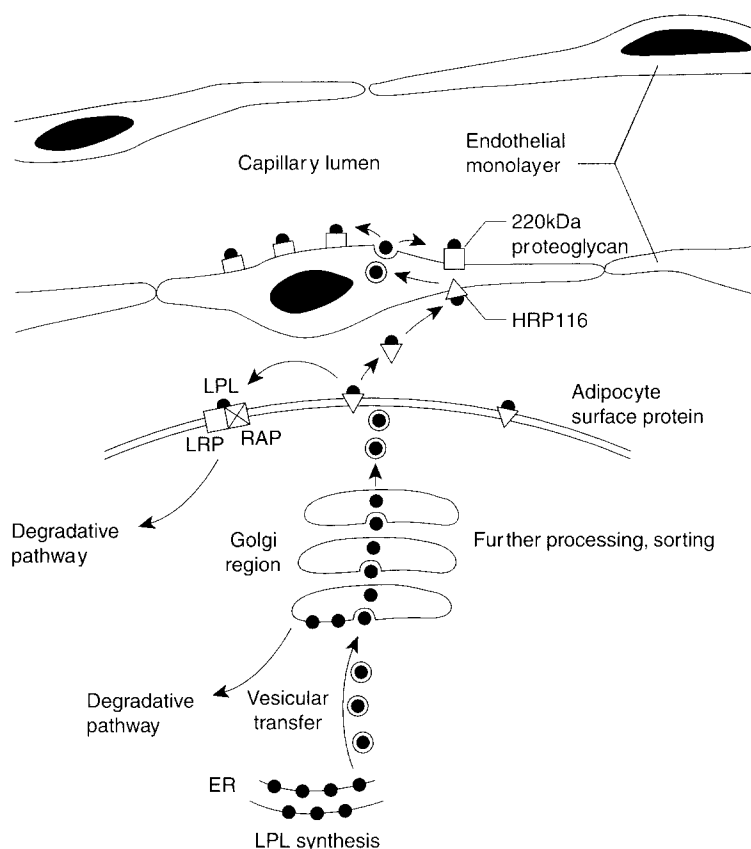


Fig. 3. Synthesis, secretion and transport of LPL from the adipocyte to the vascular endothelial surface. ER, endoplasmic reticulum. Degradative pathways from the Golgi compartment and cell surface are illustrated. RAP, receptor-associated protein; LRP, LDL-receptor-like protein. The suggested roles of HRP (heparin-released protein)-116, and 220 kDa proteoglycan are also shown.

2.3. Transport of LPL to its endothelial site

LPL, synthesized in adipocytes and myocytes, is transported out of the parenchymal cells, through the pericyte layer, and across the endothelium, before binding to functional sites on the vascular endothelial surface (Fig. 3). Mature LPL contains several polysaccharide chains, which are required for effective LPL secretion. Unesterified fatty acids and lysophosphatidylcholine increase the rate of LPL secretion from adipocytes. Adipocytes can degrade secreted LPL. Two distinct pathways are involved. One requires the 39 kDa receptor-associated protein (RAP) which binds to the LDL receptor-related protein [4]. LPL is also internalized via a proteoglycan-dependent pathway.

Subsequent stages of the activation of newly secreted LPL involve its transendothelial migration to specific binding site on the capillary vascular surface (Fig. 3). Transcytosis was recently shown to involve both the VLDL receptor protein, and proteoglycans

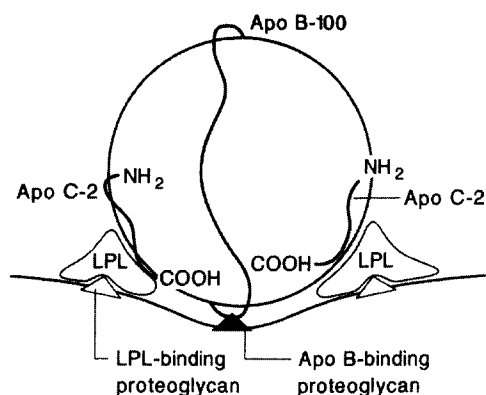


Fig. 4. Model to show interactions between the endothelial cell surface, triacylglycerol-rich lipoproteins, apo C2 and LPL. Two LPL molecules are shown reacting with the same VLDL particle. These are representative of the multiple LPLs probably reactive with each triacylglycerol-rich lipoprotein particle.

[3]. Earlier, a 116-kDa LPL-binding protein, released by heparin, had been implicated in LPL binding to endothelial cells. Microsequencing of peptides from this protein indicated it to be a fragment of apo B. The exact role of this fragment in the migration of LPL remains to be established. LPL is bound to the endothelial vascular surface via a 220 kDa proteoglycan whose functional site is probably a highly sulfated deca-saccharide [5].

2.4. Structure of the LPL–substrate complex at the vascular surface

It seems likely that LPL and its large triacylglycerol-rich lipoprotein substrates both establish multiple interactions with each other and with the capillary wall to anchor the enzyme–substrate complex to the vascular surface. Components of such a multi-protein functional complex would include LPL itself, apo C2 and apo B on the VLDL or chylomicron, the 220 kDa proteoglycan, and possibly VLDL receptor protein or another member of this family (Fig. 4).

Apo B100 has a length sufficient to make only a single circumference of VLDL. This was estimated from electron microscopic studies of apo B in the smaller lipolysis product, LDL [6]. It follows that contact between apo B and the endothelial cell must be restricted to a relatively small fraction of the primary sequence. The same considerations apply to the interaction of the larger chylomicron particles, which are stabilized by the shorter apo B variant, apo B48. Since triacylglycerols in VLDL and chylomicrons are competitive substrates for LPL in mixtures of these lipoproteins, the same LPL binding sites must accommodate either lipoprotein particle.

Kinetic data suggest that several molecules of LPL simultaneously catabolize the triacylglycerol of each VLDL or chylomicron particle. The turnover number of LPL under physiological conditions is about 10 s^{-1} . For a chylomicron containing 3×10^5 molecules of triacylglycerol, catabolism of 50% of this lipid by a single LPL molecule would take about 3 h, yet the measured $t_{1/2}$ is 10–15 min. These data suggest that several molecules of LPL become attached to the circumference of each chylomicron

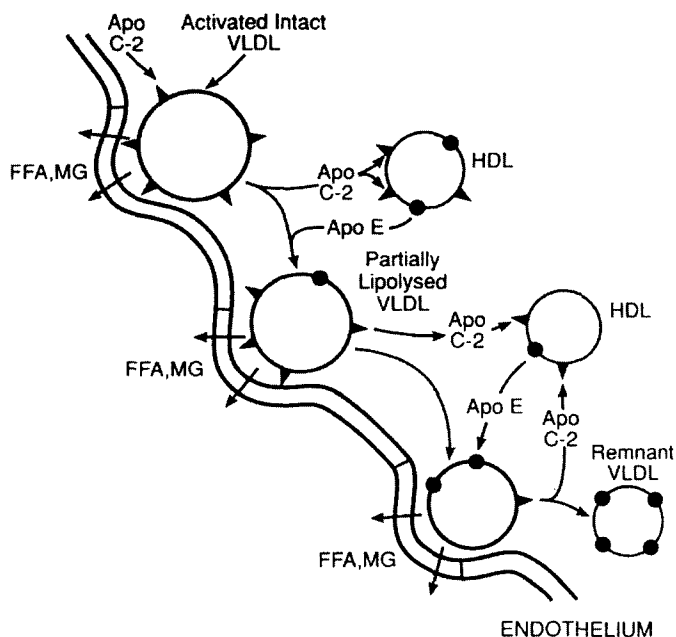


Fig. 5. Mechanism of remnant lipoprotein formation at the endothelial surface. Apo B is not illustrated. FFA, free fatty acid; MG, monoacylglycerol; apo C2, closed triangles; closed circles, apo E. This model reflects the appearance of partially lipolyzed lipoprotein particles in the circulation during LPL-mediated lipolysis of triacylglycerol-rich lipoproteins.

or VLDL particle during lipolysis, with each LPL activated by one molecule of apo C2. In Fig. 4, apo B is illustrated binding directly to the endothelial cell surface, while individual proteoglycan anchors bind LPL to the endothelium. Each apo C2 would link one LPL to the surface of the lipoprotein. If this model were correct, LPL binding sites on the capillary endothelium must be fluid yet highly organized, and able to adapt rapidly to substrates with different diameters and apo C2 contents.

2.5. Kinetics of the LPL reaction and the role of albumin

As VLDL and chylomicrons pass down their delipidation cascade, partially catabolized intermediates formed as a result of LPL activity are detected in the circulation (Fig. 5). This observation makes it likely that lipolysis does not result from a single binding event. Rather, there must be repeated dissociation and rebinding, during which lipoprotein triacylglycerol is catabolized, apo C2 is gradually lost, and LPL catalytic rate is decreased while remnant end-products are formed.

There has been considerable discussion of mechanisms by which triacylglycerol-rich lipoproteins could be reversibly displaced from the endothelial surface. The most likely would involve the transient accumulation of lysogenic lipolysis products at the lipoprotein surface within the LPL-binding surface microdomain. After dissociation of the lipoprotein particle, these lipids would diffuse away, leaving the partially lipolyzed

particle once more competent to bind to the lipase site. There are three candidate lipids for such a role: unesterified fatty acids, monoacylglycerols, and lysophosphatidylcholine.

Only a portion of the fatty acids generated by LPL are cleared locally. The rest remain in the circulation, after transfer from the surface of the substrate lipoprotein to albumin, part of which remains bound to the lipoprotein surface. Under physiological conditions, fatty acids are largely converted to their sodium and potassium salts, and can act as detergents. Monoacylglycerols are effective lysogens. Even at concentrations of 1–2 μM , they inhibit LPL activity in the isolated perfused rat heart. Monoacylglycerols do not bind to albumin, but are taken up rapidly by cells. LPL also generates lysophospholipids. These are effective lysogens but unlike monoacylglycerols, they form stable complexes with albumin. Although further research is needed, monoacylglycerols seem the most likely contributors to the transient displacement of triacylglycerol-rich lipoproteins from the vascular surface.

LPL has an important role in directing VLDL triacylglycerol to muscle tissues during fasting, and VLDL and chylomicron triacylglycerol to adipose tissue postprandially. In addition to the transcriptional regulation described above, there is evidence of posttranslational and kinetic differences between LPL sites in adipose and muscle tissues that contribute to the distribution of lipolysis products between tissues.

In adipocytes, fasting is associated with the synthesis of LPL molecules whose *N*-linked polysaccharide chains retain an unmodified high-mannose structure. In the fed state, these chains are modified by mannose trimming, and the addition of glucose, hexosamine and sialic acid units. The high mannose form of LPL has low specific activity and is retained within the adipocyte. The modified form is actively secreted. Insulin levels are an important determinant of LPL processing.

The apparent K_m of endothelial LPL in adipose tissue is relatively high, compared to that in muscle tissues such as the heart (C.J. Fielding, 1976). This means that the hydrolysis of lipoprotein triacylglycerol by LPL in adipose tissue remains proportional to substrate concentration. In contrast, LPL at the surface of muscle capillaries is saturated, even at the low circulating levels of triacylglycerol-rich lipoproteins characteristic of the fasting state.

2.6. *Later metabolism of chylomicron and VLDL triacylglycerol*

Chylomicrons recirculate until about 80% of initial triacylglycerol content has been catabolized in peripheral tissues. The chylomicron remnant is then endocytosed by hepatic receptors (Chapter 21). Chylomicron remnants retain almost the whole of their original cholesteryl and retinyl ester content. This is cleared by the liver along with remnant triacylglycerol.

The metabolism of VLDL remnants is more complex. In humans some VLDL remnants (IDL) are cleared by the liver via the LDL receptor; but a significant proportion (estimated at 50–70%) is further modified within the circulation to generate LDL. Essentially the whole of circulating LDL is formed in this way. Comparison of the composition of IDL with that of LDL indicates that this conversion involves the loss of 80–90% of IDL triacylglycerol, some phospholipid, and the dissociation of

remaining apo E. In contrast, IDL free cholesterol content is the same as that in VLDL while cholesteryl ester is increased in LDL, as a result of the activity of cholesteryl ester transfer protein (CETP) that exchanges apo B-associated triacylglycerol for apo A1-associated cholesteryl esters (see Section 4.3).

It was formerly considered that the loss of lipids from IDL was mediated mainly via the activity of hepatic lipase, a triacylglycerol hydrolase with a role in the generation of small HDL (see Section 3). Recent data make this explanation less likely. Mice in which hepatic lipase was inactivated, and mice and rabbits overexpressing hepatic lipase, had similar, normal levels of circulating total and remnant triacylglycerol, even postprandially. In contrast, IDL is an optimal substrate for CETP. While VLDL and IDL contain similar numbers of cholesteryl ester molecules per apo B, LDLs have about 50% more cholesteryl ester molecules per apo B than either. These data suggest that in normal metabolism, the conversion of IDL to LDL is driven mainly by CETP, and that the role of hepatic lipase is to hydrolyze triacylglycerol on HDL, not on IDL. If this model is correct, the loss of apo E which is part of the IDL-to-LDL conversion is probably passive, and reflects the changing surface lipid composition of IDL.

2.7. Congenital deficiencies of lipoprotein triacylglycerol metabolism

The functional pools of LPL and hepatic lipase are quantitatively released into plasma by heparin (post-heparin plasma). Genetic deficiency of LPL is associated with a massive increase in the circulating levels of chylomicrons, and an absence of LPL activity from post-heparin plasma. However, there is less increase in VLDL levels than would be predicted from the role of LPL in VLDL catabolism. An alternative, low-capacity pathway probably exists for the clearance of intact VLDL particles by the liver. Numerous mutations within the human LPL gene have now been identified. Their effects on LPL function were discussed in Section 2.2. Congenital hepatic lipase deficiency is associated with increased levels of plasma triacylglycerol compared to controls.

Because of the dominating role of apo C2 as cofactor for LPL activity, the effects of congenital apo C2 deficiency in human plasma mimic those of LPL deficiency. Mice overexpressing or deficient in LPL have been developed. Their plasma lipoprotein patterns resemble those of the corresponding human genetic deficiency, and confirm the roles of LPL and hepatic lipase in plasma lipid metabolism described above. Mice with muscle-specific overexpression of LPL developed insulin resistance along with the expected increase in muscle triacylglycerol levels [7]. This effect was associated with a decrease in insulin-stimulated glucose uptake. These findings show the power of transgenic mouse models in studying complex metabolic diseases.

3. HDL and plasma cholesterol metabolism

3.1. The origin of HDL

Unlike apo B-containing lipoproteins, HDL-containing apo A1 are formed in the extracellular space. This process involves the association of lipid-poor apo A1 with

cell-derived phospholipids and cholesterol. The association of apo A1 and phospholipid is thermodynamically favorable; phospholipid-free apo A1 has not been detected in biological fluids. Nevertheless isolated, lipid-free apo A1 is often used as a convenient surrogate for lipid-poor apo A1 in the analysis of lipid transfers from the cell surface.

Newly synthesized apo A1 made by the liver and (particularly in humans) by the small intestine is recovered loosely associated with the surface of lymphatic triacylglycerol-rich lipoproteins. The apo A1, probably in association with small amounts of phospholipid, dissociates spontaneously after entering the plasma compartment, in a reaction independent of lipolysis. Lipid-poor apo A1 can also be generated via the action of lipid transfer proteins and/or hepatic lipase (Sections 4.2 and 4.3), when these reduce the core size of mature, spherical HDL.

Both lipid-poor and lipid-free apo A1 demonstrate pre β -migration when plasma is fractionated by nondenaturing agarose gel electrophoresis. Under these conditions, the bulk of HDL, made up of spherical lipid-rich particles, has more rapid, α -migration, while LDL migrates more slowly in a β -position. This technique has proven to be useful for discriminating 'early' or lipid poor HDL from mature, lipid-rich particles in the RCT pathway. The major pre β -HDL of human plasma (pre β_1 -HDL) has a molecular weight of about 70 kDa. An increase in pre β -HDL levels has been correlated with an impairment of RCT and an increased risk of coronary artery disease in human patients.

There is little information yet on the physical structure of prebeta-HDL, although at least two inter-convertible forms, containing one and two molecules of apo A1, may be present in plasma.

3.2. Role of the ABCA1 transporter in HDL genesis

Studies in vitro have shown that the prebeta-HDL population includes avid acceptors of cell-derived cholesterol and phospholipids. These lipoprotein complexes are precursors of mature HDL. In human Tangier Disease, there is an almost complete deficiency of mature HDL. The low levels of apo A1 present (1–2% of normal) have pre β -mobility. Tangier Disease patients also have localized patches of orange, lipid-laden macrophages, classically in the tonsils. LDL levels are very low. Cultured Tangier Disease fibroblasts lack significant ability to transfer either phospholipid or free cholesterol to lipid-free apo A1, though transfer of cellular lipids to mature HDL is almost normal (G. Rogler, 1995). These data have led to the conclusion that Tangier Disease patients inherit a defect in the ability of peripheral cells to build normal mature HDL from lipid-poor, apo A1-containing precursors.

Genetic analysis of Tangier Disease families recently led to the identification of one of the key factors for HDL assembly. The DNA of these patients was found to contain deletions or other defects in the ABCA1 gene. ABCA1 is an ATP-binding cassette (ABC) transporter protein closely related to the multidrug resistance transporter, to several hepatic bile acid transporters, and other transporter proteins active in the transmembrane movement of small amphipathic solutes [8].

ABCA1 mRNA is widely expressed among tissues and in cultured cells. It has been studied most intensively in fibroblasts and macrophages, where its activity has been linked to the efflux of cholesterol and phospholipids to extracellular apo A1.

The regulation of ABCA1 expression is complex, and incompletely understood. At least three classes of mRNA transcripts have been identified, corresponding to different transcriptional start sites. These are under the regulation of different promoter sequences. In macrophages and hepatocytes, ABCA1 mRNA levels are strongly upregulated by oxysterols and retinoic acid via a LXR/RXR tandem transcription site (see Chapter 16) [9]. In a few rodent transformed macrophage cell lines, ABCA1 expression is regulated by cAMP but human cells generally are unaffected (A.E. Bortnick, 2000). ABCA1 mRNA levels are also upregulated by cholesterol itself. The molecular mechanism of the response to cholesterol has not yet been clarified. It may involve oxysterols generated intracellularly (X. Fu, 2001).

Despite the stimulus that the identification of ABCA1 has given to HDL studies, a number of key questions on its role in HDL formation remain unresolved. The first is the mechanism by which ABCA1 promotes lipid efflux, and the identity of the lipids transported. In ABCA1^{-/-} mouse embryos, a defect was identified in the catabolism of apoptotic cell bodies. At the same time, ABCA1^{-/-} cells were found to be defective in annexin V binding, an assay for exofacial phosphatidylserine in apoptotic cells (Y. Hamon, 2000). At the present time, the primary substrate for ABCA1 has not been identified. Phospholipid leaving the cell surface under the influence of ABCA1 is almost entirely phosphatidylcholine. One current hypothesis is that ABCA1 could modify the phospholipid composition, and possibly charge, of the exofacial leaflet of the membrane bilayer, thereby secondarily reducing the activation energy for efflux of phosphatidylcholine (Fig. 6). Another hypothesis is that ABCA1 directly transports phosphatidylcholine, and possibly free cholesterol. This possibility is consistent with the loss of both free cholesterol and phospholipid efflux from Tangier cells, and the restoration of both activities in cells transfected with ABCA1 cDNA. However, several recent reports suggest that ABCA1 might not play a direct role in cholesterol transport. These studies each showed that phospholipid efflux by ABCA1 was regulated independently of free cholesterol efflux [10]. These data appear more consistent with a two-step process: (1) addition of phospholipid; followed by (2) addition of FC.

3.3. *The role of caveolae in HDL genesis*

Another area of active investigation is the origin of the cellular cholesterol transferred to pre β -migrating (lipid-poor) HDL. There is general agreement that unesterified cholesterol transferred to lipid-poor apo A1 originates mainly from the plasma membrane. Caveolae (see Chapter 1) are microdomains of the cell surface implicated in cholesterol homeostasis and transport as well as signal transduction. These functions are probably related, because unesterified cholesterol levels in caveolae regulate the efficiency of signal transmission. In primary cells including fibroblasts, smooth muscle cells and endothelial cells, caveolae are implicated as the direct precursors of cholesterol in lipid-poor HDL. While caveolae contain several proteins involved in cellular cholesterol homeostasis, such as the scavenger receptor BI (see Section 4.4) there is no convincing evidence at present that the ABCA1 transporter is located there, consistent with the phospholipids and unesterified cholesterol on apo A1 originating from different membrane microdomains.

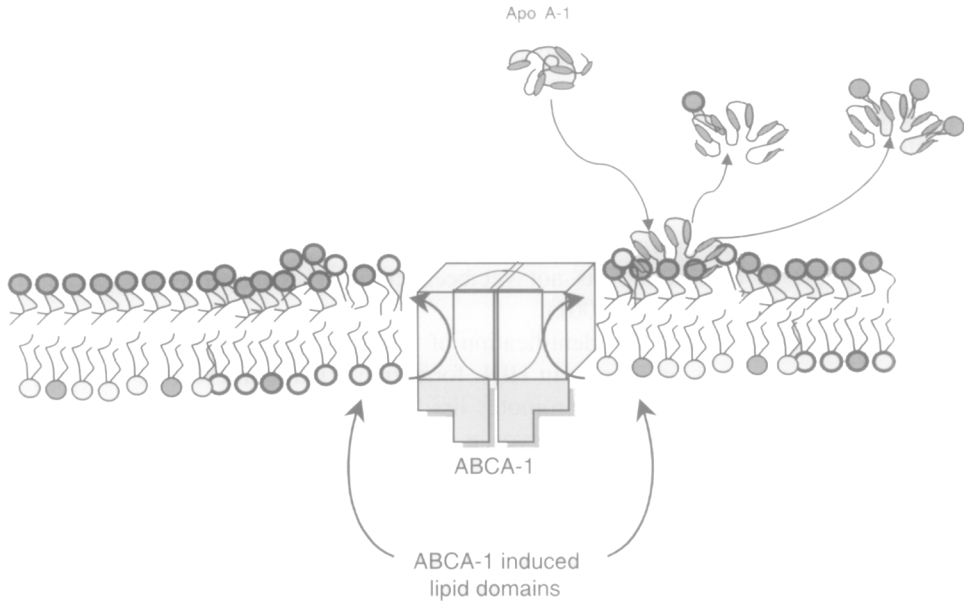
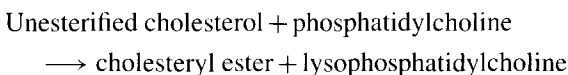


Fig. 6. The ABCA1 transporter and the formation of lipid-poor HDL reactive as acceptors of cell-derived unesterified cholesterol. The figure illustrates the role proposed for ABCA1 in the distribution of phospholipids between the exo- and cyto-facial leaflets of the membrane bilayer. The exofacial leaflet is rich in phosphatidylcholine, substrate for apo A1 at ABCA1 transporter sites. Closed circles, phospholipid; open circles, free cholesterol. Modified from G. Chimini (2002) by permission.

3.4. The role of LCAT in HDL genesis

Further growth and maturation of apo A1-HDL depend on the activity of lecithin : cholesterol acyltransferase (LCAT):



Though present in lymph, LCAT is active mainly in the plasma compartment. It had been thought until recently that the primary substrates of LCAT were phospholipid-rich, discoidal apo A1-containing particles which support maximal acyltransferase rates, and accumulate in the plasma of LCAT-deficient subjects. Recently, it was reported that LCAT could be directly reactive with lipid-poor HDL [11]. This could indicate that more than one pathway can convert lipid-poor to mature HDL.

LCAT consumes unesterified cholesterol and phospholipids to produce insoluble cholesteryl ester (Fig. 7). This is retained in the HDL core, while the water-soluble lysophosphatidylcholine formed at the same time is transferred away to albumin. In this way, LCAT maintains concentration gradients of cholesterol and phosphatidylcholine between cell and lipoprotein surfaces and the growing HDL particle. The later stages of HDL genesis probably depend entirely on diffusion of lipids from the surface of other lipoprotein particles that is independent of ABCA1 activity.

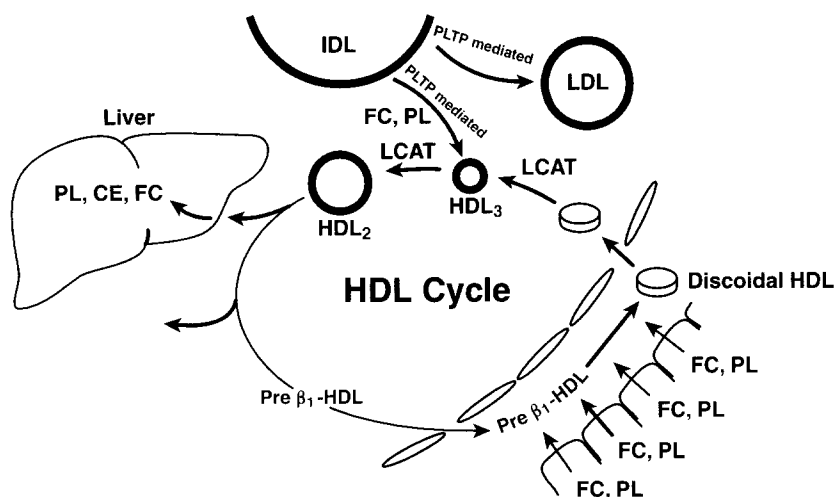


Fig. 7. The HDL cycle, showing: lipid-poor, pre β -migrating (pre β -1) particles incorporating cell- and lipoprotein-derived unesterified cholesterol and phospholipid; the formation of discoidal HDL; the role of the LCAT reaction in generating spherical, α -migrating HDL; and the generation of a new cycle of lipid-poor particles at the surface of liver cells. The transfer of excess phospholipid from VLDL and IDL as a result of lipolysis, is catalyzed by phospholipid transfer protein (PLTP).

3.5. Regeneration of pre β -migrating HDL

Lipid-poor, pre β -migrating HDL are formed when the size of the central lipid core of α -migrating HDL, which is mainly cholesteryl ester, is decreased, or when HDL surface lipids are increased (Fig. 7). The surface area occupied by HDL protein, cholesterol and phospholipids then exceeds the surface area of the core. Apo A1 dissociates from the particle in the form of lipid-poor, pre β -migrating HDL. Once released, these pre β HDL are available as acceptors of additional cell-derived lipids. They may be sufficiently primed with phospholipid to participate in a new cycle of cholesterol efflux. Pre β -HDL particles in plasma are sphingomyelin-rich, while particles newly formed as the result of ABCA1 transporter activity contain mainly phosphatidylcholine molecules. This finding is consistent with the hypothesis that once formed, pre β -HDL can recycle via LCAT, losing phosphatidylcholine to transesterification but retaining sphingomyelin, which is not a LCAT substrate.

Three pathways have been identified for pre β -HDL formation from α -HDL:

- (1) phospholipid transfer protein (PLTP) activity;
- (2) exchange of HDL cholesteryl esters for triacylglycerol in VLDL and LDL catalyzed by cholesteryl ester transfer protein (Section 4.3) concomitant with hepatic lipase-mediated lipolysis of HDL triacylglycerol.
- (3) selective uptake of cholesteryl ester from HDL catalyzed by the cell-surface scavenger receptor SR-BI.

It is not known if the pre β -HDL formed by these different pathways have the same kinetic properties as acceptors of cell-derived lipids, though it seems likely. Because triacylglycerol molecules have a similar volume to that of cholesteryl esters, CETP

alone would seem unlikely to promote pre β -HDL formation. The relative contribution of the different pathways towards the recycling of apo A1 is likely to differ significantly under different physiological conditions.

Several other plasma apolipoproteins (particularly apo A4 and apo E) have marked sequence similarity to apo A1. Lipid-poor HDL with these proteins in place of apo A1 have been identified. Their concentration is much lower than those of apo A1 particles. Also, it is not clear if apo A4 and apo E particles, two of those identified, can recycle between lipid-rich and lipid-poor populations in the way described for apo A1. As a result, apo A1 is likely to play the predominant role in transporting peripheral cell cholesterol through the plasma compartment to the liver, at least in normal metabolism.

3.6. Regulation of gene expression and structure of apo A1

The apo A1 gene codes for a 287-aa preproprotein. Following the loss of its leader sequence, and the removal of a 6-aa pro-sequence in plasma, mature apo A1 circulates as a 243-aa polypeptide. Apo A1 gene transcription rates are not highly regulated, compared to those of ABCA1 and other catalytic factors of the HDL cycle. Sp1, a 'housekeeping' transcription factor, plays a major role in regulating apo A1 transcription rates. PPARs which regulate phospholipid efflux to apo A1, are reported to have little effect on the expression of apo A1 itself. Regulation of the cholesterol transporting activities of apo A1 in plasma is probably determined for the most part by its distribution within its three structural forms, i.e. the amorphous (lipid-poor), discoidal and spheroidal HDL species.

Apo A1, like other phospholipid-binding plasma apolipoproteins, is largely made up a series of amphipathic helical segments, typically 22 amino acids in length [12] (Chapter 18). These are separated by helix-breaking proline or glycine residues. Synthetic amphipathic helical segments whose primary sequence is unrelated to that of native apo A1 can be effective mimics of native apolipoprotein in binding phospholipid, promoting cholesterol efflux from cells, and activating the formation of cholesteryl esters by the LCAT reaction. In spite of this, some repeats in native apo A1 are clearly of more significance than others. This has been seen in experiments where the position of repeats within the primary sequence was systematically varied, though the amino acid sequence of each repeat was unchanged (M. Sorci-Thomas, 1997). The biological activity of such mutant apo A1 species varied widely. These data indicate that apo A1 retains significant tertiary structure. Some generalizations are now possible. The central ('hinge') region of the polypeptide (residues 143–164) appears to be of particular significance in promoting the LCAT reaction [13]. The same domain is important in promoting cellular cholesterol efflux, in the presence or absence of LCAT activity. A C-terminal domain has been implicated in phospholipid binding.

Most of the information on apo A1 function has been obtained from synthetic discoidal recombinants of apo A1 and pure phospholipids, with a molar ratio of 1 : 200 to 1 : 500. The size and molecular properties of these particles, produced by sonication or dialysis from detergent solution, are quite similar to those of discoidal lipoproteins found in the plasma of LCAT-deficient human subjects. The particles have been shown to consist of a planar phospholipid bilayer. The edges of the bilayer are sealed from the

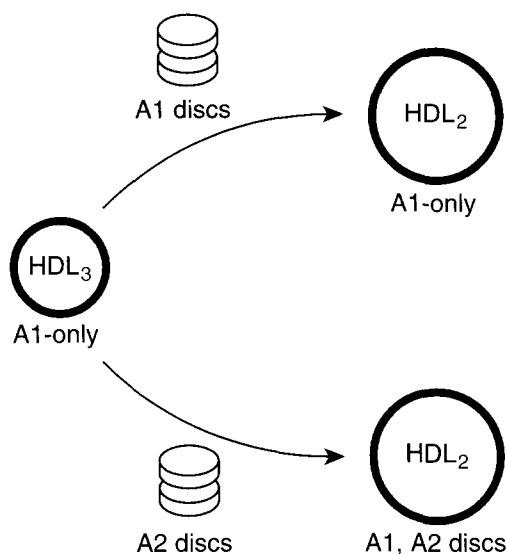


Fig. 8. HDL fusion and the formation of apo A1, apo A2 HDL. The role proposed for PLTP in the promotion of fusion is illustrated, together with the formation of apo A1, apo A2 products from alpha-HDL.

aqueous medium by apo A1. It was thought earlier that a 'picket fence' model in which the repeats were at right angles to the lipid bilayer, accurately reflected the structure of these particles (Chapter 18). A more recent 'belt' model has the repeats aligned circumferentially parallel to the bilayer [14]. The balance of evidence still suggests that discoidal HDL are the normal intermediate of the conversion of lipid-poor, prebeta HDL to mature, spherical particles. The presence of discoidal HDL in lymph is consistent with this interpretation. In any case, the end products of the action of LCAT on apo A1 complexes rich in cholesterol and phospholipid are alpha-migrating, spherical HDL particles rich in cholesteryl esters.

Most alpha HDL particles, unlike pre β - and discoidal HDL, include apo A2 as well as apo A1. Evidence recently obtained suggests that these are a product of the fusion of apo A1-only and apo A2-only HDL particles [15] (Fig. 8). This fusion could be mediated locally by the lysophosphatidylcholine formed in the LCAT reaction. In LCAT deficient plasma, apo A1 and apo A2 form distinct populations of HDL particles. Apo A2 has been considered an inhibitor of the LCAT reaction, and thus indirectly, of reverse cholesterol transport. Apo A2 might thus limit the size reached by spherical HDL. Mice transgenic for apo A2 were atherosclerosis-prone compared to normal animals of the same strain, but this effect has not been seen in mice of all genetic backgrounds.

The formation of α -HDL is accompanied by large changes in the conformation of apo A1. This was made clear by studies with monoclonal antibodies, as well as a variety of sensitive physical techniques. The unique properties of apo A1 in lipid binding and the promotion of reverse cholesterol transport reflect this elasticity.

3.7. Structure and properties of LCAT

Plasma LCAT originates mainly from hepatocytes. Hepatic levels of LCAT mRNA are determined mainly by the interplay of Sp1 and Sp3 promoter binding sites. The rate of the LCAT reaction in plasma is regulated for the most part not by changes in circulating LCAT protein levels, but by differences in its catalytic rate with the different HDL particles. Postprandially, LCAT rates are increased as unesterified cholesterol and phospholipid are transferred to HDL from triacylglycerol-rich lipoproteins; the level of LCAT protein in the circulation is unchanged.

There is enough LCAT in plasma for only about 1% of HDL particles to contain one molecule of enzyme. Either LCAT must move rapidly between HDL particles or, more likely, its substrates and products must be transferred effectively from a metabolically active HDL subfraction containing LCAT to other HDL particles. The spontaneous transfer of free cholesterol and lysophosphatidylcholine in plasma is rapid. That of phosphatidylcholine and cholesteryl esters is much slower. These transfers are stimulated by dedicated plasma lipid transfer proteins (Sections 4.2 and 4.3).

LCAT is a 416-amino acid serine hydrolase [16]. It has only limited sequence homology (<5%) to other lipases (LPL, hepatic lipase, pancreatic lipase). The amino acid residues which make up its active site triad have been identified. Several carbohydrate chains modify the reaction rate and substrate specificity of the enzyme. LCAT has not been crystallized. Efforts have been made to explain its three-dimensional structure using the coordinates obtained from X-ray diffraction analysis of triacylglycerol lipases. LCAT, like these lipases, probably has a mobile 'lid' responsive to the lipid interface of HDL. A helical domain, adjacent to the active site serine residue and partly homologous to a sequence in apo E, may be involved in lipid binding. To date, these insights have been insufficient to explain the unique selectivity of LCAT for unesterified cholesterol, rather than the hydroxyl group of water, as acyl acceptor. In the complete absence of cholesterol, LCAT is an efficient phospholipase.

Apo A1 is required for both the acyltransferase and phospholipase activities of LCAT. Three arginine residues within the 143–164 repeat of apo A1 (R₁₄₉, R₁₅₃, and R₁₆₀) are essential for its activation by apo A1, suggesting a possible role for salt-bridges between these residues and either negatively charged amino acids in LCAT, and/or phosphate groups within phosphatidylcholine [17]. In reaction with LDL, LCAT catalyzes phosphatidylcholine acyl exchange. LCAT can also hydrolyze short-chain lipid esters. This reaction is independent of the presence of apo A1, consistent with the view that the apoprotein may be needed to align the enzyme and its substrates at a phospholipid–water interface.

3.8. Congenital deficiencies of LCAT and HDL

Two variants of LCAT deficiency are recognized. In the first, LCAT synthesizes no cholesteryl esters in plasma. Cholesterol accumulates as droplets in peripheral tissues. Apo-E-rich particles accumulate in LCAT deficient plasma, indicating this alternative cholesterol transport pathway, though upregulated, cannot fully substitute for that catalyzed by LCAT. Only lipid-poor and discoidal HDL particles are present under

these conditions. In the second type of LCAT deficiency (Fish-Eye Disease) LCAT can transesterify cholesterol from VLDL and LDL, but not from exogenous HDL. LCAT deficiency and Fish-Eye Disease are the result of different mutations in the primary sequence of the LCAT protein (J.A. Kuivenhoven, 1997).

Normal HDL are absent from plasma in congenital apo A1 deficiency, and also in ABCA1 deficiency (Tangier Disease). In apo A1 deficiency, no HDL is present. In Tangier Disease, HDL present is all in the form of prebeta-HDL. Whether prebeta-HDL in Tangier Disease have the same composition as those in normal plasma has apparently not been reported.

Epidemiological studies consistently show that low HDL cholesterol is correlated with an increased risk of atherosclerotic vascular disease [18]. The relationship is usually stronger than that between the same disease and LDL. The evidence that heart disease is systematically increased in LCAT deficiency, apo A1 deficiency and Tangier Disease is equivocal at best, in spite of the fact that LCAT, apo A1 and ABCA1 play key roles in regulating cellular cholesterol content. This paradox has several possible explanations. The first is that these HDL deficiency diseases are also characterized by low levels of circulating LDL (25–50% of normal). This reduces the delivery of cholesterol to peripheral cells to partially offset reduced reverse cholesterol transport. A second possibility is that HDL cholesterol concentrations may not reflect the rate of reverse cholesterol transport. For example, mice transgenic for SR-BI (Section 4.4) increase cholesterol clearance to bile but decrease HDL cholesterol levels (K.F. Kozarsky, 1997). Other pathways, such as that involving apo E, may be able to assume part of the function of apo A1. A third possibility is that it is not HDL cholesterol as such, but a metabolically active subfraction of HDL, that is antiatherogenic. Changes in its composition could be less extreme than those of HDL cholesterol levels. Trace HDL proteins that could play such a role are the antioxidant proteins paraoxonase and platelet activating factor which are responsible for the protective role of HDL in neutralizing oxidized phospholipids in LDL (M. Navab, 2001).

4. Reactions linking the metabolism of apo A1 and apo B lipoproteins

4.1. Metabolic implications

Triacylglycerol carried by apo B lipoproteins is mainly catabolized in peripheral (that is, non-hepatic) tissues. Its fatty acids are used for oxidative metabolism or storage. In contrast, very little cholesterol is needed for growth or repair in peripheral tissues. Nevertheless there is a continuous ‘forward’ delivery of cholesterol to peripheral cells. Two main reasons can be suggested. Cholesteryl ester is needed for triacylglycerol-rich particles to be successfully secreted from the liver. Second, the recycling of free cholesterol between the liver and peripheral cells suppresses local cholesterol synthesis and the expression of lipoprotein receptors. These receptors would otherwise promote the futile uptake up large amounts of lipoprotein cholesteryl ester.

Despite the different roles of the apo A1- and apo B-lipoproteins systems, exchange reactions in plasma, catalyzed by lipid transfer proteins, have been identified. These

promote the movement of lipids between the major transport pathways. Transfer proteins are ATP-independent. Their reactions (i) are reversible; and (ii) proceed only down preexisting concentration gradients.

4.2. Phospholipid transfer protein (PLTP)

PLTP is a 476 amino acid protein showing ~20% sequence similarity to several other lipid-binding proteins, which include cholesteryl ester transfer protein and bacterial permeability inducing protein. Short, highly hydrophobic sequences in conserved regions of the primary sequence may represent the strands of a hydrophobic basket or cleft involved in lipid binding. The expression of PLTP is PPAR-gamma dependent, and may involve the LXR/RXR orphan receptor heterodimeric complex, the same factors that regulate ABCA1 expression, and phospholipid efflux from cells.

In plasma, PLTP catalyzes the transfer of phospholipids, particularly phosphatidylcholine, between lipoprotein classes [19]. The generation of excess surface phosphatidylcholine as a result of triacylglycerol lipolysis, and the consumption of phosphatidylcholine by LCAT, both ensure that a phospholipid gradient is maintained from VLDL and LDL to HDL. PLTP activity is reported to be present in all mammalian plasmas. Human genetic PLTP deficiency has not yet been unequivocally identified.

PLTP activity is needed for maximal LCAT activity because the rate of transfer of phospholipids from cells to plasma via ABCA1 transporter activity, and the spontaneous transfer of phospholipids from other lipoproteins, are both much slower than that of cholesterol. Without PLTP, reverse cholesterol transport might otherwise be limited. PLTP also plays a major role in generating prebeta-HDL, the major acceptor of cellular cholesterol. An additional role for PLTP recently identified is in the secretion of apo B lipoproteins from the liver [20]. Finally, PLTP can promote phospholipid efflux from the surface of fibroblast monolayers to preformed HDL, though not to lipid-free apo A1.

4.3. Cholesteryl ester transfer protein (CETP)

CETP is a 476-amino acid plasma protein structurally related to PLTP (Section 4.2). Like PLTP, CETP expression in hepatocytes is PPAR-dependent [21]. The C-terminus of CETP, a domain absent in PLTP, plays a key role in the transfer of both triacylglycerols and cholesteryl esters. Neither the tertiary structure nor the detailed mechanism of CETP-mediated lipid transfer are yet fully established. A model of CETP tertiary structure based on X-ray coordinates established for bacterial permeability-increasing protein has been described. Like LCAT, the activity of CETP is regulated more by the composition of substrate lipoproteins than by the circulating level of CETP protein. For example, increased CETP activity observed postprandially appears to be almost completely the consequence of increased triacylglycerol/cholesteryl ester ratios in triacylglycerol-rich dietary lipoproteins.

Like PLTP, the CETP reaction transfers lipids down a preexisting concentration gradient maintained by LCAT. CETP normally promotes transfer of CE to VLDL and LDL, at a rate typically ~50% that of LCAT. This means that much of the

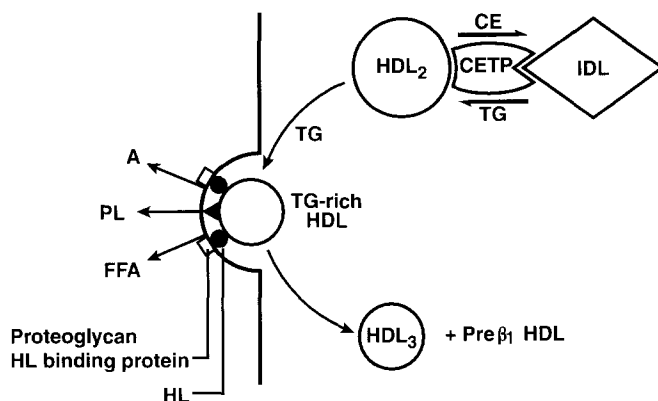


Fig. 9. Remodelling of HDL by hepatic lipase (HL). The hydrolysis of triacylglycerol (TG) transferred from VLDL and IDL via the activity of cholesteryl ester transfer protein (CETP) is shown, together with the displacement of lipid-poor (prebeta₁) HDL from the diminished surface of the spherical HDL particle. FFA, free fatty acids.

cholesteryl ester generated by LCAT is cleared directly from HDL, not from LDL after CETP-mediated lipid transfer (Section 4.4).

The net effect of CETP activity is to reduce HDL CE and increase LDL CE. In normal plasma, CE transfer is complemented by a similar and opposite transfer of triacylglycerol from VLDL and LDL to HDL. Under conditions where there is no cholesteryl ester concentration gradient between lipoproteins (for example, if VLDL secreted from the liver contains as much cholesteryl ester as HDL) CETP can still catalyze the unproductive exchange of cholesteryl esters between lipoprotein particles.

There has been considerable debate whether CETP should be considered a 'proatherogenic', pathologically neutral, or 'antiatherogenic' factor. The activities of CETP and hepatic lipase contribute to the recycling of lipid-poor apo A1, and the formation of prebeta-HDL (Fig. 9). Whether or not increased CETP activity leads to an increase in circulating LDL cholesterol levels depends on the capacity of hepatic LDL receptors. A study of Japanese subjects expressing partial CETP deficiency did not suggest any increased resistance to atherosclerosis. On the other hand, reduced CETP levels in hemodialysis patients have been linked to an increased incidence of heart disease (although other relevant activities, including LCAT, were also reduced). Several groups of thiol reagents have been described which inhibit CETP activity. As of the time of writing, CETP inhibitors have not been shown to reduce atherosclerosis in human populations, though beneficial effects in rabbits have been reported.

4.4. Scavenger receptor BI (SR-BI)

SR-BI (the human protein is also known as CL-A1) is a 409-amino acid transmembrane protein member of the scavenger B-family. Its primary sequence contains no consensus ATP binding site. As a result, lipid transfers mediated by SR-BI, like those supported by CETP and PLTP, are driven by established concentration gradients. (Fig. 10). SR-BI

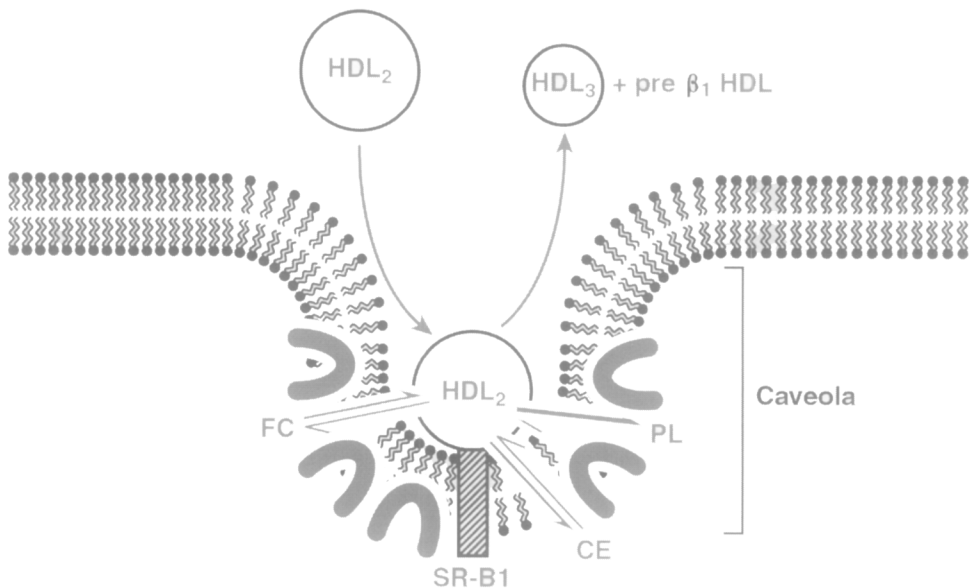


Fig. 10. Lipid transfers catalyzed by SR-BI. SR-BI is shown localized to caveolae. The selective transfers of cholesteryl ester (CE), free cholesterol (FC) and phospholipids (PL) are shown between HDL and the cell surface. The model suggests that SR-BI acts to promote facilitated (that is, protein-mediated) diffusion of lipoprotein lipids down their physiological concentration gradients, under conditions where FC efflux is driven by the LCAT reaction.

catalyzes the selective uptake of lipids, particularly CE, from lipoprotein particles, particularly HDL, though it is active with a wide array of lipids and lipoproteins. SR-BI also promotes efflux of unesterified cholesterol from cells [22]. Transcriptional regulation of SR-BI expression by transcription factor SF-1, by sterol regulatory element binding protein, and by Sp proteins 1 and 3 has been described. While there are significant species differences in the tissue specificity of SR-BI expression, SR-BI is usually expressed at high levels in tissues forming steroid hormones (such as adrenal and gonadal cells). Expression is high in mouse liver but low in human liver. This pattern is the inverse of that seen with CETP and reflects the specialization of humans and mice as 'LDL' and 'HDL' animals respectively. SR-BI has been localized to caveolae, the cholesterol-rich microdomains that are abundant in many peripheral cells.

The selective uptake from HDL requires SR-BI binding. Two positively charged residues (R_{402} , R_{418}) are important for efficient uptake of cholesteryl esters into the cell. The mechanism by which SR-BI promotes transport of unesterified cholesterol out of the plasma membrane is not yet clear. From a study of mutant SR-BI species and anti-SR-BI antibodies, it was suggested that HDL binding was essential for cholesterol efflux. In contrast, kinetic studies suggested that much of the effect of SR-BI on cholesterol efflux was indirect, possibly the result of induced local modifications in the distribution of lipids within the membrane bilayer. It is unclear to what extent cholesterol efflux from caveolae depends on the presence of SR-BI. CD-36, a second scavenger protein, also found in caveolae, has a weaker effect.

SR-BI may facilitate uptake of unesterified cholesterol from the intestinal lumen. ABCA1 has been implicated in the transport of unesterified cholesterol out of the intestine into lymph. While outside the scope of this chapter, these findings indicate that mechanisms parallel to those described in this chapter may be in place to regulate cholesterol transport to plasma from other extracellular spaces.

4.5. Animal models of human plasma cholesterol metabolism

The availability of technology to over-express or delete individual genes in mice has had a wide impact in this field. The effects of modulating the levels of a single enzyme or transport protein can be studied *in vivo* against the background of interacting factors. Many of these studies were initiated to estimate the role of each gene product in promoting or inhibiting atherogenesis. A major problem of the transgenic/knockout approach has been that in several respects plasma cholesterol metabolism and transport in mice differ significantly from that in humans. As a result, despite successful efforts to create mouse models of human lipid diseases, the quantitative role played by individual factors has sometimes been difficult to establish.

Identification of ABCA1 as the defective protein in human Tangier Disease led rapidly to the production of ABCA1^{-/-} mice [23]. The plasma lipoprotein pattern in these animals, in particular the almost complete absence of HDL, mirrors that of human ABCA1 deficiency. In the developing mouse fetus, disposal of apoptotic cell bodies was inhibited. Cellular cholesterol accumulation in ABCA1^{-/-} mice, particularly in the lungs, was dramatic. These abnormalities have not been reported in human Tangier patients.

Apo A1^{-/-} mice showed decreased HDL cholesterol levels (about 25% of normal). Apo E levels in HDL were increased, and probably as a result, atherosclerosis susceptibility was not increased. Plasma LCAT activity was almost completely inhibited in these animals. This suggests that other apolipoproteins are unable to replace apo A1 *in vivo* as LCAT cofactors. In mice transgenic for human apo A1, the human protein displaced mouse apo A1 almost completely from HDL. These animals were protected against diet-induced atherosclerosis.

LCAT^{-/-} knockout mice have a marked phenotypic resemblance to LCAT deficient human subjects. Discoidal and lipid-poor HDL species accumulate in the plasma, consistent with the identification of these as precursors of mature, alpha-migrating HDL. LCAT transgenic mice were not protected from atherosclerosis, in contrast to LCAT transgenic rabbits [24].

PLTP^{-/-} mice had reduced HDL levels, consistent with the role proposed for this transfer protein in supplying phospholipids to the LCAT reaction, but atherosclerosis susceptibility was not increased [20]. This may be linked to a concomitant reduction in the secretion of apo B lipoproteins (Section 4.2). Overexpression of PLTP in mice was associated with atherosclerosis resistance, and the appearance of increased concentrations of prebeta-HDL, consistent with the role predicted for these particles in the early steps of reverse cholesterol transport.

CETP is absent from mouse plasma. The human CETP gene was expressed in mice in a number of independent studies. Its effects were complex. In one study, the expression

of human CETP together with human LCAT, reduced atherosclerosis susceptibility. In a second study, overexpression of human CETP alone in mice led to increased levels of cholesteryl ester in apo B lipoproteins, and induction of atherosclerosis. The molecular basis of these differences is still not clear.

SR-BI deficiency was associated with reduction in the biliary cholesterol content, and an increase in circulating HDL levels. Intestinal cholesterol absorption was not increased in SR-BI^{-/-} mice. Mice transgenic for human SR-BI showed significantly lower plasma HDL levels. Paradoxically, atherosclerosis susceptibility was reduced [25].

5. Summary and future directions

Since the last edition of this volume, there have been many advances in understanding plasma lipid metabolism. Major developments in basic mechanisms include the identification of ABCA1 and SR-BI. In the field of triacylglycerol transport, the roles of new members of the LDL receptor family have been established, and new insights into the regulation and transport of LPL identified. The efflux of cholesterol from cells to plasma lipoproteins, previously considered the result of passive diffusion, is now recognized to be highly regulated by cell membrane proteins, and on a par with influx as a key determinant of cellular cholesterol homeostasis. The availability of mice overexpressing or deficient in almost all known factors of plasma lipid transport has provided key insights into regulatory pathways.

In other areas, much remains to be done. Knowledge of the tertiary structure of key proteins of plasma lipid metabolism remains very incomplete. Our understanding of the transcriptional regulation of newly identified lipid transport proteins is, not surprisingly, still in its infancy. Efforts to generate mouse models better reflective of human vascular biochemistry and pathology continue. Plasma lipoprotein metabolism has shown itself again to be much more complex, and much more highly regulated than previously thought, and still a target for further intensive research.

Abbreviations

ABCA1	ATP-binding cassette transporter A1
Apo-	apolipoprotein
CETP	cholesteryl ester transfer protein
HDL	high-density lipoprotein
IDL	intermediate-density lipoprotein
LCAT	lecithin : cholesterol acyltransferase
LDL	low-density lipoprotein
LPL	lipoprotein lipase;
PLPT	phospholipid transfer protein
RCT	reverse cholesterol transport
SR-BI	scavenger receptor BI;
VLDL	very low-density lipoprotein

Acknowledgements

Research by the authors cited in this chapter was supported by the National Institutes of Health via HL 14237, HL 57976 and HL 67294.

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Lipoprotein receptors

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1. Introduction

From a systemic lipoprotein metabolism point of view, the main task of lipoprotein receptors is the clearance of lipoproteins from the circulation, body fluids, and interstitial spaces. One can envision several reasons why lipoproteins need to be cleared from extracellular fluids into cellular compartments, for instance: (1) they are transport vehicles for components that are vital to the target cells; (2) their uptake serves signalling and/or regulatory roles in cellular metabolism; (3) they have done their job and have become dispensable; and (4) they might have deleterious effects if allowed to remain extracellular for prolonged time. Chapters 18, 19, and 20 have described the structures, syntheses, and interconversion pathways of lipoprotein particles in the circulation. Here, the mechanisms of lipoprotein transport from the plasma compartment to various types of cells of the body, which is one of the best understood aspects of receptor biology, is described. In addition, newly discovered functions of receptors thus far thought to be specialized exclusively for lipoprotein transport are described.

In the context of lipoprotein transport via cell surface receptors, it helps to recall that the two major transported lipid components of lipoproteins, triacylglycerols and cholesterol, have quite different fates. Triacylglycerols are delivered primarily to adipose tissue and muscle where their fatty acids are stored or oxidized for production of energy, respectively. Cholesterol, in contrast, is continuously shuttled among the liver, intestine, and other extra-hepatic tissues. Actually, the major transport form of cholesterol is its esterified form; within cells the cholesteryl esters are hydrolyzed and the unesterified sterol has multiple uses. Among their many functions, sterols serve as structural components of cellular membranes, as substrates for the synthesis of steroid hormones and bile acids, and they perform several regulatory functions (a classical example is the low density lipoprotein (LDL) receptor pathway, see Section 2.2). For correct targeting of lipoproteins to sites of metabolism and removal, the lipoproteins rely heavily on the apolipoproteins (apos) associated with their surface coat. Apos mediate the interaction of lipoprotein particles with enzymes, transfer proteins, and with cell surface receptors, the main topic of this chapter.

Key features of human lipoprotein metabolic pathways are schematically summarized in Fig. 1; this outline necessarily omits many of the details which are less significant for aspects of receptor-mediated removal of lipoproteins. The interwoven complex pathways can be divided into exogenous and endogenous branches, concerned with the transport of dietary and liver-derived lipids, respectively. Both metabolic sequences start with the

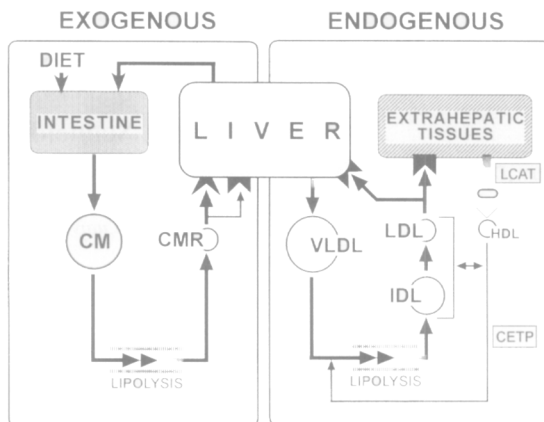


Fig. 1. Highly schematized summary of the pathways for receptor-mediated lipoprotein metabolism in humans. The liver is the crossing point between the exogenous pathway (left-hand side) dealing with dietary lipids, and the endogenous pathway (right-hand side) that begins with hepatically synthesized lipoproteins. The exogenous metabolic branch starts with the production of chylomicrons (CM) in the intestine, whereas the liver synthesizes very low density lipoprotein particles (VLDL). Abbreviations: CMR, chylomicron remnants; IDL, intermediate density lipoproteins; LDL, low density lipoproteins; HDL, high density lipoproteins; LCAT, lecithin : cholesterol acyltransferase; CETP, cholesteryl ester transfer protein; ▲, LDL receptor-related protein (LRP); and ■, LDL receptor; 'Lipolysis' denotes lipoprotein lipase-catalyzed triacylglycerol lipolysis in the capillary bed.

production and secretion of triacylglycerol-rich lipoproteins (Chapter 19). Intestinally derived chylomicrons (CM in Fig. 1) are secreted into the lymph and from there enter the bloodstream, where they function as energy carriers by providing triacylglycerol-derived fatty acids to peripheral tissues. This lipolytic extraction of fatty acids from the triacylglycerol core of the lipoprotein particles ('Lipolysis' in Fig. 1) is achieved mainly by the enzyme lipoprotein lipase, which is bound to the luminal surface of the endothelial cells lining the capillary bed. Removal of triacylglycerol in extrahepatic tissues results in decreased size of the chylomicrons and produces cholesteryl ester-rich lipoprotein particles termed chylomicron remnants. During this conversion, apoCs are lost from the surface of the particles; the remnants, having finished their task, are destined for catabolism by the liver, which occurs almost exclusively by receptor-mediated processes. Both the so-called LDL receptor-related protein (▲ in Fig. 1, and see Section 3.1) and the LDL receptor (■ in Fig. 1, and see Section 2) mediate their removal via recognition of apoE. The apoB48, which resides on chylomicrons throughout their life span, is not recognized by these receptors.

In analogy to the exogenous lipid transport branch, the endogenous pathway begins with the production and secretion of triacylglycerol-rich lipoproteins by the liver, here termed very low density lipoprotein (VLDL). In significant contrast to chylomicrons, VLDL contains apoB100, in addition to the apoCs and apoE. Lipoprotein lipase in the capillary bed hydrolyzes triacylglycerol of secreted VLDL, but less efficiently than from chylomicrons, which is likely one of the reasons for slower plasma clearance of VLDL ($t_{1/2}$, days) compared to chylomicrons ($t_{1/2}$, minutes to a few hours). Lipolysis

during the prolonged residency of VLDL particles in the plasma compartment generates intermediate density lipoproteins (IDL) (Fig. 1) and finally, LDL. In parallel, apolipoproteins and surface components (mostly phospholipids and unesterified cholesterol), but also cholesteryl esters and triacylglycerol, are subject to transfer and exchange between particles in the VLDL lipolysis pathway and certain species of high density lipoproteins (HDL). In addition, it appears that another receptor akin to the LDL receptor, the so-called VLDL receptor (see Section 3.2), might act in concert with lipoprotein lipase in delivering fatty acids to a limited set of peripheral tissues. Enzymes involved in cholesterol loading and esterification, and in interparticle-transfer reactions are, e.g., lecithin-cholesterol acyltransferase (LCAT) and cholesteryl ester transfer protein (CETP) (see Chapter 20). IDL particles (which still harbor some apoE) to a variable degree, and LDL as the end product of VLDL catabolism in the plasma (and at least in man, free of apoE), are catabolized via the LDL receptor. This receptor is found in the liver (which harbors 60–70% of all the LDL receptors in the body) as well as in extrahepatic tissues, and is a key regulatory element of systemic cholesterol homeostasis (see Section 2.2).

Thus, steady-state plasma LDL levels are not only the result of lipoprotein receptor numbers, but also are influenced by the rate of VLDL synthesis, the activity of lipoprotein lipase and other lipases, the VLDL receptor, and by other metabolic processes. As far as HDL levels are concerned, one result of the LCAT- and CETP-catalyzed reactions is the production of a dynamic spectrum of particles with a wide range of sizes and lipid compositions. The further metabolic fates of these HDL fractions are described in Chapter 20.

In addition to receptor-mediated metabolism of lipoproteins, which clearly is the predominant mechanism for removal from the plasma of intact lipoproteins, individual components of lipoproteins, in particular unesterified cholesterol, might diffuse into cells across the plasma membrane. Other minor uptake processes may include so-called fluid-phase endocytosis, which does not involve binding of lipoproteins to specific cell surface proteins, and phagocytosis, in which lipoproteins are thought to attach to the cell surface via more or less specific forces, and are engulfed by the plasma membrane.

2. Removal of LDL from the circulation

The supply of cells with cholesterol via receptor-mediated endocytosis of LDL is one of the best characterized processes of macromolecular transport across the plasma membrane of eukaryotic cells. The following sections describe this process, provide an overview of the biochemical and physiological properties of the LDL receptor, and discuss the molecular basis for the genetic disease, familial hypercholesterolemia.

2.1. Receptor-mediated endocytosis

This multi-step process, originally defined as a distinct mechanism for the cellular uptake of macromolecules, emerged from studies by M.S. Brown, J.L. Goldstein and their colleagues which they performed in the mid-1970s in order to elucidate the

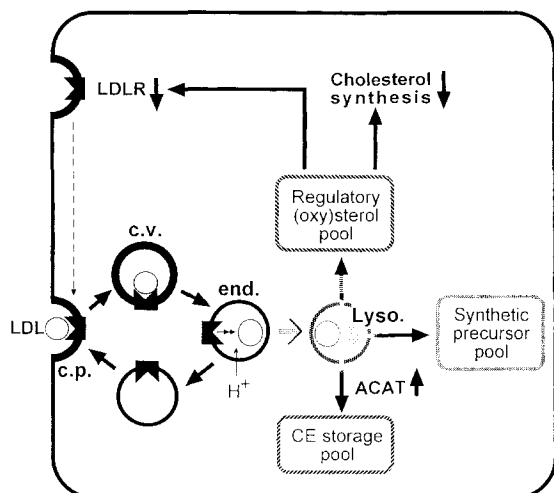


Fig. 2. The LDL receptor pathway, regulation of cellular cholesterol homeostasis. LDL receptors are synthesized in the endoplasmic reticulum and undergo post-translational modification in the Golgi compartment; from there they travel to the cell surface and collect in coated pits (c.p.). LDL particles bound to LDL receptors (■) are internalized in coated vesicles (c.v.) which become uncoated and acidified by protons (H^+) being pumped into their lumen, resulting in endosomes (end.), in which the LDL particles dissociate from the receptors due to the low pH. From there, the LDL are delivered to lysosomes (Lyso.), but almost all of the receptors travel back to the cell surface (where they become incorporated into c.p. again) within a recycling vesicle. Lysosomal degradation of LDL results in complete breakdown of apoB100 and liberation of cholesterol via hydrolysis of cholesteryl esters. The LDL-derived cholesterol has three main fates: (a) it is reconverted to cholesteryl esters via stimulation of acyl-CoA cholesterol acyltransferase (ACAT) for storage in droplets (CE storage pool; bottom); (b) it is used as biosynthetic precursor for bile acids, steroid hormones, membranes, etc. (synthetic precursor pool; right-hand side); and (c) it serves, especially if converted to oxysterols (top), several regulatory functions. The most important of these are suppression of cholesterol synthetic enzymes, and decreasing the production of LDL receptors.

normal function of LDL. The salient features of the itinerary of a LDL particle (mean diameter, ~ 22 nm) from the plasma into a normal human fibroblast are summarized in Fig. 2. First, the lipoprotein particle binds to one of the approximately 15,000 LDL receptors on the surface of the cell. LDL receptors are not evenly distributed on the cell surface; rather, up to 80% are localized to specialized regions of the plasma membrane comprising only 2% of the cell surface. These regions form pits and are lined on their cytoplasmic side with material that in electron micrographs has the appearance of a fuzzy coat. Each of these so-called 'coated pits' contains several kinds of endocytic receptors in addition to LDL receptors, but LDL particles bind only to 'their' receptors, due to their extremely high affinity and specificity. Next, the receptor/LDL complex undergoes rapid invagination of the coated pit, which eventually culminates in the release of the coated pit into the interior of the cell. At this point, the coated pit has been transformed into an endocytic 'coated vesicle', a membrane-enclosed organelle that is coated on its exterior (cytoplasmic) surface with a polygonal network of fibrous protein(s), the main structural component of which is a fascinating protein called clathrin [1]. Subsequently, the coat is rapidly removed, in concert with acidification

of the vesicles' interior and fusion with other uncoated endocytic vesicles. Transiently, LDL and the receptor are found in smooth vesicles in which the lipoprotein particle dissociates from the receptor due to the acidic environment. LDL is then delivered to lysosomes, where it is degraded, while the receptor escapes this fate and recycles back to the cell surface, homes in on a coated pit and is ready to bind and internalize new ligand molecules [1].

There are variations to the theme; not in all systems of receptor-mediated endocytosis are ligand degradation and receptor recycling coupled; however, all have in common the initial steps leading to the formation of endosomes. Then, the receptors are either degraded, recycled back to the cell surface, or are transported (for example, across polarized cells); their respective ligands can follow the same or divergent routes [1]. The reutilization of the LDL receptor via recycling constitutes an economical way to ensure efficient removal of LDL from the extracellular space.

2.2. *The LDL receptor pathway*

The LDL receptor is the key component in the feedback-regulated maintenance of cholesterol homeostasis in the body [1]. In fact, as an active interface between extra- and intra-cellular cholesterol pools, it is itself subject to regulation at the cellular level (cf. Fig. 2). LDL-derived cholesterol (generated by hydrolysis of LDL-borne cholesteryl esters) and its intracellularly generated oxidated derivatives mediate a complex series of feedback control mechanisms that protect the cell from over-accumulation of cholesterol. First, (oxy)sterols suppress the activities of 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA) synthase, and HMG-CoA reductase, two key enzymes in cellular cholesterol biosynthesis. Second, the cholesterol activates the cytoplasmic enzyme acyl-CoA: cholesterol acyltransferase (ACAT; E.C. 2.3.1.26) which allows the cells to store excess cholesterol in re-esterified form. Third, the synthesis of new LDL receptors is suppressed, preventing further cellular entry of LDL and thus cholesterol overloading. The coordinated regulation of LDL receptors and cholesterol synthetic enzymes relies on the sterol-modulated proteolysis of a membrane-bound transcription factor, SREBP, as described in Chapter 15.

The overall benefits from, and consequences of, this LDL receptor-mediated regulatory system are the coordination of the utilization of intra- and extra-cellular sources of cholesterol at the systemic level. Mammalian cells are able to subsist in the absence of lipoproteins because they can synthesize cholesterol from acetyl-CoA. When LDL is available, however, most cells primarily use the LDL receptor to import LDL cholesterol and keep their own synthetic activity suppressed. Thus, a constant level of cholesterol is maintained within the cell while the external supply in the form of lipoproteins can undergo large fluctuation.

Most of these concepts have arisen from detailed studies on cultured fibroblasts from normal subjects and from patients with the disease, familial hypercholesterolemia (FH). Lack of the above described regulatory features in FH fibroblasts led to the conclusion that the abnormal phenotype is caused by lack of LDL receptor function, and thus, disruption of the LDL receptor pathway. In particular, the balance between extracellular and intracellular cholesterol pools is disturbed. Clinically, the most important effect of

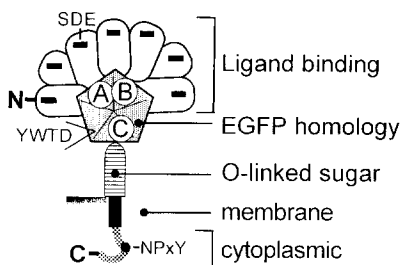


Fig. 3. Domain model of the LDL receptor. The five domains of the mature protein, from the N-terminus (bold N) to the carboxy-terminus (bold C) are as follows. (1) The ligand binding domain, characterized by seven cysteine-rich repeats, each with a cluster of negatively charged amino acids whose core consists of Ser-Asp-Glu ('SDE'); repeats 2-7 cooperatively bind apoB100 and apoE. (2) The EGF-precursor (EGFP) homology region, consisting of 400 amino acid residues (central pentagon); adjacent to the ligand binding domain and at the carboxy-terminus of this region, respectively, are located three repeats with high homology to repeat motifs found in the precursor to epidermal growth factor (encircled letters A, B, and C). The remaining portion of this domain consists of five internally homologous stretches of approximately 50 amino acid residues each of which contains the sequence Tyr-Trp-Thr-Asp (YWTD). (3) The *O*-linked sugar region, consisting of 58 amino acids with 18 serine and threonine residues containing *O*-linked carbohydrate chains. (4) A single membrane-spanning domain. (5) The cytoplasmic tail with 50 amino acid residues containing the internalization sequence Asn-Pro-Val-Tyr (NPxY; the Val is not absolutely conserved in all species).

LDL receptor deficiency is hypercholesterolemia with ensuing accelerated development of atherosclerosis and its complications (Chapter 22). In the following sections, a detailed description of the LDL receptor is provided, with emphasis on the impact of mutations on its structure and function.

2.3. Relationships between structure and function of the LDL receptor

Studies at the levels of protein chemistry, molecular biology, and cell biology have led to a detailed understanding of the biology of the LDL receptor. The mature receptor is a highly conserved integral membrane glycoprotein consisting of five domains (Fig. 3). In order of appearance from the amino terminus these domains are: (1) the ligand binding domain; (2) a domain that has a high degree of homology with the epidermal growth factor precursor (EGFP); (3) a domain that contains a cluster of *O*-linked carbohydrate chains; (4) a transmembrane domain; and (5) a short cytoplasmic region. Until direct information on the three-dimensional structure of the 839-residue receptor becomes available, an arrangement of these domains as presented in Fig. 3 may serve as a useful model.

2.3.1. The ligand binding domain

This domain mediates the interaction between the receptor and lipoproteins containing apoB100 and/or apoE [2]. The function is localized to a region at the amino terminus of the receptor, comprised of seven repeats of approximately 40 residues each. These seven repeats have six cysteines each, which presumably mediate the folding of the domain into a rigid structure with clusters of negatively charged residues on its surface (with

the signature tripeptide Ser–Asp–Glu). These clusters are thought to participate in the binding of lipoprotein(s) via positively charged residues on apoB100 or apoE.

2.3.2. *The epidermal growth factor precursor homology domain*

This region of the LDL receptor lies adjacent to the ligand binding site and is comprised of approximately 400 amino acids; the outstanding feature is the sequence similarity of this region to parts of the EGFP, i.e., three regions termed ‘growth factor repeats’. Two of these repeats (A, B in Fig. 3) are located in tandem at the amino terminus, and the other (C21) is at the carboxy-terminus of the precursor homology region of the LDL receptor. The remainder consists of five ~50-residue stretches that contain tetrapeptide sequences with a consensus of Tyr–Trp–Thr–Asp. Experimental evidence suggests an involvement of this region in the receptor’s acid-dependent dissociation from LDL and its subsequent recycling (cf. Fig. 2).

2.3.3. *The O-linked sugar domain*

The *O*-linked sugar domain of the human LDL receptor is a 58-amino acid stretch highly enriched in serine and threonine residues, located just outside the plasma membrane. Most, if not all, of the 18 hydroxylated amino acid side chains are glycosylated. The *O*-linked oligosaccharides undergo elongation in the course of receptor synthesis and maturation: when leaving the endoplasmic reticulum, *N*-acetylgalactosamine is the sole *O*-linked sugar present, and upon processing in the Golgi, galactosyl and sialyl residues are added. Despite the detailed knowledge about the structure of this region, its functional importance remains unclear.

2.3.4. *The membrane anchoring domain*

The membrane anchoring domain lies carboxy-terminally to the *O*-linked carbohydrate cluster. It consists of 22–25 hydrophobic amino acids; as expected, the deletion of this domain in certain naturally occurring mutations, or by site-directed mutagenesis, leads to secretion of truncated receptors from the cells.

2.3.5. *The cytoplasmic tail*

The cytoplasmic tail of the LDL receptor constitutes a short stretch of 50 amino acid residues involved in the targeting of LDL receptors to coated pits. Naturally occurring mutations and site-specific mutagenesis [3] have identified an ‘internalization signal’, Asn–Pro–Xxx–Tyr (NPxY; where x denotes any amino acid). Recently, the cytoplasmic domains of the LDL receptor and structural relatives have come into new focus, since they hold the key to the involvement of these receptors in signal transduction as indicated below (Section 5).

2.4. *The human LDL receptor gene — organization and naturally occurring mutations*

The ~48 kb human LDL receptor gene contains 18 exons and is localized on the distal short arm of chromosome 19. There is a strong correlation between the functional domains of the protein and the exon organization in the gene. For instance, the seven cysteine-rich repeats of the ligand binding domain are encoded by exons 2 (repeat 1), 3

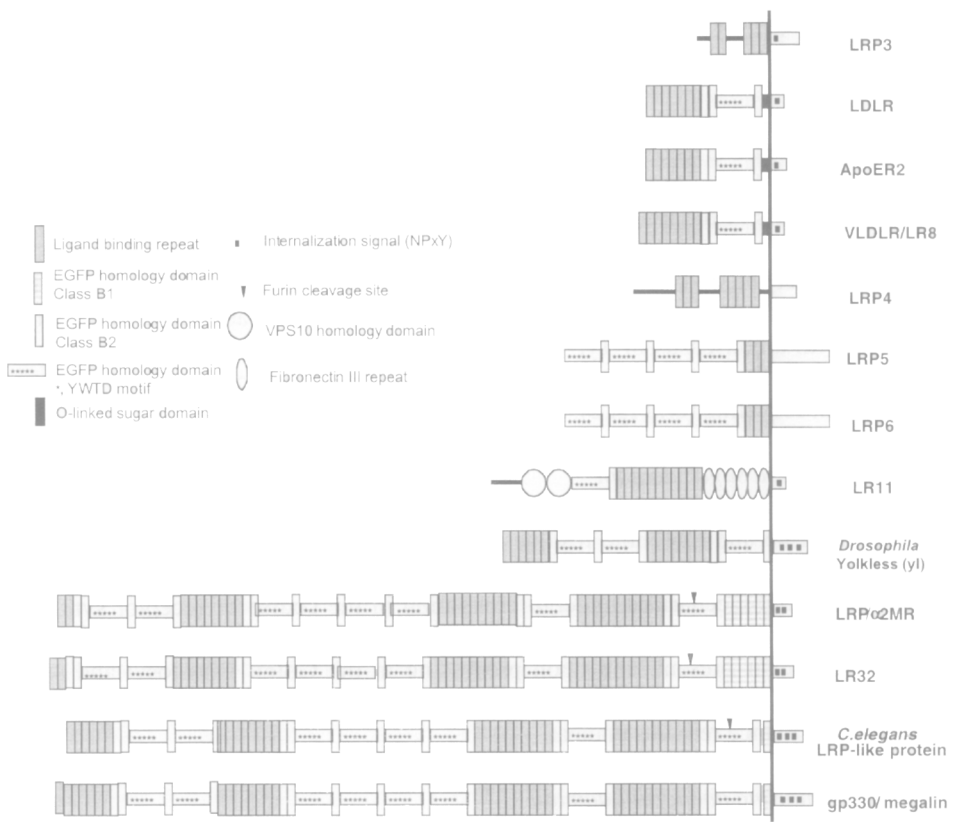


Fig. 4. The LDL receptor gene family. The structural building blocks making up these proteins are listed in the left-hand top part (for more details, see Fig. 3); presumed extracellular domains are depicted to the left of the plasma membrane (black vertical line). The standard modules are: negatively charged ligand binding repeats with six cysteines each; EGFP homology repeats (in the entire family, two subclasses with slightly different consensus sequences are distinguished, termed B1 and B2; these repeats also contain six cysteines each); the five 'YWTD' motifs within the 5-times-50 residues structure of EGFP homology domains; the O-linked sugar domains, just outside the plasma membrane, typical for LDL receptor, apoER2 and VLDL receptor/LR8; and the consensus or presumed internalization signals, NPxY. Several large members of the gene family harbor consensus furin cleavage sites; LR11 (Section 6.2) contains domains not found in other relatives, i.e., two VPS10 domains and six fibronectin type III repeats. Except for the yolk protein transport receptor of *Drosophila melanogaster* (yl), all receptors are discussed in the text.

(repeat 2), 4 (repeats 3, 4, and 5), 5 (repeat 6) and 6 (repeat 7). The EGFP homology domain is encoded by eight exons, organized in a manner very similar to the gene for the EGFP itself. The third domain is translated from a single exon between introns 14 and 15. Thus, the LDL receptor gene is a compound of shared coding sequences; in fact, many more molecules containing all or some of these elements have been discovered and likely will continue to be found. Membrane proteins with clusters of ligand binding repeats in their extracellular domains are now recognized as relatives of the so-called LDL receptor gene family (Fig. 4).

Molecular genetic studies in FH patients have identified over 600 different mutations in the LDL receptor gene. A listing of all mutations with their original literature citations can be found at <http://www.ucl.ac.uk/fh/>. In order to gain some insight into the nature of these mutations, they are grouped into five classes according to their effects on the protein [4] as follows.

2.4.1. Class 1: null alleles — no detectable receptor

These mutant alleles fail to produce receptor proteins as determined by immunoprecipitation with many anti-receptor antibodies, and thus, cells carrying these mutations do not bind any LDL. The spectrum of these mutations includes point mutations causing premature termination codons early in the protein coding region, mutations in the promoter region that block transcription, mutations that lead to abnormal splicing and/or instability of the mRNA, and large deletions.

2.4.2. Class 2: slow or absent processing of the precursor

These alleles, probably accounting for at least half of all mutant LDL receptor alleles, specify transport-deficient receptor precursors which fail to move with normal rates from the endoplasmic reticulum to and through the Golgi compartment(s) and on to the cell surface. As a consequence, the typical sudden increase in apparent M_r observed during biosynthesis of the normal receptor is lacking. While some mutations only slow down processing, most of these mutations are complete in that there is total absence of transport from the endoplasmic reticulum, and the mutant receptors never reach the cell surface.

2.4.3. Class 3: defective ligand binding

These receptors in general reach the cell surface at normal rates, but are unable to bind LDL efficiently due to subtle structural changes near the ligand binding domain. By definition, these mutant receptors undergo the normal maturation process.

2.4.4. Class 4: internalization-defective

Here, one of the prerequisites for effective ligand internalization — localization of LDL receptors to coated pits — is not met. The failure of these ‘internalization-defective’ receptors to localize to coated structures results from mutations that directly or indirectly disrupt the carboxy-terminal domain of the receptor. Variants of class 4 mutations have been identified in which large deletions lead to a lack of both the cytoplasmic and transmembrane domains; the majority of these mutant truncated proteins are, as expected, secreted.

2.4.5. Class 5: recycling-defective

The classification of these mutations into a separate class [4] is based on the observation that upon deletion of the first two EGFP domains of the human LDLR, the truncated receptor binds and internalizes ligand, but fails to release it in the acidic environment of the endosome [5]. The mutant receptor is rapidly degraded without returning to the surface in an unoccupied state. Class 5 mutants all affect the EGFP homology domain, and most often the YWTD regions thereof.

In summary, to a large extent through the delineation of natural mutations in the LDL receptor gene, structural as well as regulatory features of receptor-mediated metabolism of the major cholesterol-carrying lipoprotein in human plasma are now well understood. In the following sections, our still somewhat limited knowledge about the events involved in receptor-mediated plasma clearance of triacylglycerol-rich lipoproteins is outlined.

3. Removal of triacylglycerol-rich lipoproteins from the plasma

3.1. Catabolism of chylomicrons

Chylomicrons are too large to cross the endothelial barrier; thus, their prior lipolysis to remnants serves a dual function: transport of energy to tissues, and decrease in size to facilitate terminal catabolism. Classical experiments in hepatocytes, perfused rat livers, and transgenic and knockout mice studies have shown that chylomicron remnant transport into the liver is mediated by cell surface receptors. Although apoE, which is recognized by the LDL receptor, is the surface component that targets chylomicron remnants to their site of uptake, studies in LDL receptor-deficient model systems predicted that chylomicron remnant removal would be LDL receptor independent. Individuals with homozygous FH, who lack functional LDL receptors, show no signs of delayed clearance of chylomicron remnants. Furthermore, evidence for a separate hepatic chylomicron remnant removal mechanism came from studies in which dietary, drug, and hormonal factors were shown to regulate the number of hepatic LDL receptors without greatly affecting the clearance rate for chylomicron remnants.

Since the LDL receptor and the proposed chylomicron remnant receptor share at least one property, namely apoE binding, attempts to isolate this receptor were based on the presumed similarity of its ligand binding region to that of the LDL receptor. Indeed, homology cloning resulted in the characterization of an unusually large membrane protein, composed exclusively of structural elements found in the LDL receptor molecule; it has therefore been termed LDL receptor-related protein, or LRP [6]. As shown in Fig. 4, LRP, a 4526-amino acid integral membrane glycoprotein, contains (among other structural elements found in the LDL receptor) 31 repeats of the type forming the ligand binding domain in the LDL receptor and 22 repeats of the growth factor type. The unusually large membrane protein binds lipoproteins in apoE-dependent fashion.

Soon after its cloning, LRP was shown to be identical to the receptor for α_2 -macroglobulin (α_2 MR), a major plasma protein that functions in 'trapping', and thereby inactivating, cellular proteinases that have entered the plasma compartment. Since then, many more plasma proteins and protein complexes have been identified, which at least in vitro bind to LRP [7]. Importantly, α_2 -macroglobulin-proteinase complexes are cleared rapidly by the liver (with the same kinetics as chylomicron remnants), indicating that LRP may indeed perform multiple functions in the removal of spent vehicles of intestinal lipid transport and of potentially harmful proteinases. Another LDL receptor

relative with an even broader range of functions is introduced in the following section, and further properties of this protein are described in detail in Sections 4 and 5.

3.2. *The so-called VLDL receptor: a role in catabolism of VLDL?*

The name VLDL receptor was coined for a protein discovered in 1992 by Takahashi et al. [8]. The overall modular structure of the VLDL receptor is virtually superimposable with that of the LDL receptor, except that the ligand recognition domain contains an additional binding repeat located at the amino terminus (Fig. 4). The VLDL receptor shows an amazing degree of conservation among different species; within mammals, there is 95% identity between the corresponding proteins. Even the VLDL receptor homologs of more distant species such as the chicken and frog share 84% and 73%, respectively, of identical residues with the human VLDL receptor. In addition, VLDL receptors exist in variant forms, arising from differential splicing of exon 16 which specifies an *O*-linked sugar domain.

However, uptake of VLDL as such into tissues *in vivo* has not been conclusively shown to involve the eight ligand binding repeat receptor to a significant extent. This is despite the fact that its tissue distribution is highly suggestive of a role in triacylglycerol transport into metabolically active tissues, such as heart, skeletal muscle, and adipose tissue. In contrast to the LDL receptor, and as expected from a receptor implicated in triacylglycerol transport, the VLDL receptor is not regulated by cellular sterols, but its level appears to be influenced by hormones such as estrogen and thyroid hormone. On the other hand, its expression pattern is not congruent with that of lipoprotein lipase with which the VLDL receptor would be expected to act in concert. Nevertheless, numerous studies suggest that the VLDL receptor is, at least in part, involved in the delivery of fatty acids derived from VLDL-triacylglycerols to peripheral tissues, such as adipose tissue [9].

Subsequent to the revelation that despite the proposed and/or intuitive physiological function, the VLDL receptor likely has only a limited role in lipoprotein metabolism, our view of VLDL receptor function has dramatically changed. In 1999, it was discovered by elegant experimentation that it clearly plays a role in neuronal migration in the developing brain, via binding of a ligand quite distinct from lipoproteins. These important results are described in Section 5. However, there is a homologue of the VLDL receptor which indeed does deserve this name, since it has a very well defined function in lipoprotein metabolism, as described in the following section.

4. *Multifunctional receptors in the chicken*

A particularly interesting VLDL receptor homolog is that of the chicken (termed LDL receptor relative with eight binding repeats, or LR8; Fig. 4), as its functions are documented by both biochemical and genetic evidence: it mediates a key step in the reproductive effort of the hen, i.e., oocyte growth via deposition of yolk lipoproteins [10,11]. This conclusion is based on studies of a non-laying chicken strain carrying a single mutation at the *lr8* locus that disrupts LR8 function (the 'restricted ovulator',

R/O, strain) [12]. As a consequence of the mutation, the hens fail to deposit into their oocytes VLDL and the lipophosphoglycoprotein vitellogenin, which are produced at normal levels in the liver, and the mutant females develop severe hyperlipidemia and features of atherosclerosis. The phenotypic consequences of the single-gene mutation in R/O hens revealed the extraordinary multifunctionality of LR8, i.e., that the receptor recognizes in essence over 98% of all the yolk precursors that eventually constitute the mass of the fully grown oocyte [11]. Obviously, R/O hens, which represent a unique animal model for an oocyte-specific receptor defect leading to familial hypercholesterolemia (Section 2.2), are sterile due to non-laying.

Those tissues which express the VLDL receptor in mammals, i.e., heart, skeletal muscle, brain, and adipose tissue, but not the liver, also express this receptor in chicken, albeit at very low levels compared to the oocytes [13]. One difference between the structures of the major VLDL receptors in mammals and chicken LR8 is the presence (in mammalian tissues) and absence (in chicken oocytes) of the *O*-linked sugar domain, respectively [13]. Here, the larger form is termed LR8⁺, and the smaller one, LR8⁻. It was found that in chicken, the somatic cells and tissues, in particular the granulosa cells surrounding the oocytes, heart, and skeletal muscle express predominantly LR8⁺ (at very low levels, as indicated above), while the oocyte is by far the major site of LR8-expression. In the male gonad, the same expression dichotomy exists in that somatic cells express the larger, and spermatocytes the shorter, form of LR8 [14].

The properties of LR8 and its central role in reproduction strengthen the hypothesis that the avian receptor is the product of an ancient gene with the ability to interact with many, if not all, ligands of more recent additions to the LDL receptor gene family (Fig. 4). In this context, vitellogenin, absent from mammals, and apoE, not found in birds, possess certain common biochemical properties and regions of sequence similarities, and have been suggested to be functional analogues [15]. Even high density lipophorin, an abundant lipoprotein in the circulatory compartment of insects, is likely endocytosed in a variety of tissues via an LR8 homologue with very high similarity to the VLDLR/LR8 group [16]. Presumably, binding of lipophorin to this receptor is mediated by apolipophorins I and II, which share sequence homology with mammalian apoB, and thus may behave similarly to the major yolk precursor proteins.

Furthermore, studies in the chicken have revealed that members of the LDL receptor gene family from different animal kingdoms have common structures, and share a growing list of physiological roles, including the most recently discovered function(s) in signal transduction, as described in the following section.

5. VLDL receptor and apoE receptor type 2 (apoER2) as signal transducers

5.1. ApoER2 — a close relative of the VLDL receptor

The structure of apoER2, which was discovered by homology cloning, is highly reminiscent of that of the VLDL receptor (Fig. 4). However, the produced proteins are now known to harbor a cluster of either three, four, five, seven, or eight binding

repeats, dependent on the species and organ expressing apoER2 [17]. Murine apoER2 mRNA may contain an additional small exon following that encoding repeat 8, giving rise to a variant containing a furin consensus cleavage site at the carboxy-terminal end of the ligand binding domain. This may lead to the secretion of a soluble receptor fragment constituting the ligand binding domain, which could act as a dominant negative ligand-trapping receptor. Most interestingly, this variant is the only one detectable in the placenta, showing that some of the splice events are tissue-specific.

ApoER2 is predominantly found in brain, placenta, and testis. This is in contrast to other members of the LDL receptor family, which are all expressed to a small extent in the brain, but most prominently in a variety of other organs and cells. Besides the liver, the brain is also the most prevalent site of apoE expression in mammals, and it is widely believed that apoE serves a role in local lipid transport in the central nervous system [18]. Ligand binding studies with the human apoER2 demonstrated high affinity of the receptor for β -VLDL, indicating that the receptor might be involved in apoE-mediated transport processes in the brain. In addition, at least the apoER2 splice variant containing eight binding repeats can act as receptor for α_2 -macroglobulin (also a ligand of LRP, see Section 3.1) in brain, which suggests a role in the clearance of α_2 -macroglobulin–proteinase complexes from the cerebrospinal fluid and from the surface of neurons. In turn, proteinases may play a role in synaptic plasticity [19], and the balance between proteolytic activity and its inhibition might be controlled by proteinase inhibitors and their receptors. Despite these intriguing possibilities, an even more important, recently delineated, function of apoER2 — and of the VLDL receptor — is described below.

5.2. Genetic models reveal new roles for apoER2 and VLDL receptor

In addition to their potential to bind and endocytose ligands in a variety of metabolic and cellular contexts, these receptors are now known to be involved in signal transduction [20], which likely is independent of ligand internalization. Surprisingly, targeted disruption of both the VLDL receptor and the apoER2 genes in mice (so-called double-knockout mice) display a dramatic phenotype, essentially identical to that of mice lacking the extracellular matrix glycoprotein reelin [20]; single-knockout mice of either receptor gene show only very subtle phenotypes. Reelin is secreted by Cajal–Retzius cells in the outermost layer of the developing cerebral cortex and orchestrates the migration of neurons along radial fibers, thus forming distinct cortical layers in the cerebrum.

The reason for the grossly abnormal phenotype of the double-knockout mice, i.e., disturbed foliation of the cortical layers, is that reelin normally interacts with the extracellular domains of both the VLDL receptor and apoER2 [21,22], but the ensuing vital signal cascade remains inactivated when the receptors are missing. The current model suggests that reelin binding to the VLDL receptor and apoER2 leads to phosphorylation of the cytoplasmic adapter protein Disabled-1 (mDab-1), which is associated with the NPxY-motifs present in the receptors' tails. Reelin-triggered tyrosine-phosphorylation of mDab-1 may then start kinase cascade(s) controlling cellular motility and shape by acting on the neuronal cytoskeleton. The specificity of the reelin signalling via

apoER2 and VLDL receptor seems to be achieved by selective binding of reelin to these receptors and not to other members of the LDL receptor family. This is an important aspect, since mDab-1 not only binds to the VLDL receptor and apoER2, but also to LRP and the LDL receptor [23].

6. Other relatives of the LDL receptor family

In the past few years, several additional LDL receptor gene family members have been identified at the molecular level. Since all of these, by definition, contain LDL receptor ligand binding repeat clusters and may therefore play roles in lipid-related metabolism, they are also described in this chapter. For simplified schematized structures of these membrane proteins, please refer to Fig. 4.

6.1. Small and mid-sized LDL receptor relatives: LRP 3, 4, 5, and 6

These rather new additions to the LR gene family were discovered more or less serendipitously. Degenerative probes corresponding to the highly conserved amino acid sequence WRCDGD, found in LDL receptor ligand repeats, were used to screen a rat liver cDNA library, resulting first in the cloning of LRP3, and then of its human homologue from a HepG2 cDNA library [24]. The same approach, using a murine heart cDNA library, resulted in cloning of LRP4. LRP3 is a 770 residue membrane protein with clusters of two and three binding repeats, respectively. Murine LRP4 contains two clusters of LDL receptor binding repeats with three and five modules, respectively. Future studies will help to clarify whether these proteins are capable of binding ligands that have been shown to interact with other LDL receptor relatives, whether they are endocytotically competent, and of course, what their relevant in-vivo functions are.

Two other new members of the LDLR family, LRP5 [25] and LRP6 [26] have been discovered (Fig. 4), quite surprisingly, in the course of attempts to identify the nature of the insulin-dependent diabetes mellitus locus IDDM4 on chromosome 11q13. Human and mouse LRP5 and LRP6 are type I membrane proteins, approximately 1600 residues long (about twice as large as the LDL receptor), and their extracellular domains are organized exactly as a portion of LRP (Fig. 4). The cytoplasmic domains of LRP5 and LRP6 do contain motifs (dileucine, and aromatic-X-X-aromatic/large hydrophobic) similar to those known to be functional in endocytosis of other receptors.

Probably more importantly, they harbor serine- and proline-rich stretches that may serve as ligands for Src homology 3 (SH3) and WW (a variant of SH3) domains, properties that relate these receptors to signal transduction pathways, albeit different ones from those of apoER2 and the VLDL receptor described above. Indeed, LRP6 has been shown to be a co-receptor for proteins of the Wnt family, which trigger signalling pathways important for correct development of anterior structures. Most recently, LRP6 has been demonstrated to interact with proteins called Dickkopf and Axin, respectively [27]. Dickkopf inhibits Wnt signalling by releasing receptor-bound Wnt. Axin is one of the components in the cascade that regulates the activation of gene expression in the nucleus of target cells. The interplay of Wnt-, Dickkopf- and Axin-binding to LRP5/6

may hold the key to important developmental signals, similar to the role of VLDL receptor and apoER2 in neuronal migration (Section 5.2).

6.2. *The unusual one: LR11*

Also by homology cloning, LR11, a novel and unusually complex member of the LDL receptor gene family (Fig. 4) has been discovered first in rabbit, and subsequently in man, mouse and chicken [28]. Significantly, overall sequence identities between the ~250 kDa proteins range from 80% (man vs. chicken) to 94% (man vs. rabbit). The predicted amino acid sequence of LR11 suggests that the polypeptide is made up of seven distinct domains (Fig. 4), among which is a cluster of eleven LDL receptor ligand repeats. Unusual features are a large domain (~400 residues) highly homologous to a yeast receptor for vacuolar protein sorting, VPS10p, and modules found in cellular adhesion molecules (six tandem fibronectin type III repeats). The membrane spanning and cytoplasmic domains are extremely highly conserved; e.g., the presumed internalization signal is FANSHY in all LR11s known to date.

LR11 is found mainly in the nervous system, and depending on the species, also in testis, ovary, adrenal glands, and kidney. It appears that LR11 is developmentally regulated, and several studies have demonstrated induced expression during morphogenetic processes. For instance, LR11 levels are increased during proliferation, but become downregulated following differentiation of neuroblastoma cells. LR11 is also markedly increased in arterial intimal smooth muscle cells during atherogenesis and in the proliferative phase of smooth muscle cells in culture. Thus, to date, the limited information available suggests that LR11 is involved in cellular proliferation during development and possibly, in pathological processes.

However, knowledge about the cell biology of LR11 is still highly rudimentary. Only 10% of the receptors are found on the cell surface in transfected cells, and thus far identified ligands of LR11, such as apoE-containing lipoproteins, have not offered significant insights into LR11 functions [29]; finally, endocytic competence of the receptor has not been demonstrated unambiguously to date.

6.3. *Large LDL receptor relatives: megalin and LR32*

6.3.1. *Megalín, a true lipid transport receptor*

Encoded by a different gene from that for LRP, this 600-kDa protein is another large member of the LDL receptor gene family containing four LDL receptor ligand binding repeat clusters. Although many proteins which bind to LRP are also ligands for megalin (with the exception of apoJ, also known as clusterin), its expression pattern and specificity for certain ligands account for physiological roles distinct from those of LRP. Megalín is essential for development of the forebrain by taking up apoB-containing lipoproteins into the embryonic neuroepithelium (reviewed in Willnow et al. [7]). Another important function is its involvement in the metabolism of certain lipophilic vitamins. For instance, in the kidney, vitamin B12/transcobalamin complexes are recaptured from the ultrafiltrate directly by binding to megalin expressed on proximal tubule cells [30]. Furthermore, megalin mediates the reabsorption from the proximal

tubules of 25-(OH) vitamin D3/vitamin D binding protein complexes, which constitutes a key step in converting the precursor into active vitamin D3 in the kidney [7].

6.3.2. LR32 (*LRP1B*)

This 4599-residue type I membrane protein contains 32 LDL receptor ligand binding repeats in its extracellular portion (Fig. 4). Among all of its relatives, LR32 shows the highest homology to LRP, containing one additional ligand binding repeat and an insertion of 33 amino acids in the cytoplasmic domain compared to the sequence of LRP (cf. Fig. 4). This newly discovered receptor molecule may thus have a role in lipoprotein metabolism.

However, homology searches revealed that LR32 is identical to the product of a candidate tumor suppressor gene, *lrp1b* [31]. The human LR32 gene locus was mapped to chromosome 2q21 by fluorescence in-situ hybridization. The receptor is expressed mainly in brain and skeletal muscle, and its regulation has been studied in smooth muscle cells derived from rabbit arteries and in an established smooth muscle cell-line. In both systems, LR32 expression is induced during the exponential phases of cellular proliferation. Peaks of expression seem to occur at later time points than those observed for LR11 induction (see Section 6.2), consistent with different roles of LR11 and LR32. The future will tell us more about the true physiological role(s) of this close relative of LRP.

7. Scavenger receptors

In addition to the type of scavenger receptors (SRs) described in Chapter 20, there is also a growing list of hepatic and extrahepatic SRs with potential disease-related functions. For many of these, the criterion to be considered as SR is their broad spectrum of ligands which includes diverse polyanionic compounds and, importantly, modified lipoproteins, such as oxidized LDL. The more than a dozen currently known SRs are classified according to their primary structure, tissue distribution, and proposed function(s) into six groups (SR-A to -F). The most prominent and probably best understood SRs in the context of lipoprotein metabolism are the SR class A (SR-A) and a class E SR, LOX-1 (for a concise review of SRs, see Terpstra et al. [32]).

7.1. Class A SRs

SR-As are trimeric membrane proteins characterized structurally by the presence of a small amino terminal intracellular region, an extracellular coiled-coil collagen-like stalk, and a cysteine-rich carboxy-terminal domain [33]. Three isoforms of this receptor are produced from the same gene by differential splicing. SR-A isoforms are expressed at different levels in tissue macrophages, Kupffer cells, and various extrahepatic endothelial cells. SR-A expression is induced by some of its ligands, which include, in addition to modified lipoproteins and polyanions, Gram-positive bacteria, heparin, lipoteichoic acid, and a precursor of lipid A from LPS of Gram-negative bacteria.

The potential role of SR-A in atherosclerotic plaque development was demonstrated in a study on apoE- and SR-A double-knockout mice. Mice deficient in apoE develop

severe plaques, but simultaneous absence of SR-A leads to a reduction in plaque size by 58%. This reduction may be related to the greatly reduced uptake of acetylated LDL and oxidized LDL that can be observed in in-vitro uptake studies using macrophages and liver cells of SR-A knockout mice.

7.2. *Lectin-like oxidized LDL receptor (LOX)-1*

This 50-kDa transmembrane protein shows no structural similarity to other SRs. It belongs to the C-type lectin family of molecules, and its carboxy-terminal cytoplasmic tail contains several potential phosphorylation sites [34]. It can act as endocytic receptor for atherogenic oxidized LDL, but in contrast to SR-A, it interacts only weakly, if at all, with acetylated LDL. Thus, LOX-1 differs from other SRs in that binding of oxidized LDL is inhibited by polyinosinic acid and delipidated oxidized LDL, but not by acetylated LDL, maleylated bovine serum albumin, or fucoidin. LOX-1 is found in thoracic and carotid vessels, and highly vascularized tissues such as placenta, lungs, brain, and liver. Its expression is apparently not constitutive, but can be induced by proinflammatory stimuli; it is then detectable in cultured macrophages and in activated smooth muscle cells. LOX-1 also mediates the recognition of aged red blood cells and apoptotic cells. However, little is known about whether LOX-1 has similar functions in clearance of damaged cell in vivo.

In summary, SRs are a widely expressed and highly diverse group of proteins that are appropriately named for their recognition of a broad array of ligands. At least some of this intriguing group of receptors, as indicated here, may play roles in the metabolism of modified lipoproteins, and thus may well be related to lipid anomaly disorders.

8. *Future directions*

It is exciting to realize that multifunctionality of LDL receptor gene family members can no longer be viewed as being limited to their extracellular moiety. The originally established concepts regarding their functions in lipoprotein metabolism remain valid, but have been extended significantly by the discovery that these membrane proteins also play a role in important signalling pathways. Given the wealth of signalling pathways, future efforts will be directed towards developing concepts that define the contributions of the individual intracellular domains of LDL receptor relatives. These efforts can also be expected to add alternative aspects to the ongoing quest to delineate the evolutionary history of the LDL receptor family. In fact, based on the most recently gained knowledge, an evolutionary theory will have to consider the combinatorial events arising from the multitude of both intra- and extra-cellular domains of these proteins in the generation of new signalling and/or transport units.

Another point to consider is the functional redundancy of receptors involved in a multitude of metabolic pathways and events. This aspect has been addressed here not only for the LDL receptor gene family, but also for the growing group of scavenger receptors, which show broad overlapping ligand specificities (Section 7). In the case of the well understood LDL receptor gene family, when need arises (e.g., when one

or more receptors are unfunctional), certain members can substitute for others, or are at least sufficiently active for preserving life. For example, VLDLR^{-/-} mice, and apoER2^{-/-} mice, have phenotypes that are at first sight indistinguishable from normal; but double-knockout mice are grossly abnormal, as described in Section 5.2. Moreover, (1) LRP can physiologically substitute for the LDL receptor in hepatic clearance of chylomicron remnants (which also bind to the LDL receptor *in vitro*), (2) VLDLR, not expressed in the liver and normally not significantly involved in lipoprotein metabolism, can, when expressed in the liver, substitute for the LDL receptor, and (3) in chickens, an oocyte-specific LRP mediates growth of oocytes, although not sufficient for egg laying, in the LR8-deficient R/O hens (see Section 4). Thus, functional redundancy of LDL receptor relatives can be due to their simultaneous expression on the same cells in a given organ. In turn, different functions despite overlapping ligand spectra may arise from their expression in different cell types within a tissue or organism. LDL receptor family members presumably have *in-vivo* access to different ligands in different environments, and in addition, their cytoplasmic domains interact with cell-type-specific adaptor proteins in order to mediate a spectrum of signal transduction pathways.

As a consequence, specific functions of the gene family must be defined at two levels: the cellular level, in order to delineate molecular events, and in physiological studies including state-of-the-art genetic manipulations, which should reveal the functional relevance of receptor redundancy.

Abbreviations

α_2 MR	α_2 -macroglobulin receptor
ACAT	acyl-CoA : cholesterol acyltransferase
Apo	apolipoprotein
CETP	cholesteryl ester transfer protein
CM(R)	chylomicron (remnants)
EGFP	epidermal growth factor precursor
FH	familial hypercholesterolemia
HDL	high density lipoprotein
HMG-	3-hydroxy-3-methylglutaryl-
LCAT	lecithin-cholesterol acyltransferase
LOX-1	lectin-like oxidized LDL receptor
LR	LDL receptor relative
LRP	LDL receptor-related protein
(V)LDL	(very) low density lipoprotein

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Lipids and atherosclerosis

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1. Introduction

Atherosclerotic vascular disease is the cause of heart attacks, stroke, aortic aneurysms, and peripheral vascular disease, which together represent the most frequent causes of death in the industrialized world. Indeed, the aging of the population and the 'westernization' of world diet is predicted to increase the impact of atherosclerosis worldwide over the next few decades despite continuing advances in plasma lipid-lowering therapy (E. Braunwald, 1997).

Atherosclerosis progresses in a series of stages, although some lesions at each stage may not progress further or may even regress if inciting events, such as hypercholesterolemia, diabetes, smoking, or hypertension, are controlled [1–4] (Fig. 1). The initial

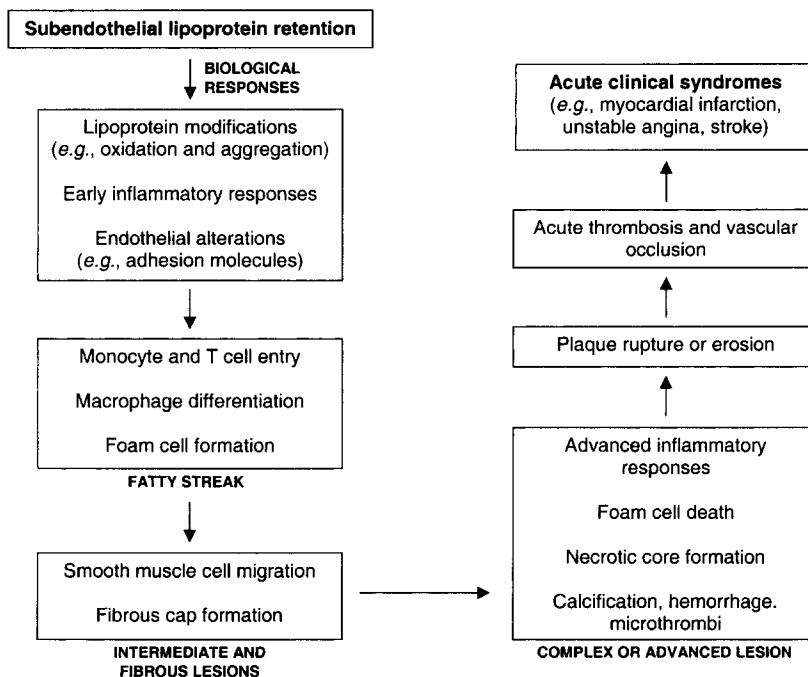


Fig. 1. Progression of atherosclerotic lesions. As noted in the text, only a portion of lesions at any stage progress and, under the proper conditions, lesion regression may occur.

stage involves the accumulation of subendothelial lipoproteins in focal areas of the arterial tree, usually at branch points with disturbed laminar flow. In response to this retention, a series of biological responses ensue, including (1) lipoprotein oxidation, (2) endothelial alterations, (3) inflammatory responses including T cell recruitment, cytokine secretion, monocyte chemotaxis, and subendothelial macrophage accumulation, (4) and intracellular cholesterol accumulation in macrophages. Much of the cholesterol is stored as cholesteryl fatty acid esters (CE) in cytoplasmic lipid droplets surrounded by a monolayer of phospholipid. These cytoplasmic droplets give the macrophages a foamy appearance when viewed by microscopy, and thus these cells are referred to as 'foam cells' (Fig. 2C). The presence of macrophage foam cells defines the earliest pathological lesion, referred to as the 'fatty streak'. Interestingly, fetuses of hypercholesterolemic mothers have been observed to have fatty streaks (W. Palinski, 1997). These fatty streaks disappear soon after birth, but virtually all Westerners have fatty streaks by the teenage years.

Although sensitive tests of endothelial function show abnormalities in vasodilatation in the very earliest phases of atherosclerosis (R.A. Vogel, 1998), fatty streaks are not occlusive and cause no overt symptoms. However, some fatty streaks may progress over years to more complex lesions that can give rise to chronic symptoms or, more importantly, acute events. An important event in the progression of fatty streaks involves the migration of smooth muscle cells from the media to the intima and the secretion of large amounts of collagen and other matrix proteins by these cells. In addition, macrophages proliferate and continue to accumulate more lipid. Smooth muscle cells can also accumulate lipid and become foam cells. These events give rise to so-called fibrous lesions, which are eccentric lesions consisting of lipid-loaded macrophages and smooth muscle cells covered by a fibrous cap. Further progression to complex lesions involves the accumulation of extracellular lipid, which results from a combination of aggregation and fusion of matrix-retained lipoproteins and release of lipid droplets from dying foam cells. Calcification, hemorrhage, and microthrombi can also be observed in these complex lesions [1].

At this stage, several fates of the lesion are possible [1,5,6]. Plasma cholesterol lowering can result in lesion regression, particularly of the foam cells. In this case, the macrophages begin to lose their cholesterol via the process of cholesterol efflux, and the number of macrophages decreases, probably through a combination of decreased monocyte entry, decreased macrophage proliferation, and increased macrophage egress and apoptosis. Alternatively, the complex lesions can progress. If arterial occlusion increases gradually, the patient may experience exercise-induced ischemia, but collateral vessel formation often prevents additional clinical symptoms. However, if the lesions rupture or erode before they become large and occlusive, acute vascular events such as unstable angina, heart attacks, sudden death, or strokes can occur. Rupture involves the abrupt disruption of the fibrous cap, followed by exposure of thrombogenic material and acute thrombosis. Importantly, rupture mostly occurs in lipid-rich and macrophage-rich 'shoulder' regions of the plaque and is probably triggered by the degradation of the fibrous cap by proteases secreted by macrophages or released from dying foam cells. Physical stresses related to pools of soft lipid underneath a thin fibrous cap also contribute to plaque rupture. These pools of lipid and cellular debris, often referred

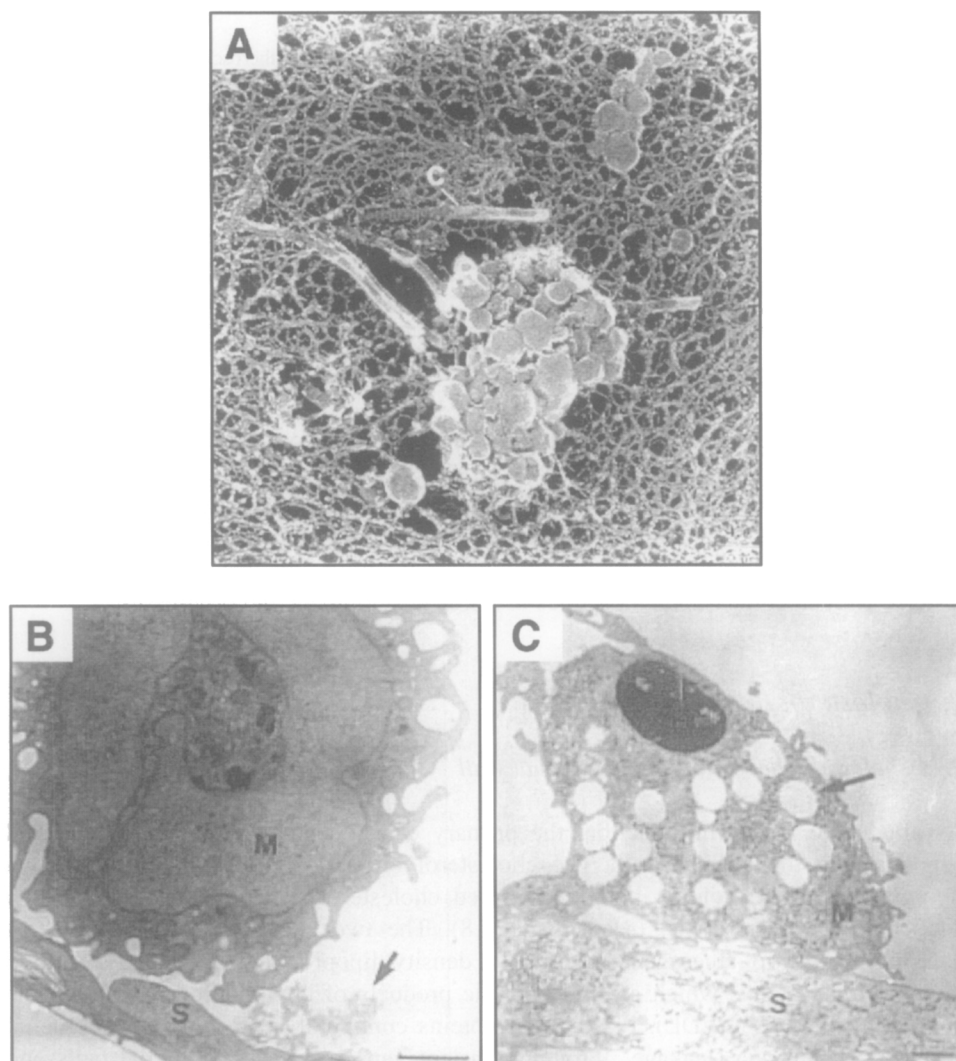


Fig. 2. Lipoprotein aggregation and macrophage foam cell formation. (A) Freeze-etch replica-plated electron micrograph of rabbit aorta subendothelium 2 hours after intravenous injection of LDL (from Nievelstein et al. (1991) *Arterio. Thromb.* 11: 1795–1805). (B) A J774 murine macrophage (M) immediately after plating on sphingomyelinase-induced LDL aggregates (arrow) formed on the surface matrix of smooth muscle cells (S). (C) After 24 hours of incubation, the aggregates have been internalized by the macrophage, which now has large cytoplasmic neutral lipid droplets consisting mostly of cholesteryl ester (arrow). The cytoplasmic droplets are characteristic of lesional foam cells. Bars in B and C, 1 μ m. Panels B and C are from Tabas et al. (1993) *J. Biol. Chem.* 268: 20419–20432.

to as ‘necrotic’ or ‘lipid’ cores, result from the death of macrophage foam cells (M.J. Mitchinson, 1995).

As is evident from this overview, lipids are the sine qua non of atherosclerosis. Indeed, the ‘athero’ of ‘atherosclerosis’ is derived from the Greek word for ‘gruel’,

Table 1
Percent weight of major lesion lipids in four progressive stages of atherosclerotic lesions

Lipid	Fatty streak	Intermediate lesion	Fibrous lesion	Advanced lesion
Cholesterol	9.6	21.1	22.5	31.5
Triglyceride	2.8	4.4	5.2	6.0
Cholesteryl ester	77.0	55.0	55.5	47.2
Phospholipid	10.1	19.6	16.8	15.3
Phosphatidylcholine	4.8	7.6	4.5	4.3
Sphingomyelin	5.6	11.0	11.7	10.1
Lysophosphatidylcholine	0.3	1.0	0.6	0.9

Adapted from Katz et al. [16].

which refers to the massive accumulation of lipids in these vascular lesions. The major types of lipids that accumulate during the various stages of atherosclerosis are shown in Table 1. In addition, there are many lipids that are minor in quantity but, because of their biological activities, are thought to have a major impact on atherogenesis. This chapter will cover the properties and activities of many of the lipids that occur in atherosclerotic lesions, with an emphasis on their roles in lesion development and progression.

2. Cholesterol and atherosclerosis

2.1. Cholesterol deposition in the arterial wall

As alluded to in the Introduction, the primary event in atherogenesis is cholesterol deposition in the arterial wall. The cholesterol originates from circulating plasma lipoproteins, which contain both unesterified cholesterol ('free' cholesterol, or FC) and cholesteryl ester (CE) (see Chapter 18). The two classes of lipoproteins that contribute most to atherogenesis are low density lipoprotein (LDL) and so-called remnant lipoproteins, which are the lipolytic products of chylomicrons and very low density lipoprotein (VLDL). Plasma lipoproteins continually enter the subendothelial space of vessels via 'leakage' through transient gaps between endothelial cells and probably also via endothelial transcytosis. Under normal conditions, lipoproteins are not retained in the subendothelium and simply re-enter the circulation. In certain focal areas of the arterial tree, however, lipoprotein retention by subendothelial extracellular matrix is increased, leading to their net accumulation in the arterial wall. This retained material elicits a series of biological responses, leading to the cellular and extracellular processes that constitute atherosclerotic lesion formation ([4,7]; see also Section 1). Because a high concentration of circulating atherogenic lipoproteins promotes the accumulation of these lipoproteins in the arterial wall, this model explains the well-established relationship between plasma cholesterol levels and atherosclerosis in both experimental animal models and humans.

The fate of the FC and CE moieties of retained lipoproteins includes both extracellular and intracellular processes. Extracellular matrix-retained lipoproteins are modified

by lipases, proteases, and oxidation reactions (P.T. Kovanen, 2000) [8]. These reactions can lead to the generation of lipid vesicles that are rich in FC but poor in protein and CE (H. Kruth, 1985). The biological or pathological significance of these FC-rich vesicles is not known. Other reactions lead to the generation of modified lipoproteins that act as extracellular signaling molecules on lesional cells or that are avidly internalized by macrophages and smooth muscle cells. Thus, these modified lipoproteins are responsible for foam cell formation and a variety of cell signaling events.

2.2. Cholesterol accumulation in lesional macrophages: lipoprotein internalization

The major cell type that internalizes subendothelial lipoproteins is the macrophage [2,3]. Lesional macrophages are derived from circulating monocytes that enter the arterial wall in response to chemokines; the chemokines are secreted by endothelial cells in response to both underlying retained lipoproteins and T cell-derived cytokines. Under the influence of other molecules secreted by endothelial cells, notably macrophage colony stimulating factor, subendothelial monocytes differentiate into macrophages. The differentiated macrophages then engage and internalize subendothelial lipoproteins and thus accumulate lipoprotein-derived cholesterol in the form of intracellular cholesteryl ester droplets (foam cell formation). As outlined in the Section 1, this cellular event is the hallmark of early lesion development and also contributes to late lesional complications.

Two key issues in the area of macrophage foam cell formation include the cell-surface processes and receptors involved in lipoprotein internalization and the metabolic fate of lipoprotein-derived cholesterol following internalization [9]. Most studies examining macrophage–lipoprotein interactions use an experimental system in which monolayers of cultured macrophages are incubated with soluble, monomeric lipoproteins dissolved in tissue culture media. These studies have revealed that native LDL is poorly internalized by macrophages, suggesting that LDL undergoes modification in the arterial wall. While a variety of LDL modifications have been proposed, two types, namely oxidation and aggregation, have received the most attention [8,10,11].

LDL particles with oxidative modifications of both its protein and lipid moieties are known to exist in atherosclerotic lesions and are readily internalized by macrophages. A number of receptors have been implicated in oxidized LDL uptake by macrophages, including class A and B scavenger receptors (e.g., CD36) and lectin-like oxidized LDL receptor-1 (LOX-1) (Chapter 21). While internalization of oxidized LDL by macrophages may have important implications in atherogenesis, it is unlikely that all of the hallmarks of macrophage intracellular cholesterol metabolism that are known occur in lesions can be explained by this process alone [8].

As stated above, lipoproteins in the subendothelium are also known to be aggregated and fused, which may result from oxidation, lipolysis, or proteolysis [8,12] (Fig. 2A). For example, hydrolysis of the sphingomyelin on LDL particles to ceramide by sphingomyelinase leads to LDL aggregates that appear similar to those that exist in lesions, and there is evidence that LDL in the arterial wall is hydrolyzed by a form of sphingomyelinase secreted by arterial-wall cells [13] (Fig. 2B). Aggregated lipoproteins, like oxidized LDL, are readily internalized by macrophages. When aggregated LDL is

added in tissue culture medium to monolayers of cultured macrophages, the LDL receptor (Chapter 21) seems to participate in a phagocytic-like process to internalize these particles. In vivo, however, most of the aggregated lipoproteins are bound to extracellular matrix, and newer experimental systems that attempt to mimic the uptake of retained and aggregated LDL have revealed that multiple receptors in addition to or instead of the LDL receptor are involved. Most importantly, macrophage internalization of aggregated lipoproteins leads to massive CE accumulation, which is the key intracellular cholesterol metabolic event that is known to occur in macrophages in lesions [9] (Fig. 2C).

Remnant lipoproteins are also important in atherogenesis (R.W. Mahley, 1985; R.J. Havel, 2000). These particles can be internalized by macrophages in their native form, although both oxidation and aggregation of these particles occur and probably further enhance macrophage uptake. The receptor or receptors involved in the uptake of remnant particles is not definitively known, but the likely candidates are the LDL receptor and the LDL receptor-related protein (LRP), which interact with the apolipoprotein E moiety of the remnant lipoproteins (Chapter 21). Remnant lipoproteins, like aggregated lipoproteins, lead to massive cholesteryl ester accumulation in macrophages. Finally, it is worth mentioning that another lipoprotein called lipoprotein(a), in which a large glycoprotein called apolipoprotein(a) is covalently attached to the apolipoprotein B100 moiety of LDL, has been implicated in atherogenesis (A.M. Scanu, 1998). Although macrophage receptors for lipoprotein(a) have been described, neither the mechanism of atherogenicity nor the role of lipoprotein(a) lipids in macrophage cholesterol loading and lesion development are known.

2.3. Cholesterol accumulation in lesional macrophages: intracellular trafficking of lipoprotein-derived cholesterol

The fate of lipoprotein cholesterol after internalization is a key issue in understanding the biology and pathology of lesional macrophages. After internalization by receptor-mediated endocytosis or phagocytosis, the lipoproteins are delivered to late endosomes or lysosomes, where hydrolysis of proteins and lipids occurs. Most importantly, the large lipoprotein-CE stores are hydrolyzed by a lysosomal enzyme called lysosomal acid lipase. The liberated FC then trafficks to the plasma membrane and other cellular sites [9].

The trafficking of lipoprotein-derived cholesterol from lysosomes has been a major area of focus in the field of intracellular cholesterol metabolism, and many of the cellular and molecular events are not known (Chapter 17). By analyzing cells with mutations in cholesterol transport, investigators have identified roles for two proteins, called *npc1* and *npc2* (HE1), in lysosomal and/or endosomal cholesterol transport (E.J. Blanchette-Mackie, 2000; P. Lobel, 2000). In addition, the lipid lysobisphosphatidic acid may also play a role in these processes (J. Gruenberg, 1999). The mechanisms by which these molecules are involved in cholesterol transport, however, are poorly understood. One current model suggests that there is an initial *npc1*-independent phase consisting of rapid cholesterol transport from late endosomes or lysosomes to the plasma membrane, probably by vesicular transport (T.Y. Chang, 2000; Y. Lange, 2000).

According to this model, the cholesterol is then internalized into a 'sorting organelle', from which cholesterol is distributed to peripheral cellular sites in an npc1-dependent manner. This transport process probably also occurs via vesicular transport. It must be emphasized, however, that until the mechanism of action of the molecules mentioned above and other molecules are elucidated and the cellular sites in the itinerary identified, this model must be considered hypothetical. It is also likely that different cell types and different conditions in the same cell type may result in different cholesterol trafficking patterns.

From the point of view of atherosclerosis, the two most important peripheral trafficking pathways are those to the endoplasmic reticulum, where cholesterol is esterified by acyl-CoA:cholesterol acyltransferase (ACAT), and to the plasma membrane, where cholesterol can be transferred to extracellular acceptors in a process known as cholesterol efflux (Chapter 20). The former process leads to the massive cholesteryl ester accumulation seen in foam cells [9,14,15]. The ACAT reaction utilizes primarily oleoyl-CoA, and so ACAT-derived CE is rich in oleate. In contrast, plasma lipoprotein-CE tends to be rich in linoleate. As expected, therefore, the cholesteryl oleate:cholesteryl linoleate ratio in foam cell-rich fatty streak lesions is relatively high (1.9) [16]. However, the ratio in advanced lesions is only 1.1, suggesting an increase in lipoprotein-CE in advanced atheromata due to poor cellular uptake of lipoproteins or to defective lysosomal hydrolysis following uptake by lesional cells. Further discussion of the cholesterol esterification pathway appears in Chapter 15, and cholesterol efflux, which is an important mechanism that may prevent or reverse foam cell formation, is covered in Chapter 20.

2.4. Accumulation of free cholesterol in lesional macrophages

Interestingly, foam cells in advanced atherosclerotic lesions accumulate large amounts of FC [16], some of which is in crystalline form and may be deposited in the extracellular space when foam cells die (Fig. 3). For example, while 2 of 13 abdominal aortic and femoral artery fatty streak lesions contained cholesterol crystals, all of 24 advanced lesions had these structures [16]. The mechanism of FC accumulation is not known, but could involve either defects in cholesterol trafficking to ACAT or a decrease in ACAT activity itself. Because much of the FC accumulating in the cells appears to be associated with lysosomes, it is tempting to speculate that defects in lysosomal cholesterol transport arise in advanced foam cells. In this context, macrophages exposed to oxidized LDL can internalize a substantial amount of cholesterol, but there is relatively little stimulation of ACAT-mediated cholesterol esterification [8]. According to one model, oxysterol-induced inhibition of lysosomal sphingomyelinase leads to accumulation of lysosomal sphingomyelin, which binds cholesterol and thus inhibits transport of the cholesterol out of lysosomes (M. Aviram, 1995).

Free cholesterol accumulation in macrophages may be an important cause of macrophage death in advanced atherosclerotic lesions [17] (Fig. 4). Death induced by intracellular free cholesterol excess probably involves both necrosis and apoptosis. Necrotic death may result from the malfunction of critical plasma membrane proteins exposed to a microenvironment with a high cholesterol:phospholipid ratio. Intracellular cholesterol crystal accumulation may also contribute to this form of death.

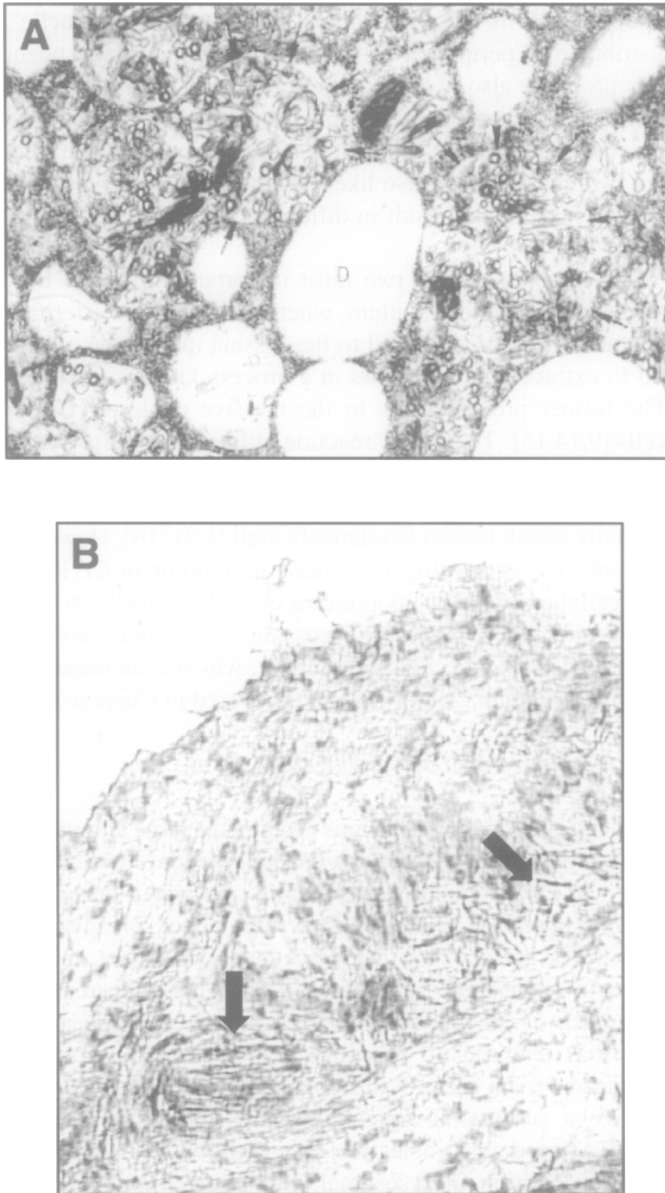


Fig. 3. (A) Intracellular free cholesterol accumulation in a lesional foam cell. Electron micrograph of the cytoplasm of a foam cell isolated from an advanced aortic atherosclerotic lesion in a cholesterol-fed rabbit. The cell was treated with filipin, which forms spicules with unesterified cholesterol. Multiple spicules are observed in vesicles, shown to be lysosomes (depicted by arrows). Bar, 0.5 μm . (From Shio et al. (1979) *Lab. Invest.* 41: 160–167.) (B) Extracellular cholesterol crystals in an advanced atherosclerotic lesion. The section is from the proximal aorta of a fat-fed apolipoprotein E knockout mouse. This mouse model is often used to study atherosclerosis *in vivo* because the high plasma levels of remnant lipoproteins resulting from absence of apolipoprotein E leads to a much greater degree of atherosclerosis lesion development than observed in wild-type mice. The arrows depict the areas of cholesterol crystals.

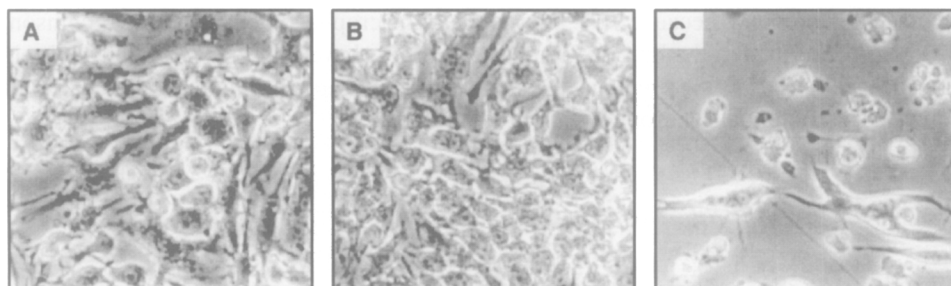


Fig. 4. FC-induced death in macrophages. Cultured macrophages were cultured under control conditions (A) or conditions leading primarily to CE loading (B) or FC loading (C). Note that FC-loaded macrophages, but not those loaded mostly with CE, have obvious signs of cytotoxicity, including detachment from the dish and altered morphology. (From Tabas et al. (1996) *J. Biol. Chem.* 271: 22773–22781.)

Free cholesterol-induced apoptosis in cultured macrophages involves both activation of Fas ligand and release of cytochrome *c* from mitochondria, which is associated with increased levels of Bax in these cells (I. Tabas, 2000 and 2001).

2.5. Cholesterol accumulation in lesional smooth muscle cells

Smooth muscle cells in atherosclerotic lesions also accumulate large amounts of cholesteryl ester, although the mechanisms involved are poorly understood [18]. As with macrophages, native LDL is a poor inducer of foam cell formation, but substantial cholesterol accumulation has been induced in cultured smooth muscle cells by aggregated LDL (L. Badimon, 1998). Cytokine treatment of cultured smooth muscle cells leads to the induction of the type A scavenger receptor, but there are no data specifically showing that oxidized LDL can cause foam cell formation in smooth muscle cells either in vitro or in vivo. Finally, remnant lipoproteins, including β -VLDL, cationized LDL, and cholesteryl ester emulsions, can induce cholesterol accumulation in cultured smooth muscle cells, but their roles in vivo are not known.

2.6. The fate of foam cell cholesterol in atheromata

Cholesteryl esters, which exist in membrane-bound droplets in the macrophage cytoplasm, undergo a continuous cycle of hydrolysis by neutral cholesteryl ester hydrolase and re-esterification by ACAT (M.S. Brown and J.L. Goldstein, 1980). If extracellular cholesterol acceptors, like HDL or apolipoprotein A-I, are available, some of this cholesterol can leave the cell, enter the circulation, and be transported to the liver in a process known as reverse cholesterol transport (Chapter 20). The fatty acyl and neutral lipid composition of foam cell droplets may influence this process by affecting the fluidity of the droplets. It is also possible that the foam cells themselves can leave lesions, although this event has been difficult to document. Finally, as described above, foam cells in atherosclerotic lesions die, and thus cellular stores of cholesteryl ester and free cholesterol, including cholesterol crystals, can be released into the lesions. This process undoubtedly contributes to the formation of the necrotic, or lipid, core of

advanced atheromata, because such areas contain macrophage debris (M.J. Mitchinson, 1995). As described in the Introduction, necrotic cores have important pathophysiologic significance because they predispose lesions to plaque rupture, the proximate cause of acute vascular clinical syndromes.

3. Oxysterols and atherosclerosis

3.1. Origins of oxysterols

Oxysterols arise from dietary sources, non-enzymatic oxidation, and enzymatic oxidation reactions [19]. The structure of some of the oxysterols that may be involved in atherosclerosis are shown in Fig. 5. Dietary oxysterols are incorporated into chylomicrons and include 7-ketocholesterol (7K), 7 α - and 7 β -hydroxycholesterol (7OH), and α - and β -5,6-epoxycholesterol (EPOX). 7OH, 7K, 24-hydroxycholesterol (24OH), 25-hydroxycholesterol (25OH), and 27-hydroxycholesterol (27OH) can be formed in vivo, but it is not clear whether non-enzymatic or enzymatic mechanism are involved. Specific enzymatic reactions include the formation of 7 α OH by cholesterol 7 α -hydroxylase in liver (Chapter 16) and 27OH and 3 β -hydroxy-5-cholestenoic acid by 27-hydroxylase in liver and macrophages.

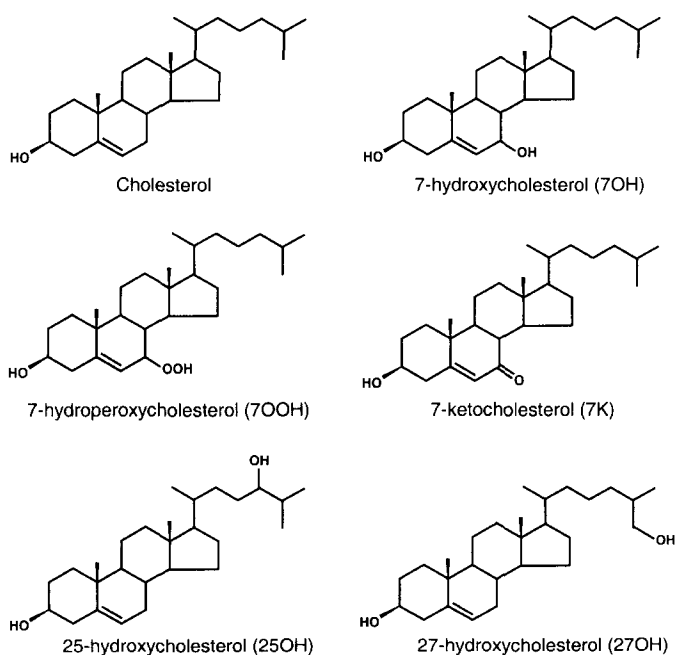


Fig. 5. Structures of some oxysterols that have been implicated in atherogenesis. (Adapted from Brown and Jessup [19].)

3.2. Oxysterols in plasma, lipoproteins, and atherosclerotic lesions

The most abundant oxysterols in human plasma are 27OH, 24OH, and 7 α OH (Fig. 5), and most of these are esterified to fatty acids at the 3 β position by lecithin:cholesterol acyltransferase [19]. Both free and esterified oxysterols partition in lipoproteins similar to cholesterol and cholesteryl ester, respectively, although 27OH is generally not found in VLDL, and unesterified 25OH can be associated with albumin. With the possible exception of 7 β OH, there is no clear relationship between plasma levels of oxysterols and atherosclerosis.

Oxysterols are also found in copper-oxidized LDL and consist predominantly of 7K, 7-hydroperoxy-cholesterol (7OOH), 7OH, and EPOX. LDL oxidized by more physiologic means, such as during contact with macrophages or by incubation with lipoxygenase, accumulates 7OOH (Fig. 5). Myeloperoxidase-treated cholesterol leads to the formation of unique chlorinated cholesterol derivatives, which can give rise to cholesterol epoxides [19].

The predominant oxysterols in atherosclerotic lesions include 27OH and 7K, where the levels are approximately 1% of cholesterol, with smaller amounts of 7OH; as in plasma, the vast majority are in the esterified form [19]. These oxysterols are found mostly in macrophage foam cells, which probably reflects the abundance of 27OH (Section 3.3). Moreover, 27-hydroxylated 7K is found in human atherosclerotic lesions, probably via macrophage sterol 27-hydroxylase acting on 7K internalized by macrophages (W. Jessup, 2000). When cultured macrophages are incubated with copper-oxidized LDL, 50% of the accumulated sterols are oxysterols, many of which accumulate in lysosomes as non-ACAT-derived oxidized fatty acid esters of oxysterols (W. Jessup, 2000). Although copper-oxidized LDL contains substantial amounts of 7OOH, macrophages accumulate very little of this lipid probably due to conversion to 7OH by phospholipid hydroperoxide glutathione peroxidase [19].

3.3. Physiologic significance of oxysterols in atherosclerosis

The proposed roles of oxysterols in atherosclerosis are based primarily on the results of cell-culture experiments. There are a number of *in vivo* studies in which investigators have exposed animals to oxysterols through diet or injection, but the overall results are not conclusive. For example, in a review by Brown and Jessup of 13 oxysterol dietary studies, six showed an increase in atherosclerosis, but four demonstrated a decrease in lesion size and three showed no effect [19]. Differences in animal models and types and doses of oxysterols undoubtedly account for some of these differences.

There are four major areas of oxysterol biology that have emerged from cell-culture studies. These include oxysterol effects on the regulation of intracellular cholesterol metabolism, cellular cytotoxicity, sterol efflux from macrophages, and activation of nuclear transcription factors. Issues related to oxysterols and intracellular cholesterol metabolism are covered in detail in Chapters 15 and 16. In brief, both cholesterol and certain oxysterols, such as 25OH and 7K, suppress the proteolytic activation of sterol response element binding protein. This, in turn, leads to transcriptional down-regulation of the LDL receptor and certain enzymes in the cholesterol biosynthetic

and fatty acid synthesis and metabolism pathways. 25OH and 7K, like cholesterol, can also promote the degradation of 3-hydroxyglutaryl-3-methylglutaryl CoA reductase, a rate-limiting enzyme in isoprenoid and cholesterol biosynthesis, and 25OH can activate ACAT and suppress neutral cholesteryl ester hydrolase activity. Cells possess a protein that binds oxysterols, called oxysterol binding protein, but its role in cellular responses to oxysterols is not known. Importantly, the physiologic role of oxysterols in cellular cholesterol metabolism as it relates to atherosclerosis is far from certain, particularly because the concentrations of oxysterols used in most macrophage cell-culture studies far exceeds those found in macrophage foam cells *in vivo* [19].

Oxysterols have diverse roles in cholesterol efflux, a critical topic in foam cell biology. On the one hand, cells incubated with 7K and 25OH have decreased cholesterol efflux. Possible mechanisms include inhibition of membrane desorption of cholesterol or phospholipids or, as mentioned above, inhibition of lysosomal sphingomyelinase leading to lysosomal sequestration of cholesterol (M. Aviram, 1995). On the other hand, the conversion of cholesterol by macrophage sterol 27-hydroxylase to 27OH and 3 β -hydroxy-5-cholestenoic acid, which are efficiently effluxed from cells, has been proposed to promote sterol efflux from foam cells (I. Björkhem, 1994). Indeed, 27OH is a prominent oxysterol found in lesions, and some studies have shown an inverse correlation between 27-hydroxylase levels and atherosclerosis (N.R. Cary, 2001). Future studies with 27-hydroxylase-deficient mice in general, and mice with a macrophage-specific deficiency of 27-hydroxylase in particular, should shed further light on the physiologic importance of this pathway.

A major area of investigation has been on the cytotoxic effects of certain oxysterols on cultured endothelial cells, macrophages, and smooth muscle cells [19]. The most potent cytotoxic oxysterols include 27OH, 7 β OOH, 7 α OH, and 7K (G.M. Chisolm, 1996). These sterols damage cells through a variety of mechanisms including cholesterol starvation, membrane perturbation, cellular lipid peroxidation, and activation of apoptotic pathways. Although death of endothelial cells, macrophages, and smooth muscle cells does occur in atherosclerotic lesions and might be expected to promote lesion development and complications, the role of isolated oxysterols or oxysterols in oxidized LDL in these events is far from certain. The conditions used in many cell-culture studies, notably high concentrations of oxysterols and/or serum-free medium, may not reflect the situation *in vivo*.

Exciting recent work has revealed that certain oxysterols are activators of nuclear transcription factors [20]. In particular, four oxysterols found *in vivo*, 24,25 EPOX, 24OH, 22OH, and 27OH, but not cholesterol, activate LXR α , LXR β , and FXR (Chapter 16). Once activated, these molecules heterodimerize with activated RXR, forming active transcription factors which translocate to the nucleus and induce several genes important in atherosclerosis. In particular, a set of genes important in the reverse cholesterol transport pathway is activated by this pathway [21]. The proteins encoded by these genes include (1) macrophage ABCA1 and apolipoprotein E, which promote cholesterol efflux from foam cells, (2) plasma cholesteryl ester transfer protein, which transfers HDL-cholesterol to lipoproteins that can be internalized by hepatocytes, (3) and liver cholesterol 7 α -hydroxylase, which is the key enzyme that converts hepatocyte cholesterol into bile acids for excretion. Recent studies with genetically manipulated

mice have demonstrated the importance of the LXR pathway *in vivo*. For example, activation of RXR reduces atherosclerosis in apolipoprotein E knockout mice (J. Auwerx, 2001), and the livers of cholesterol-fed LXR α knockout mice accumulate very large amounts of cholesterol (D.J. Mangelsdorf, 1998).

In summary, oxysterols are known to exist in atherosclerotic lesions and have been demonstrated in cell-culture experiments to have profound cellular effects that could influence the development, progression, and reversal of atherosclerosis. The key question in this field of research, however, is whether the concentrations of oxysterols *in vivo* are high enough to influence atherogenesis. Thus far, only the oxysterol-activated nuclear transcription pathway has been directly supported by *in vivo* data, and even in this case the precise roles and identification of the activating oxysterols *in vivo* have not yet been elucidated.

4. Triglycerides and atherosclerosis

There are two major issues that arise when considering the role of triglycerides in atherosclerosis: the effect of triglyceride-containing lipoproteins in the plasma on atherosclerotic lesion development, and the direct role of arterial-wall triglycerides in atherogenesis. The association between triglyceride-rich lipoproteins and atherosclerotic vascular disease is often difficult to assess due to complex metabolic relationships between these lipoproteins and other risk factors for atherosclerosis, including low plasma HDL and hyperfibrinogenemia [22]. Certainly, those triglyceride-containing lipoproteins that also have a high content of cholesterol, such as remnant lipoproteins, have been shown to be associated with atherosclerotic disease and probably function largely by delivering large amounts of cholesterol to the subendothelial space. The possibility that triglycerides and triglyceride-derived fatty acids also contribute to the atherogenicity of these lipoproteins, however, must also be considered (Section 5). Interestingly, metabolic disorders resulting in severe increases in plasma triglyceride, such as lipoprotein lipase deficiency, are not associated with increased risk of atherosclerosis. In these disorders, the triglyceride-rich lipoproteins are so large that they cannot enter the arterial wall (D.B. Zilversmit, 1989).

Triglycerides constitute a measurable proportion of the lipid content of atherosclerotic lesions, although considerably less than that of cholesterol. In one study, for example, the weight-percentages of triglycerides in fatty streaks and advanced lipid-rich lesions were 2.8 and 6.0, respectively; the corresponding total cholesterol percentages were 9.6 and 31.5, respectively [16]. However, both extracellular and intracellular triglycerides could play an important role in atherogenesis by serving as a source of free fatty acids following hydrolysis by extracellular and intracellular lipases. Indeed, the relatively low content of lesional triglyceride may be partially due to this process. As discussed in Section 5, free fatty acids are precursors of potentially important bioactive lipids.

Lesional cells in general, and macrophages in particular, can accumulate triglycerides via the uptake of triglyceride-rich lipoproteins and by intracellular triglyceride synthesis. Although triglycerides are a relatively minor component of neutral lipid droplets in lesional foam cells, even small percentages can lower the melting temperature of these

droplets [16]. Polyunsaturated fatty acids in the cholesteryl esters and triglycerides of lipid droplets also lower their melting temperature. Liquid neutral lipid droplets in foam cells are hydrolyzed at a more rapid rate than liquid crystalline droplets, and thus foam cells with a higher content of triglycerides may have an increased rate of cholesterol efflux (J.M. Glick, 1989).

5. Fatty acids and atherosclerosis

5.1. Direct effects of fatty acids

Fatty acids may have direct effects on atherogenesis and are also the precursors of specific bioactive lipids that may have important roles in lesion development. Lesions contain up to 0.4 mg of free fatty acids per gram of wet tissue (L. Robert, 1976). In terms of direct effects, high concentrations of extracellular free fatty acids could, in theory, lower the pH of focal areas in lesions, thus enabling the action of certain enzymes, such as lysosomal hydrolases that are secreted or leak from cells. When taken up by cells, fatty acids stimulate cholesteryl ester, phospholipid, and triglyceride syntheses. Individual types of fatty acids can have specific effects. For example, oleate, but not linoleate, is a potent stimulator of the ACAT reaction [15], and neutral lipids esterified to polyunsaturated fatty acids have a lower melting temperature, which tends to promote neutral lipid hydrolysis and lipid efflux.

5.2. Oxidation of long-chain polyunsaturated fatty acids: introduction

The major bioactive products of free fatty acid metabolism relevant to atherosclerosis are those that result from enzymatic or non-enzymatic oxidation of polyunsaturated long chain fatty acids. In most cases, these fatty acids are derived from phospholipase A₂-mediated hydrolysis of phospholipids (Chapter 11) in cellular membranes or lipoproteins, or from lysosomal hydrolysis of lipoproteins after internalization by lesional cells. In particular, arachidonic acid is released from cellular membrane phospholipids by arachidonic acid-selective cytosolic phospholipase A₂. In addition, there is evidence that group II secretory phospholipase A₂ (Chapter 11) hydrolyzes extracellular lesional lipoproteins, and lysosomal phospholipases and cholesterol esterase release fatty acids from the phospholipids and cholesteryl esters of internalized lipoproteins. Indeed, Goldstein and Brown recently surmised that at least one aspect of the atherogenicity of LDL may lie in its ability to deliver unsaturated fatty acids, in the form of phospholipids and cholesteryl esters, to lesions (J.L. Goldstein and M.S. Brown, 2001).

5.3. Oxidative metabolites of arachidonic acid

An important fate of arachidonic acid is enzymatic conversion to prostaglandins by one of two prostaglandin G/H synthases [23]. As described Chapter 13, these enzymes have both cyclooxygenase and hydroperoxidase activities, and are often referred to as COX-1 and COX-2. Although atherosclerotic lesions express both isoforms, mature

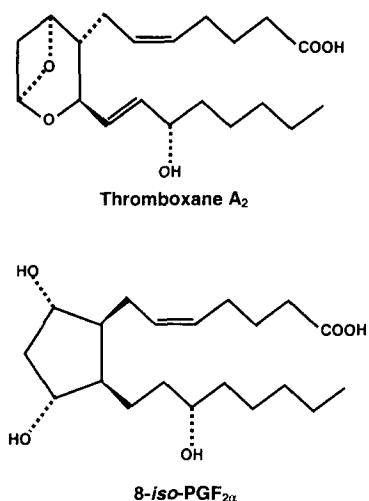


Fig. 6. Structures of two arachidonic acid derivatives proposed to play important roles in thrombosis and atherogenesis. Thromboxane A₂ is a potent inducer of platelet aggregation that contributes to acute thrombosis in advanced atherosclerosis in vivo. The isoprostane 8-iso-PGF_{2α} is being investigated as a marker of oxidative stress in atherosclerosis and may also have direct atherogenic effects on platelets and smooth muscle cells.

human platelets express only COX-1. In this regard, the most well-documented role of a cyclooxygenase product on atherothrombotic vascular disease is platelet-derived thromboxane A₂ (Fig. 6). Thromboxane A₂ is a potent inducer of platelet aggregation and vasoconstriction, and aspirin-induced inhibition of platelet COX-1 accounts for its benefit in the secondary prevention of strokes and myocardial infarction. The roles of other prostaglandins synthesized in lesions are uncertain. For example, vascular endothelial cells synthesize prostacyclin, which blocks platelet aggregation, cellular interactions, and vascular smooth muscle cell proliferation in vitro and in vivo. However, there is no evidence using drug-induced inhibition of prostacyclin synthesis or prostacyclin receptor-deficient mice that prostacyclin has a net effect on atherosclerotic lesion development in vivo (G.A. Fitzgerald, 2001). A third COX-derived prostaglandin that has received recent interest is 15-deoxy-D12,14-prostaglandin J₂ (15d-PGJ₂), which, at least in vitro, is an agonist of peroxisomal proliferator-activated receptor-γ (PPARγ) (C.K. Glass, 1998). PPARγ is expressed in atherosclerotic lesions and in cultured endothelial cells, vascular smooth muscle cells, and monocyte/macrophages (B. Staels, 2000). Although cell-culture studies have revealed a variety of biological effects that may be pro- or anti-atherogenic, recent in vivo studies suggest that PPARγ plays an anti-atherogenic role (C.K. Glass, 2000). The physiologic significance of 15d-PGJ₂, which also acts on IκB kinase, in PPARγ biology remains to be determined.

A relatively new class of oxidized arachidonic acid derivatives with potential relevance to atherosclerosis are F₂ isoprostanes [24] (Fig. 6). These compounds form as a result of non-enzymatic, free-radical attack of the fatty acid moieties of cellular or lipoprotein phospholipids, followed by release of the isoprostanes from the phospho-

lipids by a phospholipase. 8-*iso*-PGF₂ may also be formed by the action of COX-1 or COX-2 in platelets or monocytes, respectively, but the significance of COX-dependent 8-*iso*-PGF₂ formation *in vivo* is unproven. F₂ isoprostanes circulate in the plasma and appear in the urine as free compounds or esterified to phospholipids, and 8-*iso*-PGF₂ is found in atherosclerotic lesions in association with macrophages and smooth muscle cells. The potential significance of isoprostanes to atherosclerosis are their effects on platelets and vascular cells, as demonstrated in cell-culture studies, and their potential usefulness as a non-invasive marker of oxidant stress. 8-*iso*-PGF₂ induces platelet aggregation, DNA synthesis in vascular smooth muscle cells, and vasoconstriction. Moreover, elevated levels of F₂ isoprostanes are found in cigarette smokers, diabetics, and subjects with hypercholesterolemia, where they may serve as an indicator of increased lipid peroxidation.

Arachidonic acid can also be oxidized by 5-, 12-, and/or 15-lipoxygenases to various mono-, di-, and trihydroxyderivatives called leukotrienes (Chapter 13), some of which are present in atherosclerotic lesions [25]. The monohydroxylated leukotrienes 12(*S*)- and 15(*S*)-HETEs can be produced by human arterial endothelial cells and can promote monocyte adherence, an important early event in atherogenesis. Although 15(*S*)-HETE is found at relatively high levels in human atherosclerotic lesions, its role *in vivo* is not known. Atheromatous tissue has the capacity to synthesize the dihydroxylated leukotrienes LTC₄ and LTB₄ (R. De Caterina, 1988; C. Patrono, 1992). LTC₄ can be made by monocytes, macrophages, and endothelial cells, and LTB₄ is synthesized by activated monocytes. In theory, LTC₄ could promote vasoconstriction, and LTB₄ could contribute to atherosclerosis-related endothelial alterations, such as increased permeability and adhesiveness. Moreover, LTB₄ is also an activator of PPAR α , which appears to promote atherogenesis *in vivo* (C.F. Semenkovich, 2001). Finally, the trihydroxylated derivatives of arachidonic acid, the lipoxins, possess some anti-inflammatory properties, but it has been speculated that their ability to induce monocyte chemotaxis might promote atherogenesis [25].

Another fate of arachidonic acid with potential relevance to atherosclerosis is cytochrome P450 monooxygenase-derived metabolism to epoxyeicosatrienoic acids (EETs), which may also be formed nonenzymatically by the interaction of arachidonic acid with free radicals [25]. EET synthesis in cultured endothelial cells can be induced by LDL, and EETs are found both in LDL and in human atherosclerotic lesions. Biological effects of EETs include potentially anti-atherogenic effects, such as vasodilatation and prevention of platelet aggregation, and atherogenic responses, such as increased monocyte adhesion.

5.4. Atherogenic and anti-atherogenic effects of other long-chain polyunsaturated acids

Dietary intake of n-6 fatty acids such as linoleic acid, and n-3 fatty acids, such as the fish oils eicosapentanoic acid and docosahexaenoic acid, lower plasma cholesterol and antagonize platelet activation, but the fish oils are much more potent in this regard [26]. In particular, n-3 fatty acids competitively inhibit thromboxane synthesis in platelets but not prostacyclin synthesis in endothelial cells. These fatty acids have also been shown to have other potentially anti-atherogenic effects, such as inhibition of monocyte cytokine

synthesis, smooth muscle cell proliferation, and monocyte adhesion to endothelial cells. While dietary intake of n-3 fatty acid-rich fish oils appears to be atheroprotective, human and animal dietary studies with the n-6 fatty acid linoleic acid have yielded conflicting results in terms of effects on both plasma lipoproteins and atherosclerosis. Indeed, excess amounts of both n-3 and n-6 fatty acids may actually promote oxidation, inflammation, and possibly atherogenesis (M. Toberek, 1998). In this context, enzymatic and non-enzymatic oxidation of linoleic acid in the *sn*-2 position of LDL phospholipids to 9- and 13-hydroxy derivatives is a key event in LDL oxidation (Section 6.2).

6. Phospholipids and related lipids

6.1. Introduction

Phospholipids compose the outer monolayer of lesional lipoproteins and the membranes of lesional cells. In lipoproteins, the phospholipid monolayer provides an amphipathic interface between the neutral lipid core and the aqueous external environment, and it provides the structural foundation for the various apolipoproteins (Chapter 18). In the specific context of atherosclerosis, the phospholipids of lesional lipoproteins are modified by various oxidative reactions that could have important pathological consequences. In lesional cells, membrane phospholipids not only play structural roles but also are precursors to important phospholipase-generated signaling molecules that may play important roles in atherogenesis.

6.2. Oxidative modification of phosphatidylcholine in lesional lipoproteins

Oxidation of LDL and probably other lesional lipoproteins occurs in atherosclerotic lesions and may contribute to lesion pathology at various stages of atherogenesis [10,11]. Based on in-vitro studies and, in some cases, genetically altered mutant mouse models, LDL oxidation may be triggered by oxidative enzymes secreted by lesional cells, including myeloperoxidase, inducible nitric oxide synthase, and, 15-lipoxygenase. In the above sections, oxidative modifications of cholesterol, cholesteryl ester, and free fatty acids in LDL were discussed. The phospholipids of LDL also undergo oxidative modification, and the products of these reactions are found in atherosclerotic lesions and have potentially important atherogenic effects on lesional cells.

The most abundant and important oxidative changes in LDL phospholipids are those that occur to the unsaturated fatty acids in the *sn*-2 position. Witztum and Berliner described several products of phospholipid oxidation that result from this process [27] (Fig. 7). An early event is the addition of oxygens to these fatty acids, resulting in the generation of hydroxy fatty acids, hydroperoxy fatty acids, and isoprostanes. In one model, lipoxygenase-generated hydroperoxyoctadecadienoic acid (HPODE) and hydroperoxyeicosatetraenoic acid (HPETE) in LDL surface phospholipids act as 'seeding' molecules. These hydroperoxy fatty acids then trigger the oxidation of arachidonate-containing phospholipids, notably, 1-palmitoyl-2-arachidonyl-*sn*-glycero-3-phosphorylcholine. This early series of events occurs before any oxidative

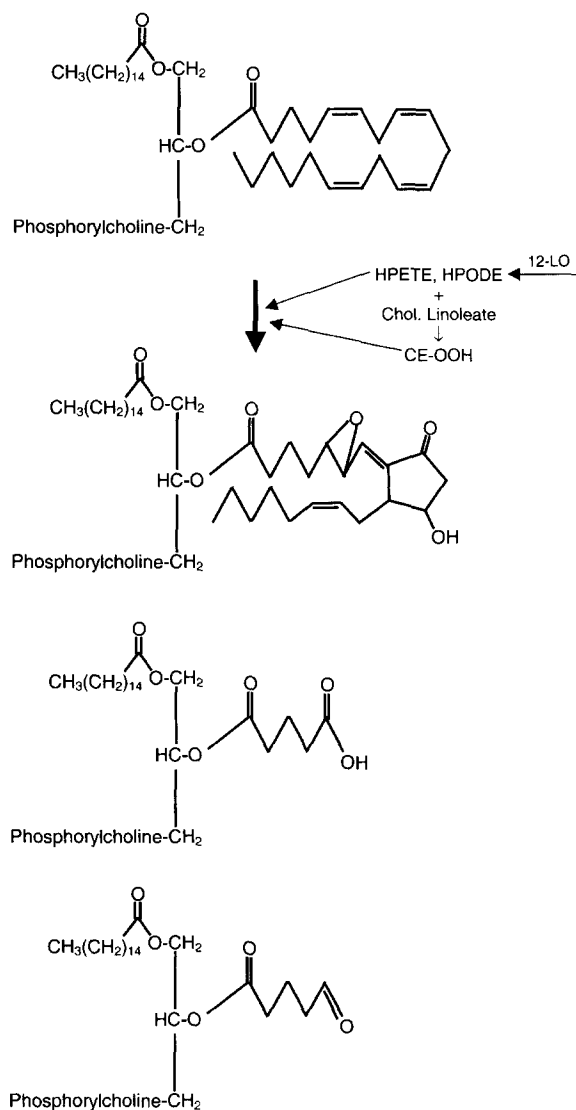


Fig. 7. Oxidation of LDL phospholipids in the generation of minimally modified LDL. 'Seeding' molecules like HPETE, HPODE, and cholesteryl linoleate hydroperoxide (CE-OOH) are proposed to trigger the oxidation of 1-palmitoyl-2-arachidonoyl phosphatidylcholine in LDL. This leads to the generation of three oxidized phosphatidylcholine species that confer atherogenic activity to minimally modified LDL. (Adapted from Navab et al. [28].)

modification of apo B100 and result in the generation of so-called 'minimally modified' LDL. Minimally modified LDL is found in atherosclerotic lesions and promotes monocyte binding to endothelial cells and monocyte chemotaxis in cultured cell studies [28]. The oxidized arachidonate-containing phospholipids, which can account for

much of the biological activity of minimally modified LDL, include those in which the arachidonyl group is modified to 5-oxovaleroyl, glutaroyl, or 5,6-epoxyisoprostane phosphatidylcholine. The potentially atherogenic molecules induced in endothelial cells and smooth muscle cells by these oxidized phospholipids include E-selectin, VCAM-1, monocyte chemoattractant protein-1, macrophage colony stimulating factor, P-selectin, and interleukin-8. These cellular effects, like those of LTB₄ (Section 5.3), likely involve the activation of PPAR α .

Another possible consequence of oxidation of *sn*-2 unsaturated fatty acids in oxidized LDL is fragmentation of the fatty acid, resulting in a phospholipid with a short acyl group in the *sn*-2 position [27]. If the *sn*-2 fatty acid were arachidonate, the truncated acyl group would be a 5-carbon aldehyde or a 5-carbon carboxylic acid. If the *sn*-2 group were linoleate, the products would be a 9-carbon aldehyde or carboxylic acid. Phospholipids containing these shortened *sn*-2 acyl chains have biological activity similar to that of platelet-activating factors, where an ether-linked fatty acyl group occupies the *sn*-1 position and an acetyl group is in the *sn*-2 position. By interaction with a G-protein-coupled receptor on a variety of cell types, platelet activating factor activates both platelets and leukocytes and increases vascular permeability. In terms of atherosclerosis, studies with PAF receptor antagonists have suggested that the chemotactic activity of minimally modified LDL may be mediated through platelet activating factor-like phospholipids acting directly on monocytes (P.D. Reaven, 1997). Similarly, the smooth muscle cell mitogenic activity of oxidized LDL can be mimicked by PC containing a 5-carbon carboxylic acid and can be blocked by a platelet activating factor receptor antagonist (S.M. Prescott, 1995).

Phospholipase A₂-mediated hydrolysis of oxidized phospholipids can result in the release of either intact oxygenated free fatty acids by phospholipase A₂, discussed in Section 5, or fragmented fatty acids, such as malondialdehyde, which can lead to protein modification. The other product of this reaction is lysophosphatidylcholine (D. Steinberg, 1988) [29]. In-vitro studies have revealed multiple effects of lysophosphatidylcholine on lesional cells, including expression of adhesion molecules on endothelial cells, monocyte chemotaxis and macrophage scavenger receptor expression, growth factor expression by smooth muscle cells and macrophages, cellular cytotoxicity, and inhibition of T cell activation. Although it is not known whether enough lysophosphatidylcholine exists in lesions to cause these effects in vivo, a receptor for lysophospholipids, called G2A, was recently identified, and studies with G2A-deficient mice suggest a possible role for lysophospholipids in the regulation of T cell activation in vivo (O.N. Witte, 2001). Finally, adducts of oxidized phospholipids and apo B100 can result from LDL oxidation. These adducts are recognized by macrophages and thus can mediate cellular uptake of oxidized LDL, and they form a potent epitope that can elicit cellular and humoral immune responses that may play important roles in atherogenesis (J.L. Witztum, 1999).

Navab et al. have suggested that one of the anti-atherogenic mechanisms of HDL may be its ability to prevent LDL oxidation or reduce its atherogenic activity [28]. Multiple mechanisms may be involved, including removal of the 'seeding' molecules HPODE and HPETE and degradation of the oxidized phospholipids themselves by paraoxonase, an esterase/peroxidase, and platelet-activating factor-acetylhydrolase, a

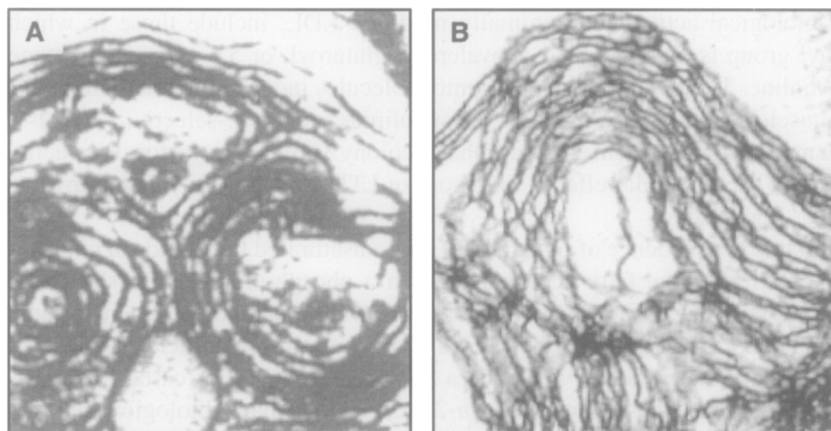


Fig. 8. Phospholipids whorls in cholesterol loaded macrophages. (A) Electron micrograph showing a membrane whorl in the cytoplasm of a foam cell. The cell was isolated from an advanced aortic atherosclerotic lesion in a cholesterol-fed rabbit. (From Shio et al. (1979) *Lab. Invest.* 41: 160–167.) (B) Electron micrograph showing a membrane whorl in the cytoplasm of an FC-loaded J774 macrophage. The cell was in the 'adaptive' stage, i.e., before the onset of FC-induced death. (From Shiratori et al. (1994) *J. Biol. Chem.* 269: 11337–11348.)

lipoprotein-bound phospholipase A₂-like enzyme that can cleave oxidized acyl groups from the *sn*-2 position of oxidized phospholipids. In apo E knockout mice, targeted disruption of the paraoxonase gene increases lipoprotein oxidation and atherosclerosis (A.J. Lusis, 2000).

6.3. The phospholipids of lesional cells

Phosphatidylcholine is the major phospholipid of lesional cells and, as mentioned above, serves both structural and signaling functions. In terms of cellular membrane structure, the cholesterol:phospholipid ratio in lesional cells must be kept within a certain limit in order for the proper functioning of membrane proteins [9]. Cholesterol-rich foam cells isolated from atherosclerotic lesions have intracellular phospholipid whorl-like structures, and PC biosynthesis is increased in lesional areas of the arterial wall (Fig. 8). Cell culture studies have revealed that free cholesterol (FC) loading of macrophages directly leads to the activation of CTP:phosphocholine cytidyltransferase and an increase in phosphatidylcholine biosynthesis and mass. Proof that this is an adaptive response to FC excess comes from a study in which the cytidyltransferase α gene was disrupted in macrophages, which resulted in accelerated FC-induced death (I. Tabas, 2000). Thus, activation of phosphatidylcholine biosynthesis in cholesterol-loaded lesional macrophages may help to protect these cells from the toxicity of FC excess [9].

Cellular phospholipids, particularly phosphatidylinositol and phospholipids containing unsaturated fatty acids in the *sn*-2 position, are precursors to a variety of signaling molecules (Chapter 12). These include (1) diacylglycerol and inositol trisphosphate by

phosphatidylinositol-specific phospholipase C-induced hydrolysis of phosphatidylinositol, (2) phosphatidic acid by phospholipase D, (3) fatty acids and lysophosphatidylcholine by phospholipase A₂, and (4) platelet activating factor-like molecules by oxidation. Diacylglycerol activates protein kinase C, and inositol triphosphate leads to intracellular calcium release. Both of these reactions are involved in a variety of signaling processes that occur in lesional smooth muscle cells, macrophages, and endothelial cells, including responses to cytokines and growth factors. Oxidized LDL has been shown to activate phospholipase D in cultured vascular smooth muscle cells by a tyrosine kinase-mediated mechanism, and phosphatidic acid could mimic the proliferative effects of oxidized LDL in these cells (S. Parthasarathy, 1995). Finally, given the potential importance of apoptosis of macrophages and smooth muscle cells in atherosclerosis (above), phosphatidylserine is an important phospholipid in lesional cells. Phosphatidylserine is normally a component of the inner leaflet of the plasma membrane, but it becomes externalized during apoptosis and acts as a recognition signal and ligand for phagocytes. Interestingly, both CD36 and scavenger receptor B-1 on macrophages are receptors for phosphatidylserine on apoptotic cells as well as for oxidized LDL. Thus, it is possible that phagocytosis and clearance of apoptotic cells in lesions may be competitively inhibited by oxidized LDL (D. Steinberg, 1999).

6.4. Sphingomyelin and ceramide

Sphingomyelin is an important component of both the phospholipid monolayer of lesional lipoproteins and of membranes of lesional cells. In atherogenic lipoproteins like LDL, hydrolysis of sphingomyelin to ceramide results in lipoprotein aggregation and fusion, resulting in the formation of large aggregates that appear similar to those that occur in extracellular regions of the subendothelium of atherosclerotic lesions [8]. The mechanism of sphingomyelinase-induced aggregation and fusion, which is dependent on lipoprotein ceramide content, probably lies in both the physical effects of ceramide on lipoprotein structure and on hydrogen bonding between ceramide on one particle and phospholipids on a neighboring particle. Tabas and coworkers have provided evidence that extracellular hydrolysis of LDL-sphingomyelin by sphingomyelinase occurs in the subendothelium of atherosclerotic lesions and is catalyzed by a form of acid sphingomyelinase, called S-sphingomyelinase, that is secreted by endothelial cells and macrophages [13]. Although the overall importance of this reaction in vivo remains to be determined, its potential importance is that subendothelial lipoprotein retention promotes lipoprotein retention in the arterial wall and is a potent substrate for macrophage and possibly smooth muscle cell foam cell formation. Lipoproteins with a high sphingomyelin:phospholipid ratio are particularly good substrates for S-sphingomyelinase. In this context, lipoproteins isolated from atherosclerotic lesions have a very high sphingomyelin (as well as ceramide) content. Moreover, a recent analysis of plasma samples from a case:control study showed that a high sphingomyelin:phospholipid ratio in plasma lipoproteins was an independent risk factor for coronary artery disease in humans (X.C. Jiang, 2000).

HDL also contains sphingomyelin. Because sphingomyelin avidly binds cholesterol, sphingomyelin may increase the ability of HDL to act as an extracellular acceptor for

cholesterol effluxed from cells (G. Rothblat, 1997). However, HDL-sphingomyelin has also been shown to inhibit the binding of lecithin:cholesterol acyltransferase to the lipoprotein, and so this effect may balance the effect of HDL-sphingomyelin-induced cholesterol efflux on reverse cholesterol transport (A. Jonas, 1996).

Sphingomyelin in cellular membranes may have several important roles related to atherogenesis. Because sphingomyelin interacts strongly with cholesterol, accumulation of sphingomyelin in cellular sites that are involved in cholesterol trafficking may influence cellular cholesterol distribution. For example, the defective intracellular trafficking and ACAT-mediated esterification of oxidized LDL-derived cholesterol in macrophages may be due to the inhibition of acid sphingomyelinase by oxidized LDL lipids (M. Aviram, 1995). Moreover, the sphingomyelin accumulation that occurs in acid sphingomyelinase-deficient macrophages leads to defective cholesterol trafficking and efflux (I. Tabas, 2001). On the other side of this issue is the response of sphingomyelin biosynthesis to FC loading of cells. Investigators have shown that the synthesis and mass of cellular sphingomyelin is increased in advanced atherosclerotic lesions [16]. In cultured FC-loaded macrophages, sphingomyelin biosynthesis is increased, suggesting that sphingomyelin, like phosphatidylcholine, plays a role in the adaptation of cells to FC excess.

Another area of sphingomyelin biology with potential relevance to atherosclerosis is related to cell signaling [30]. Hydrolysis of cellular sphingomyelin by either neutral or acid sphingomyelinases results in the generation of intracellular ceramide, which is involved in a variety of cell-signaling reactions (Chapter 14). In terms of atherosclerosis, ceramide-mediated signaling may play roles in smooth muscle cell proliferation and apoptosis and macrophage apoptosis [30]. Alterations in ceramide synthesis and ceramide hydrolysis by cellular ceramidases may also influence these events. In this context, ceramidase-generated sphingosine can be phosphorylated to sphingosine-1-phosphate, which is another signaling molecule. For example, Ross and colleagues showed that sphingosine-1-phosphate blocks the migration of vascular smooth muscle cells induced by platelet-derived growth factor (R. Ross, 1995).

6.5. *Glycosphingolipids*

Sugar transferases can convert ceramide to a variety of glycosphingolipids, including neutral glycosphingolipids such as glucosylceramide and lactosylceramide, and polar glycosphingolipids such as gangliosides, which contain ceramide, sugars, and sialic acid and/or *N*-glycolylneuraminic acid (Chapter 14). Glycosphingolipids are found both in plasma lipoproteins and in the cells and extracellular regions of atherosclerotic lesions [30]. Chatterjee and colleagues have proposed that lactosylceramide, synthesized from glucosylceramide by the enzyme UDP-galactose:glucosylceramide- β 1-4 galactosyltransferase activity (GalT-2), is a lipid second messenger that is involved in the proliferation of vascular smooth muscle cells by oxidized LDL [30]. In cultured smooth muscle cells, oxidized LDL stimulated GalT-2 activity and lactosylceramide synthesis. Proliferation induced by oxidized LDL in these cells was blocked by an inhibitor of GalT-2, and exogenous lactosylceramide was able to stimulate proliferation in the absence of oxidized LDL. The mechanism may involve 5-oxovaleroyl

phosphatidylcholine-mediated stimulation of NADPH oxidase by lactosylceramide, leading to signaling cascade triggered by superoxide radicals and involving Ras activation and p44-mitogen-activated protein kinase. Interestingly, native LDL was shown to actually decrease GalT-2 activity and lactosylceramide synthesis in smooth muscle cells in an LDL receptor-dependent manner. While these cell-culture studies have provided a potentially interesting role for GalT-2 activity and lactosylceramide in atherosclerosis, the physiologic significance of these findings overall must await future in-vivo studies.

Table 2
Summary of proposed roles of lesional lipids in atherosclerosis

Lipid	Overall effects in atherosclerosis	Specific examples
Free cholesterol	Accumulation in and alteration of macrophages and smooth muscle cells, including gene regulation and, in excess, death	Stimulation of ACAT Repress transcription of LDL receptor gene FC-induced macrophage death
Cholesteryl ester	Accumulation in macrophages and smooth muscle cells Substrate for oxidation	Foam cell formation Cholesterol linoleate hydroperoxide as a pro-oxidant
Oxysterols	Regulation of cellular cholesterol metabolism Cytotoxicity Sterol efflux pathways Activation of nuclear transcription factors	Stimulation of ACAT 7K-induced macrophage death Efflux of 27OH Activation of LXR by 22OH
Triglycerides	Source of fatty acids Affect neutral lipid droplet fluidity in foam cells	Liquid crystalline → liquid neutral transformation of foam cell droplets
Fatty acids	Stimulate CE, triglyceride, and phospholipid synthesis Polyunsaturated fatty acids are sources of bioactive eicosanoids	Thromboxane A ₂ → platelet aggregation Isoprostanes → smooth muscle cell proliferation
Phospholipids (other than sphingolipids)	Structural roles in lipoproteins and lesional cells Source of signaling molecules Substrate for oxidative modification into bioactive molecules	Part of adaptive response to FC-induced cytotoxicity Lysophosphatidylcholine → monocyte chemotaxis Oxidized phospholipids → induction of endothelial adhesion molecules
Sphingolipids	Source of signaling molecules Involve on lipoprotein aggregation Influence intracellular cholesterol trafficking	Ceramide → lesional cell death and proliferation Sphingomyelinase-induced LDL aggregation Lactosylceramide → smooth muscle cell proliferation

7. Future directions

The potential roles of the many types of lesional and lipoprotein lipids in atherogenesis is staggering (Table 2). Not surprisingly, most studies investigating these roles have been conducted on cultured cells where the concentrations of the lipids and the overall state of cells may be very different from that occurring in atherosclerotic lesions. Thus, one of the most important, and difficult, areas in future studies will be to sort out these effects *in vivo* through the use of inhibitory compounds or genetic manipulations in mice. In some cases, such studies have provided impressive results, such as the decrease in atherosclerosis observed in 15-lipoxygenase knockout mice (C.D. Funk, 1999). On the other hand, the effects of anti-oxidants in both humans and animal models have yielded conflicting results (R. Stocker, 2001). Further understanding of the molecular basis of lipid synthesis and catabolism, and of the action of bioactive lipids in cells, will help in the design of improved *in-vivo* models. The most important of these lipids include cholesterol, oxidized phospholipids and fatty acids, oxysterols, and sphingolipid derivatives. Key areas for investigating the cellular effects of bioactive lipids include (1) inflammatory responses in endothelial cells, T cells, and macrophages, (2) secretion of atherogenic and anti-atherogenic molecules by lesional cells, (3) proliferation of macrophages and smooth muscle cells, (4) and apoptotic and necrotic death in lesional cells. Moreover, the mechanism and consequences of macrophage and smooth muscle cell lipid accumulation, particular cholesteryl ester and free cholesterol accumulation, represent fundamental areas in lesional cell biology that require further investigation. New advances in genomics and proteomics have already begun to aid in these effort and will increasingly do so. Ultimately, the goal of these studies is to elucidate novel targets for drug or gene therapy that can complement plasma cholesterol-lowering therapy in the fight against the leading cause of mortality worldwide.

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